bq769x0 3 节至 15 节串联电池监控系列 用于锂离子和磷酸盐电池应用

1 介绍

1.1 特性

- 模拟前端 (AFE) 监控特性
  - 纯数字接口
  - 内部模数转换器 (ADC) 测量电池电压、芯片温度和外部热敏电阻
  - 一个单独的、内部 ADC 测量电池组电流（库伦电荷计数器）
  - 直接支持多达三个热敏电阻 (103AT)
- 硬件保护特性
  - 放电过流 (OCD)
  - 放电短路 (SCD)
  - 过压 (OV)
  - 欠压 (UV)
  - 次级保护器故障检测
- 附加特性
  - 集成电池均衡场效应晶体管 (FET)
  - 充电、放电低侧 NCH FET 驱动器
  - 到主机微控制器的警报中断
  - 2.5V 或 3.3V 输出电压稳压器
  - 无需 EEPROM 编程
  - 高电源电压最大绝对值（高达 108V）
  - 简单 I²C™兼容接口（循环冗余校验 (CRC) 选项）
  - 随机电池连接耐受

1.2 应用范围

- 轻型电动车辆 (LEV)：电动自行车 (eBike)，电动踏板车 (eScooter)，脚踏电动自行车 (Pedelec) 和踏板辅助自行车
- 电动和园艺工具
- 后备电池和不间断电源 (UPS) 系统
- 无线基站后备系统
- 12V，18V，24V，36V 和 48V 电池组

1.3 说明

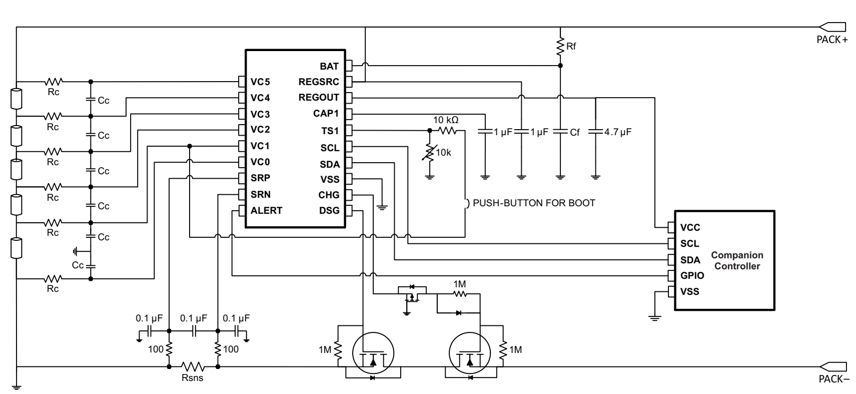
bq769x0 系列的稳健耐用型模拟前端 (AFE) 器件通常用作针对下一代高功率系统（例如，轻型电动车辆、电动工具和不间断电源）的完整电池组监控和保护解决方案的一部分。bq769x0 在设计时充分考虑了低功耗要求，不仅可通过使能/禁用集成电路 (IC) 中的子模块来控制整个芯片的电流消耗，而且还可以利用 SHIP 模式将电池组轻松切换至超低功耗状态。

器件信息[(1)]

器件型号	封装	封装尺寸（标称值）
bq76920	薄型小外形尺寸封装 (TSSOP) (20)	6.50mm x 4.40mm
bq76930	TSSOP (30)	7.80mm x 4.40mm
bq76940	TSSOP (44)	11.00mm x 4.40mm

(1) 要了解所有可用封装，请见数据表末尾的可订购产品附录。

1.4 简化系统图



bq769x0 3节至15节串联电池监控系列 用于锂离子和磷酸盐电池应用

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  - 一个单独的、内部 ADC 测量电池组电流(库伦 电荷计数器)
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- 硬件保护特性
  - 放电过流 (OCD)
  - 放电短路 (SCD)
  - 过压 (OV)
  - 欠压 (UV)
  - 次级保护器故障检测
- 附加特性
  - 集成电池均衡场效应晶体管 (FET)
  - 充电、放电低侧NCHFET 驱动器
  - 到主机微控制器的警报中断
  - 2.5V 或3.3V输出电压稳压器
  - 无需EEPROM 编程
  - 高电源电压最大绝对值(高达 108V)
  - 简单I²CTM兼容接口(循环冗余校验(CRC) 选 项)
  - 随机电池连接耐受

1.2 应应用用范范围围

- 轻型电动车辆 (LEV):电动自行车 (eBike),电动踏板车 (eScooter),脚踏电动自行车 (Pedelec) 和踏板辅助自行车

板车 (eScooter),脚踏电动自行车 (Pedelec) 和踏
<board>辅助自行车</board>

- 电动和园艺工具
- 备用电池和不间断电源 (UPS) 系统
- 无线基站后备系统
- 12V,18V,24V,36V 和 48V 电池组

1.3 说说明明

bq769x0系列的稳健耐用型模拟前端 (AFE) 器件通常用作针对下一代高功率系统（例如，轻型电动车辆、电动工具和不间断电源）的完整电池组监控和保护解决方案的一部分。bq769x0 在设计时充分考虑了低功耗要求，不仅可通过使能/禁用集成电路 (IC) 中的子模块来控制整个芯片的电流消耗，而且还可以利用 SHIP 模式将电池组轻松切换至超低功耗状态。

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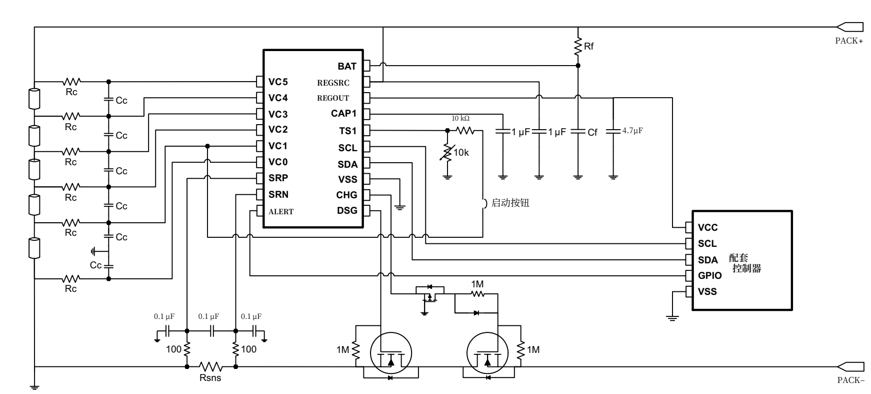
bq76930	TSSOP (30)	7.80mm x 4.40mm
bq76940	TSSOP (44)	11.00mm x 4.40mm

(1) 要了解所有可用封装,请见数据表末尾的可订购产品附录。

1.4 简化系统图



内容

2 修订历史记录

注：之前版本的页码可能与当前版本有所不同。

Changes from Revision E (November 2014) to Revision F — **Page**

- Changed bq7693002 From: Product Preview To Production in the *Device Comparison Table* 4
- Added bq7693007 device to the *Device Comparison Table* 4
- Changed table note to group ground reference in *bq76930 Pin Functions* 7
- Changed table note to group ground reference in *bq76940 Pin Functions* 9
- Changed 10th cell to 11th cell in the Description of pin 29 10
- Changed table formats for online data sheet 11
- Changed *Handing Ratings* table to *ESD Ratings* 11
- 已更改 note for R1 on 图 7-3 26
- 已添加 more description to the *Communications Subsystem* section 29
- 已更改 "SHUTA" to "SHUT_A" and "SHUTB" to "SHUT_B" in the *SHIP Mode* section 30
- 已更改 CAUTION verbiage (editorial) 31
- 已更改 the SHUT_A, SHUT_B bit descriptions in the *SYS_CTRL1 (0x04)* table 35

Changes from Revision D (July 2014) to Revision E — **Page**

- Added a note to the *Absolute Maximum Ratings* table 11
- Changed the *Handling Ratings* 11
- 已添加 the cross-reference to a new table note at V_{ALERT_IH} in *Electrical Characteristics* 16
- 已添加 a new table note 16
- 已添加 the *Typical Characteristics* section 18
- 已更改 the *Alert* section 28
- 已更改 verbiage in *Communications Subsystem* 29
- 已删除 the READ/WRITE RSVD register in the *Register Map* 32
- 已删除 the READ ONLY register information in the *Register Map* 32
- 已更改 the reset for Bit 3 in the PROTECT3 register 37
- 已更改 wording in the ADCGAIN bit descriptions 40
- 已添加 *Application and Implementation* note 42

bq76920, bq76930, bq76940
ZHCSCE2F –2013年10月–修订于2015年11月
TEXAS INSTRUMENTS
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5 9 6 4 76940 11 . bq 引脚图.电气规格.
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2 修订修订历史记录录

注：之前版本的页码可能与当前版本有所不同。

从修订版 E（2014 年 11 月）到修订版 F 页的变更

7693002
7693007
76930
76940
10 11 29
1 7 3
4 设备比较表中将 bq 从：产品预览 更改为生产.在设备比较表中添加 bq 设备.
4 将表格注释更改为在 bq
7 引脚功能 中分组接地参考.将表格注释更改为在 bq
9 引脚功能 中分组接地参考.在引脚描述中将 th 单元格更改为 th 单元格.
10 更改了在线数据表的表格格式.
11 将处理等级表更改为 ESD 等级表.
11 已更改图 - 中 R 的注释.
26 已向通信子系统部分添加更多描述.
29 在 SHIP 模式部分中将 "SHUTA" 更改为 "SHUT_A" 和 "SHUTB" 更改为 "SHUT_B".
30 已更改警告文本（编辑）.
31 35 已更改 SYS_CTRL1（0x04）表中 SHUT_A、SHUT_B 位的描述.

从修订版D（2014年7月）到修订版E的变更 Page

在“绝对最大额定值”表格中添加了注释.11
更改了“处理额定值”. 11
已添在V处添加了指向新表格注释的交叉引用 ALERT_IH in 电气特性 16
已添加一个新的表格笔记. 16
已添加典型特性部分. 18
已更改警报部分
28 已更改通信子系统的措辞.29
已删除寄存器映射中的 READ/WRITE RSVD 寄存器.32
已删除 Register Map 中的 READONLY 寄存器信息.32
已更改 PROTECT3 寄存器中 Bit 3 的复位.37 已更改 ADCGAIN 位描述中的措辞.40 已添加 应用和实现说明.42

2 修订历史记录
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- 已添加 *Design Requirements* 45
- 已添加 the *Detailed Design Procedure* 46
- 已更改 the *Layout Guidelines* 50
- 已更改 the *Layout Example* 50
- 已添加 a *Caution* 50
- 已更改 the Good Layout figure 51
- 已更改 the Weak Layout figure 52

Changes from Revision C (May 2014) to Revision D **Page**

- 已更改 table reference in *Integrated Hardware Protections* 25
- 已更改 paragraph 4 verbiage of *Integrated Hardware Protections* 25

Changes from Revision B (April 2014) to Revision C **Page**

- 已更改 文档格式 1
- Changed some devices from Product Preview to Production Data 4
- 已更改 a bit name in the PROTECT1 register 36
- 已更改 a bit name in the ADCGAIN2 register 41

Changes from Revision A (December 2013) to Revision B **Page**

- Changed Ordering Information table to 4
- Changed some devices to Product Preview 4
- Changed R_f max value in the *Recommended Operating Conditions* table 12
- 已更改 verbiage in *mmunications Subsystem* 29
- 已更改 SYS_STAT D6 bit name in the *Register Map* 32

Changes from Original (October 2013) to Revision A **Page**

- 已更改 数据手册的标题 1
- Changed some devices from Product Preview to Production Data 4
- 已更改 the $t_{INDCELL}$ test condition in *Electrical Characteristics* 14
- 已删除 duplicate CELLBAL3 table 34
- 已更改 bq76940 with bq783xx Companion Controller IC Schematic 45

- 已添加 设计要求. 45
- 已添加 详细设计流程.46
- 已更改 布局指南. 50
- 已更改 布局示例. 50
- 已添加 注意事项. 50
- 已更改 布局良好图. 51
- 已更改 布局不佳图. 52

从修订版 C（2014 年 5 月）到修订版 D 的变更 **Page**

- 已更改 集成硬件保护中的表格引用.25
- 已更改 集成硬件保护的第 4 段文字.25

从修订版 B（2014 年 4 月）到修订版 C 的变更 **Page**

- 已更改 文档格式. 1
- 将某些设备从产品预览更改为生产数据.4
- 已更改 PROTECT1 寄存器中的 a bit name.36
- 已更改 ADCGAIN2 寄存器中的 a bitname.41

从修订 A（2013 年 12 月）到修订 B 的变更 **Page**

- 已更改 Ordering Information 表格为. 4
- 已更改部分设备为 Product Preview.4
- 已更改 $R_f$max 值在 Recommended Operating Conditions 表格中.12 已更改 Communications Subsystem 中的 verbiage.29
- 已更改 Register Map 中的 SYS_STAT D6 位名称.32

从原始版本（2013年10月）到修订版 A 的变更 **Page**

- 已更改 数据手册的标题. 1
- 将某些设备从产品预览更改为生产数据.4
- 已更改 Electrical Characteristics 中的 $t_{INDCELL}$ test 条件.14
- 已删除 duplicate CELLBAL3 表格. 34
- 已更改 bq76940 withbq783xx 伴侣控制器 IC 原理图.45

3 说明（继续）

bq76920 器件支持多达 5 节串联电池或者典型值为 18V 的电池组；bq76930 可处理多达 10 节串联电池或者典型值为 36V 的电池组；而 bq76940 支持多达 15 节串联电池或者典型值为 48V 的电池组。这些 AFE 可用于管理各种电池化学成分，例如锂离子、磷酸铁锂，等等。

通过 I²C，主机控制器可以使用 bq769x0 来执行很多电池组管理功能，诸如监视（电池电压、电池组电流、电池组温度），保护（控制充电/放电 FET），以及均衡功能。集成模数 (A/D) 转换器可实现对关键系统参数的纯数字读取，这些参数的校准在 TI 制造过程中处理。

4 Device Comparison Table

TUBE	TAPE & REEL	CELLS	I²C ADDRESS (7-Bit)	LDO (V)	CRC	PACKAGE
bq7692000PW	bq7692000PWR			2.5	No	
bq7692001PW(1)	bq7692001PWR(1)		0x08		Yes	
bq7692002PW(1)	bq7692002PWR(1)	3–5			No	20-TSSOP (PW)
bq7692003PW	bq7692003PWR			3.3	Yes	
bq7692006PW	bq7692006PWR		0x18		No	
bq7693000DBT	bq7693000DBTR			2.5	No	
bq7693001DBT	bq7693001DBTR		0x08		Yes	
bq7693002DBT	bq7693002DBTR	6–10			No	30-TSSOP (DBT)
bq7693003DBT	bq7693003DBTR			3.3	Yes	
bq7693006DBT	bq7693006DBTR		0x18		No	
bq7693007DBT	bq7693007DBTR				Yes	
bq7694000DBT	bq7694000DBTR			2.5	No	
bq7694001DBT	bq7694001DBTR		0x08		Yes	
bq7694002DBT	bq7694002DBTR	9–15			No	44-TSSOP (DBT)
bq7694003DBT	bq7694003DBTR			3.3	Yes	
bq7694006DBT	bq7694006DBTR		0x18		No	

(1) Product Preview only

Texas Instruments pre-configures the bq769x0 devices for a specific I²C address, LDO voltage, and more. These settings are permanently stored in EEPROM and cannot be further modified.

Contact Texas Instruments for other options not listed above, as well as any options noted as "Product Preview only."

3 说明((继续))

bq76920 器件支持多达 5 节串联电池或者典型值为 18V 的电池组;bq76930 可处理多达 10 节串联电池或者典型值为 36V的电池组;而 bq76940 支持多达 15 节串联电池或者典型值为 48V 的电池组。这些 AFE 可用于管理各种电池化学成分,例如锂离子、磷酸铁锂,等等。

通过 I²C,主机控制器可以使用 bq769x0 来执行很多电池组管理功能,诸如监视(电池电压、电池组电流、 电池组温度),保护(控制充电/放电 FET),以及均衡功能。集成模数 (A/D) 转换器可实现对关键系统参数 的纯数字读取,这些参数的校准在TI制造过程中处理。

4 设备对比表

TUBE	卷带式	电池	I²C 地址 (7 位)	LDO (V)	CRC	封装
bq7692000PW	bq7692000PWR			2.5	No	
bq7692001PW(1)	bq7692001PWR(1)		0x08		Yes	
bq7692002PW(1)	bq7692002PWR(1)	3–5			No	20-TSSOP (PW)
bq7692003PW	bq7692003PWR			3.3	Yes	
bq7692006PW	bq7692006PWR		0x18		No	
bq7693000DBT	bq7693000DBTR			2.5	No	
bq7693001DBT	bq7693001DBTR		0x08		Yes	
bq7693002DBT	bq7693002DBTR	6–10			No	30-TSSOP (DBT)
bq7693003DBT	bq7693003DBTR			3.3	Yes	
bq7693006DBT	bq7693006DBTR		0x18		No	
bq7693007DBT	bq7693007DBTR				Yes	
bq7694000DBT	bq7694000DBTR			2.5	No	
bq7694001DBT	bq7694001DBTR		0x08		Yes	
bq7694002DBT	bq7694002DBTR	9–15			No	44-TSSOP (DBT)
bq7694003DBT	bq7694003DBTR			3.3	Yes	
bq7694006DBT	bq7694006DBTR		0x18		No	

(1) 仅限产品预览

德州仪器公司预配置了 bq769x0 设备用于特定的 I²C 地址、LDO 电压等。这些设置永久存储在 EEPROM 中且无法进一步修改。

如需了解上述未列出的其他选项，或任何标有"仅限产品预览"的选项，请联系德州仪器。

5 Pin Configuration and Functions

5.1 Versions

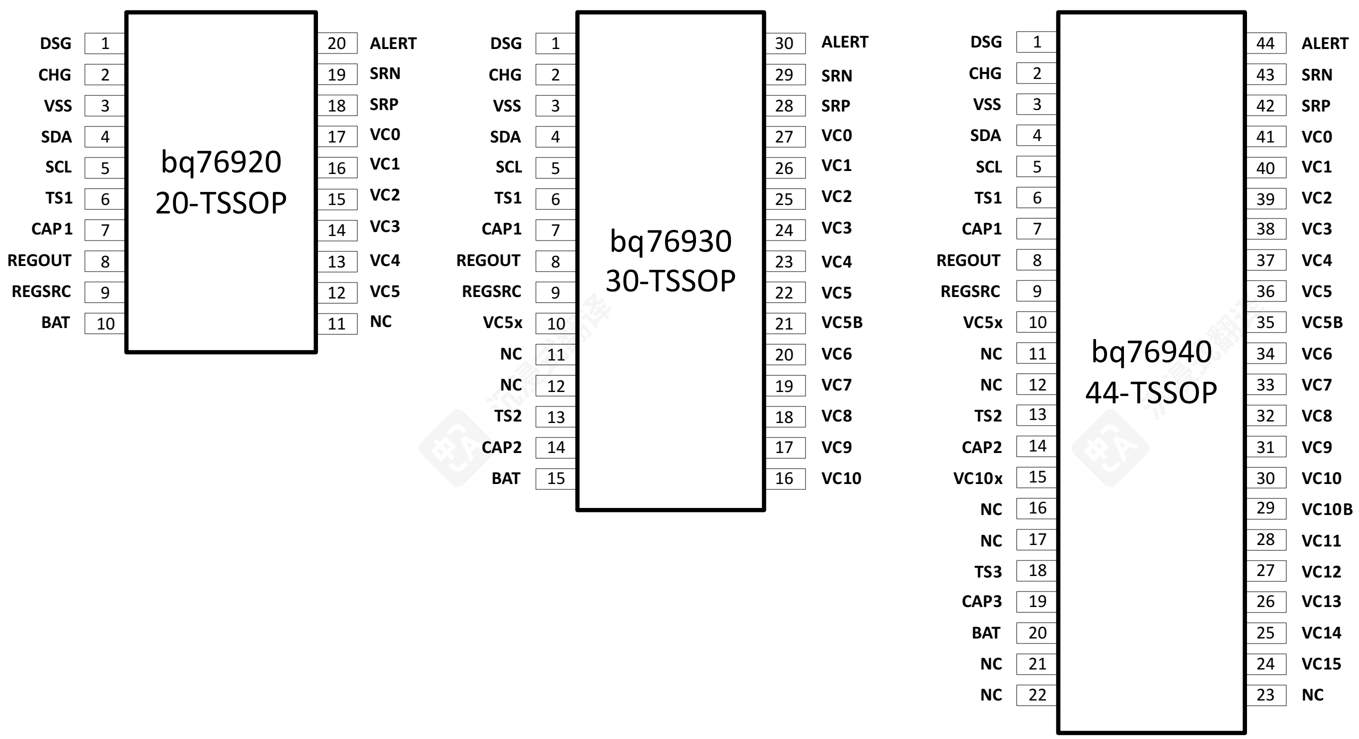


Figure 5-1. Pin Versions

bq76920: 3–5 Series Cells (20-TSSOP)
- 6.5 mm x 4.4 mm x 1.2 mm

bq76930: 6–10 Series Cells (30-TSSOP)
- 7.8 mm x 4.4 mm x 1.2 mm

bq76940: 9–15 Series Cells (44-TSSOP)
- 11.3 mm x 4.4 mm x 1.2 mm

5 Pin 配置和功能

5.1 版本

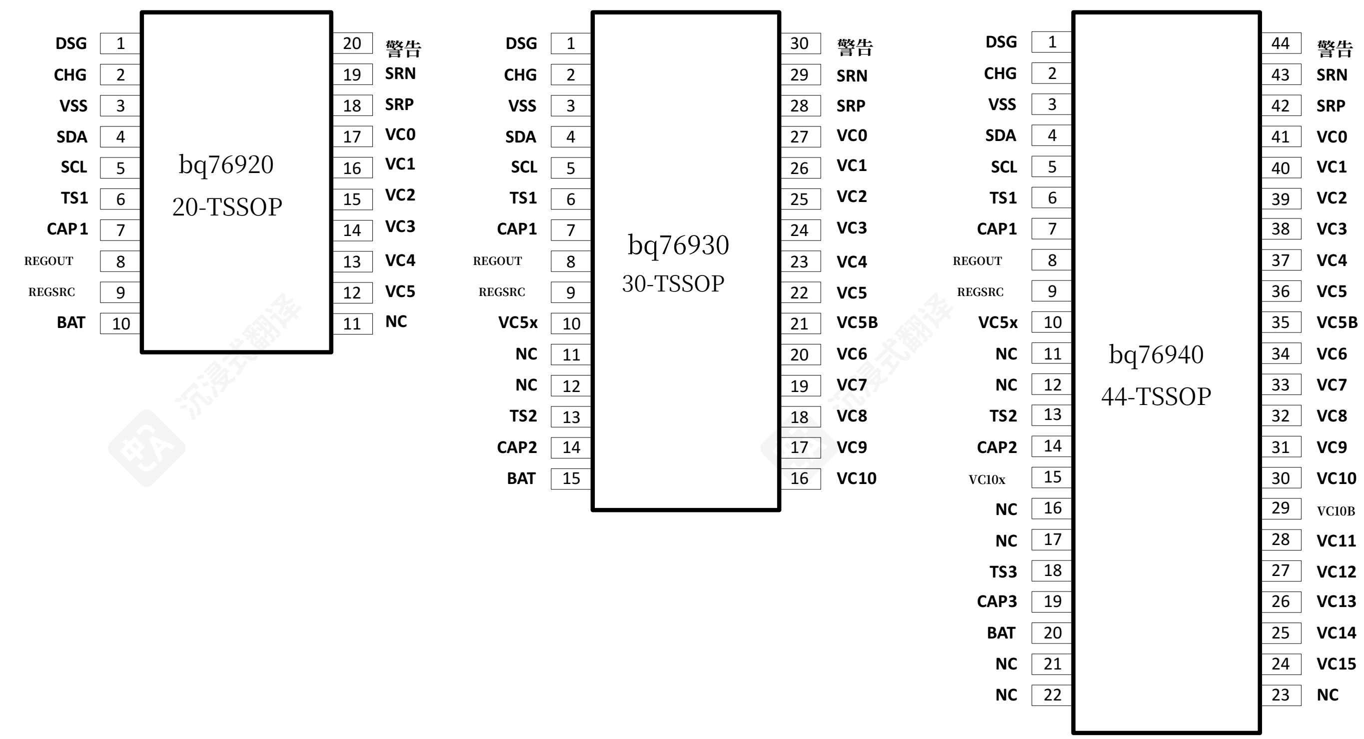


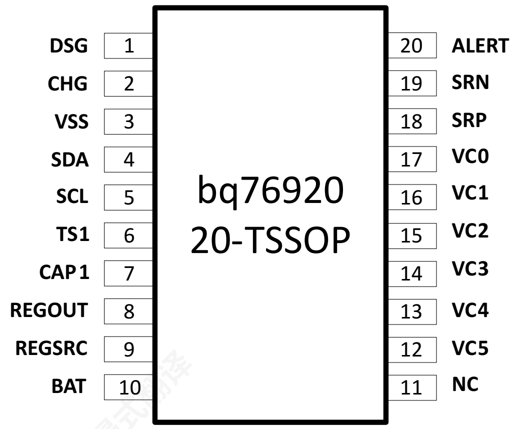
图 5-1. 引脚版本

bq76920: 3–5 系列电芯（20-TSSOP）
- 6.5 毫米 x 4.4 毫米 x 1.2 毫米

bq76930: 6–10 系列电芯（30-TSSOP）
- 7.8 毫米 x 4.4 毫米 x 1.2 毫米

bq76940: 9–15系列电芯（44-TSSOP）
- 11.3毫米 x 4.4毫米 x 1.2毫米

5.2 bq76920 Pin Diagram



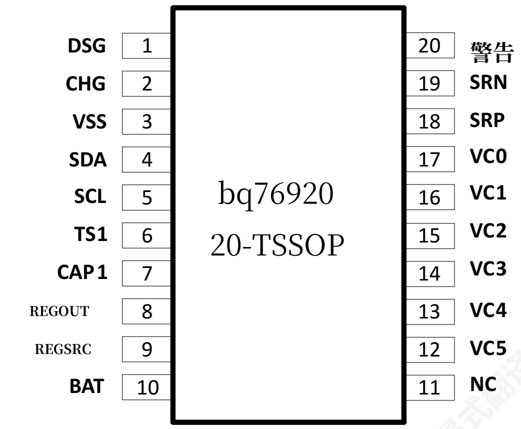
5.2.1 bq76920 Pin Map

bq76920 Pin Functions

PIN	NAME	TYPE	DESCRIPTION
1	DSG	O	Discharge FET driver
2	CHG	O	Charge FET driver
3	VSS	—	Chip VSS
4	SDA	I/O	I^2C communication to the host controller
5	SCL	I	I^2C communication to the host controller
6	TS1	I	Thermistor #1 positive terminal(1)
7	CAP1	O	Capacitor to VSS
8	REGOUT	P	Output LDO
9	REGSRC	I	Input source for output LDO
10	BAT	P	Battery (top-most) terminal
11	NC	—	No connect
12	VC5	I	Sense voltage for 5th cell positive terminal
13	VC4	I	Sense voltage for 4th cell positive terminal
14	VC3	I	Sense voltage for 3rd cell positive terminal
15	VC2	I	Sense voltage for 2nd cell positive terminal
16	VC1	I	Sense voltage for 1st cell positive terminal
17	VC0	I	Sense voltage for 1st cell negative terminal
18	SRP	I	Negative current sense (nearest VSS)
19	SRN	I	Positive current sense
20	ALERT	I/O	Alert output and override input

(1) If not used, pull down to VSS with a 10-kΩ nominal resistor.

5.2 bq76920 引脚图



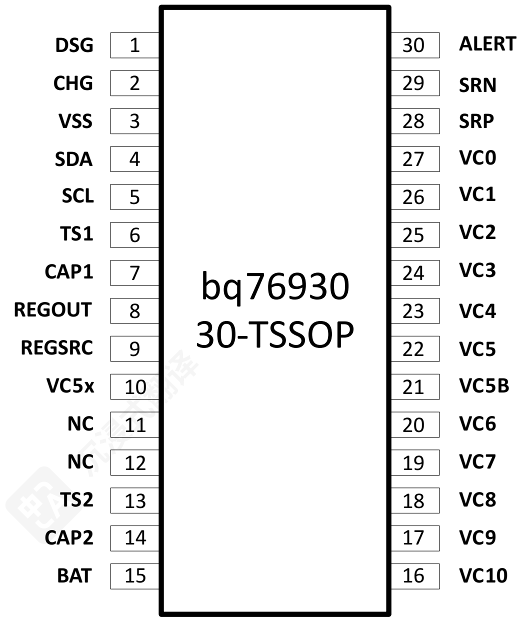
5.2.1 bq76920 引脚映射

bq76920 引脚功能

PIN	NAME	TYPE	描述
1	DSG	O	放电 FET 驱动器
2	CHG	O	充电 FET 驱动器
3	VSS	—	芯片VSS
4	SDA	I/O	I^2C通信到主机控制器
5	SCL	I	I^2C通信到主机控制器
6	TS1	I	热敏电阻#1正极(1)
7	CAP1	O	电容到VSS
8	REGOUT	P	输出LDO
9	REGSRC	I	输出LDO的输入源
10	BAT	P	电池（最顶部）端子
11	NC	—	无连接
12	VC5	I	第5节电池正极的检测电压
13	VC4	I	第4个电池正极的检测电压
14	VC3	I	第3个电池正极的检测电压
15	VC2	I	第2个电池正极的检测电压
16	VC1	I	第1个电池正极的检测电压
17	VC0	I	第1个电池负极的检测电压
18	SRP	I	负极电流检测（最近VSS）
19	SRN	I	正向电流检测
20	警报	I/O	警报输出和覆盖输入

(1) 若未使用，请用10-kΩ 的标称电阻下拉至VSS。

5.3 bq76930 Pin Diagram



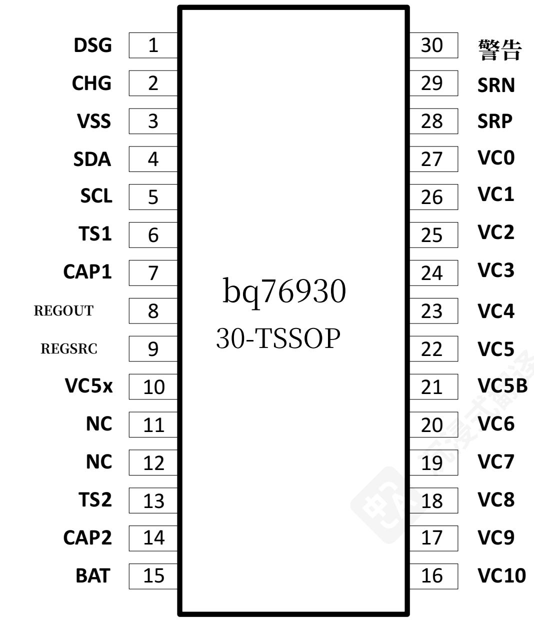
5.3.1 bq76930 Pin Map

bq76930 Pin Functions

PIN	NAME	TYPE	DESCRIPTION
1	DSG	O	Discharge FET driver
2	CHG	O	Charge FET driver
3	VSS	—	Chip VSS
4	SDA	I/O	I^2C communication to the host controller
5	SCL	I	I^2C communication to the host controller
6	TS1	I	Thermistor #1 positive terminal(1)
7	CAP1	O	Capacitor to VSS
8	REGOUT	P	Output LDO
9	REGSRC	I	Input source for output LDO
10	VC5X	P	Thermistor #2 negative terminal
11	NC	—	No connect (short to CAP2)
12	NC	—	No connect (short to CAP2)
13	TS2	I	Thermistor #2 positive terminal(1)
14	CAP2	O	Capacitor to VC5X
15	BAT	P	Battery (top-most) terminal
16	VC10	I	Sense voltage for 10th cell positive terminal
17	VC9	I	Sense voltage for 9th cell positive terminal
18	VC8	I	Sense voltage for 8th cell positive terminal
19	VC7	I	Sense voltage for 7th cell positive terminal
20	VC6	I	Sense voltage for 6th cell positive terminal
21	VC5B	I	Sense voltage for 6th cell negative terminal
22	VC5	I	Sense voltage for 5th cell positive terminal
23	VC4	I	Sense voltage for 4th cell positive terminal

(1) If not used, pull down to group ground reference (VSS for TS1 and VC5X for TS2) with a 10-kΩ nominal resistor.

5.3 bq76930 引脚图



5.3.1 bq76930 引脚映射

bq76930 引脚功能

PIN	NAME	TYPE	描述
1	DSG	O	放电FET驱动器
2	CHG	O	充电FET驱动器
3	VSS	—	芯片VSS
4	SDA	I/O	I^2C通信到主机控制器
5	SCL	I	I^2C通信到主机控制器
6	TS1	I	热敏电阻#1正极(1)
7	CAP1	O	电容到VSS
8	REGOUT	P	输出LDO
9	REGSRC	I	输出LDO的输入源
10	VC5X	P	热敏电阻#2的负极
11	NC	—	无连接（短接到CAP2）
12	NC	—	无连接（短接到CAP2）
13	TS2	I	热敏电阻#2正极(1)
14	CAP2	O	电容至VC5X
15	BAT	P	电池（最上方）端子
16	VC10	I	第10个单元正极的检测电压
17	VC9	I	第9个单元正极的检测电压
18	VC8	I	第8个单元正极的检测电压
19	VC7	I	第7个电池正极的检测电压
20	VC6	I	第6个电池正极的检测电压
21	VC5B	I	第6个电池负极的检测电压
22	VC5	I	第5个电池正极的检测电压
23	VC4	I	第4个电池正极的检测电压

(1) 若未使用，请用10-kΩ的标称电阻下拉至组地参考（TS1的VSS和TS2的VC5X）。

bq76930 Pin Functions (continued)

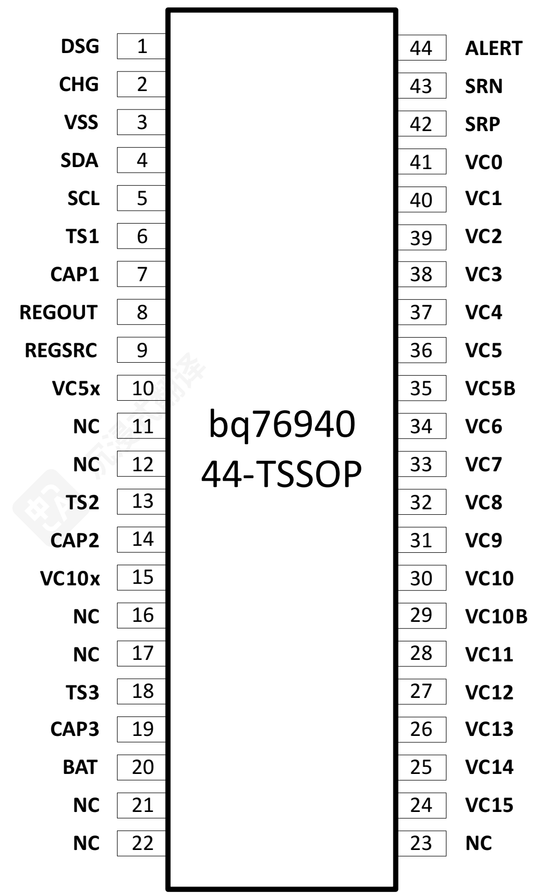
PIN	NAME	TYPE	DESCRIPTION
24	VC3	I	Sense voltage for 3rd cell positive terminal
25	VC2	I	Sense voltage for 2nd cell positive terminal
26	VC1	I	Sense voltage for 1st cell positive terminal
27	VC0	I	Sense voltage for 1st cell negative terminal
28	SRP	I	Negative current sense (nearest VSS)
29	SRN	I	Positive current sense
30	ALERT	I/O	Alert output and override input

bq76920, bq76930, bq76940
ZHCSCE2F –2013年10月–修订于2015年11月
www.ti.com.cn

bq76930 引脚功能（续）

PIN	NAME	TYPE	描述
24	VC3	I	第3个电池正极的检测电压
25	VC2	I	第2个电池正极的检测电压
26	VC1	I	第1个电池正极的检测电压
27	VC0	I	第1个单元负极电压检测
28	SRP	I	负极电流检测（最近VSS）
29	SRN	I	正极电流检测
30	警告	I/O	警告输出和覆盖输入

5.4 bq76940 Pin Diagram



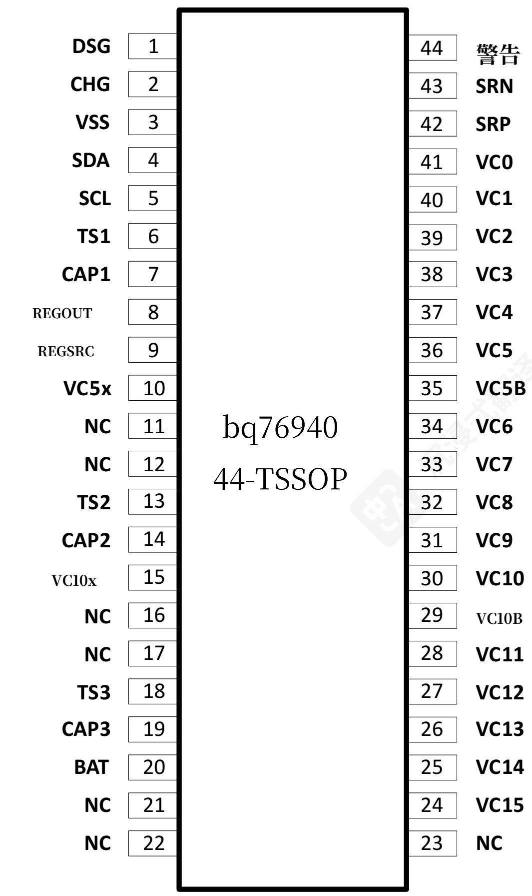
5.4.1 bq76940 Pin Map

bq76940 Pin Functions

PIN	NAME	TYPE	DESCRIPTION
1	DSG	O	Discharge FET driver
2	CHG	O	Charge FET driver
3	VSS	—	Chip VSS
4	SDA	I/O	I^2C communication to the host controller
5	SCL	I	I^2C communication to the host controller
6	TS1	I	Thermistor #1 positive terminal(1)
7	CAP1	O	Capacitor to VSS
8	REGOUT	P	Output LDO
9	REGSRC	I	Input source for output LDO
10	VC5X	P	Thermistor #2 negative terminal
11	NC	—	No connect (short to CAP2)
12	NC	—	No connect (short to CAP2)
13	TS2	I	Thermistor #2 positive terminal(1)
14	CAP2	O	Capacitor to VC5X
15	VC10X	P	Thermistor #3 negative terminal

(1) If not used, pull down to group ground reference (VSS for TS1, VC5X for TS2, and VC10X for TS3) with a 10-kΩ nominal resistor.

5.4 bq76940 引脚图



5.4.1 bq76940 引脚映射

bq76940 引脚功能

PIN	NAME	TYPE	描述
1	DSG	O	放电FET驱动器
2	CHG	O	充电FET驱动器
3	VSS	—	芯片VSS
4	SDA	I/O	与主机控制器进行I^2C通信
5	SCL	I	与主机控制器进行I^2C通信
6	TS1	I	热敏电阻 #1 正极(1)
7	CAP1	O	电容至 VSS
8	REGOUT	P	输出 LDO
9	REGSRC	I	输出 LDO 的输入源
10	VC5X	P	热敏电阻#2负极
11	NC	—	无连接（短接到CAP2）
12	NC	—	无连接（短接到CAP2）
13	TS2	I	热敏电阻#2正极(1)
14	CAP2	O	电容到VC5X
15	VC10X	P	热敏电阻#3负极

(1) 若未使用，则通过 10-kΩ 的电阻拉低至组地参考（TS1 为 VSS，TS2 为 VC5X，TS3 为 VC10X）

bq76940 Pin Functions (continued)

PIN	NAME	TYPE	DESCRIPTION
16	NC	—	No connect (short to CAP3)
17	NC	—	No connect (short to CAP3)
18	TS3	I	Thermistor #3 positive terminal (1)
19	CAP3	O	Capacitor to VC10X
20	BAT	P	Battery (top-most) terminal
21	NC	—	No connect
22	NC	—	No connect
23	NC	—	No connect
24	VC15	I	Sense voltage for 15th cell positive terminal
25	VC14	I	Sense voltage for 14th cell positive terminal
26	VC13	I	Sense voltage for 13th cell positive terminal
27	VC12	I	Sense voltage for 12th cell positive terminal
28	VC11	I	Sense voltage for 11th cell positive terminal
29	VC10B	I	Sense voltage for 11th cell negative terminal
30	VC10	I	Sense voltage for 10th cell positive terminal
31	VC9	I	Sense voltage for 9th cell positive terminal
32	VC8	I	Sense voltage for 8th cell positive terminal
33	VC7	I	Sense voltage for 7th cell positive terminal
34	VC6	I	Sense voltage for 6th cell positive terminal
35	VC5B	I	Sense voltage for 6th cell negative terminal
36	VC5	I	Sense voltage for 5th cell positive terminal
37	VC4	I	Sense voltage for 4th cell positive terminal
38	VC3	I	Sense voltage for 3rd cell positive terminal
39	VC2	I	Sense voltage for 2nd cell positive terminal
40	VC1	I	Sense voltage for 1st cell positive terminal
41	VC0	I	Sense voltage for 1st cell negative terminal
42	SRP	I	Negative current sense (nearest VSS)
43	SRN	I	Positive current sense
44	ALERT	I/O	Alert output and override input

bq76940 引脚功能（续）

PIN	NAME	TYPE	描述
16	NC	—	不连接（短接到 CAP3）
17	NC	—	不连接（短接到 CAP3）
18	TS3	I	热敏电阻 #3 正极 (1)
19	CAP3	O	电容至VC10X
20	BAT	P	电池（最上方）端子
21	NC	—	无连接
22	NC	—	无连接
23	NC	—	无连接
24	VC15	I	第15节电池正极的检测电压
25	VC14	I	第14个电池正极的检测电压
26	VC13	I	第13个电池正极的检测电压
27	VC12	I	第12个电池正极的检测电压
28	VC11	I	第11个电池正极的检测电压
29	VC10B	I	第11个电池负极的检测电压
30	VC10	I	第10个电池正极的检测电压
31	VC9	I	第9个电池正极的检测电压
32	VC8	I	第8个电池正极的检测电压
33	VC7	I	第7个电池正极的检测电压
34	VC6	I	第6个电池正极的检测电压
35	VC5B	I	第6个电池负极的检测电压
36	VC5	I	第5个电池正极的检测电压
37	VC4	I	第4个电池正极的检测电压
38	VC3	I	第3个电池正极的检测电压
39	VC2	I	第2个电池正极的检测电压
40	VC1	I	第1个电池正极的检测电压
41	VC0	I	第1个电池负极的检测电压
42	SRP	I	负电流检测（最近VSS）
43	SRN	I	正电流检测
44	警报	I/O	警报输出和覆盖输入

6 Electrical Specifications

6.1 Absolute Maximum Ratings

Over-operating free-air temperature range (unless otherwise noted)

				MIN	MAX	UNIT
V_{BAT}	Supply voltage range	(BAT–VSS)	bq76920	–0.3	36	V
		(BAT–VC5x), (VC5x–VSS)	bq76930			
		(BAT–VC10x), (VC10x–VC5x), (VC5x–VSS)	bq76940			
V_I	Input voltage range	(VCn–VSS) where n = 1..5	bq76920	–0.3	(n × 7.2)	V
		(VCn–VSS) where n = 1..5, (VCn-VC5x) where n = 6..10	bq76930			
		(VCn–VSS) where n = 1..5, (VCn–VC5x) where n = 6..10, (VCn–VC10x) where n = 11..15	bq76940			
		Cell input pins, differential (VCn–VCn–1) where n = 1..15/10/5 (bq76940/bq76930/bq76920, respectively)		–0.3	9	V
		SRN, SRP, SCL, SDA		–0.3	3.6	V
		(VC0–VSS), (CAP1–VSS), (TS1–VSS)(1)	bq76920			
		(VC0–VSS), (VC5b–VC5x), (CAP2–VC5x), (CAP1–VSS), (TS2–VC5x), (TS1–VSS)(1)	bq76930			
		(VC0–VSS), (VC5b–VC5x), (VC10b–VC10x), (CAP3–VC10x), (CAP2–VC5x), (CAP1–VSS), (TS3–VC10x), (TS2–VC5x), (TS1–VSS)(1)	bq76940			
		REGSRC		–0.3	36	
V_O	Output voltage range	REGOUT, ALERT		–0.3	3.6	V
		DSG		–0.3	20	
		CHG		–0.3	$V_{CHGCLAMP}$	
I_{CB}	Cell balancing current (per cell)		bq76920	70		mA
			bq76930, bq76940	5		mA
I_{DSG}	Discharge pin input current when disabled (measured into terminal)			7		mA
T_{STG}	Storage temperature range			–65	150	°C
	Lead temperature (soldering, 10 s)				300	

(1) The Absolute Maximum Ratings for (TS1–VSS) apply after the device completes POR and should be observed after $t_{BOOTREADY}$ (10 ms), following the application of the boot signal on TS1. Prior to completion of POR, TS1 should not exceed 5 V.

6.2 ESD Ratings

			VALUE	UNIT
$V_{(ESD)}$	Electrostatic discharge	Human body model (HBM) ESD stress voltage(1)	±2	kV
		Charged device model (CDM) ESD stress voltage(2)	±500	V

(1) JEDEC document JEP155 states that 500-V HBM allows safe manufacturing with a standard ESD control process.
(2) JEDEC document JEP157 states that 250-V CDM allows safe manufacturing with a standard ESD control process.

6.3 Recommended Operating Conditions

Over-operating free-air temperature range (unless otherwise noted). See 节 8.1.1 for more information on cell configurations. All voltages are relative to VSS, except "Cell input differential."

				MIN	TYP	MAX	UNIT
V_{BAT}	Supply voltage range	(BAT–VSS)	bq76920	6		25	V
		(BAT–VC5x), (VC5x–VSS)	bq76930				
		(BAT–VC10x), (VC10x–VC5x), (VC5x–VSS)	bq76940				

6 电特性

6.1 绝对最大额定值

过温自由空气温度范围（如无其他说明）

				MIN	MAX	UNIT
V_{BAT}	电源电压范围	(BAT–VSS)	bq76920	–0.3	36	V
		(BAT–VC5x), (VC5x–VSS)	bq76930			
		(BAT–VC10x), (VC10x–VC5x), (VC5x–VSS)	bq76940			
V_I	输入电压范围	(VCn–VSS) 其中 n = 1..5	bq76920	–0.3	(n × 7.2)	V
		(VCn–VSS) 其中 n = 1..5，(VCn-VC5x) 其中 n =6..10	bq76930			
		(VCn–VSS) 其中 n = 1..5，(VCn–VC5x) 其中 n =6..10, (VCn–VC10x) 其中 n = 11..15	bq76940			
		单元输入引脚，差分 (VCn–VCn–1) 其中 n = 1..15/10/5 (分别对应 bq76940/bq76930/bq76920)		–0.3	9	V
		SRN，SRP，SCL，SDA		–0.3	3.6	V
		(VC0–VSS), (CAP1–VSS), (TS1–VSS)(1)	bq76920			
		(VC0–VSS), (VC5b–VC5x), (CAP2–VC5x), (CAP1–VSS), (TS2–VC5x), (TS1–VSS)(1)	bq76930			
		(VC0–VSS),(VC5b–VC5x), (VC10b–VC10x), (CAP3–VC10x), (CAP2–VC5x), (CAP1–VSS), (TS3–VC10x), (TS2–VC5x), (TS1–VSS)(1)	bq76940			
		REGSRC		–0.3	36	
V_O	输出电压范围	REGOUT, ALERT		–0.3	3.6	V
		DSG		–0.3	20	
		CHG		–0.3	$V_{CHGCLAMP}$	
I_{CB}	电池均衡电流（每节电池）		bq76920	70		mA
			bq76930, bq76940	5		mA
I_{DSG}	禁用时的放电引脚输入电流（测量至端子）			7		mA
T_{STG}	存储温度范围			–65	150	°C
	引脚温度（焊接，10秒）				300	

(1) 设备完成上电复位 (POR) 后，(TS1–VSS) 的绝对最大额定值适用，应在 $t_{BOOTREADY}$ (10 ms) 后观察并遵守，此时应在 TS1 上施加启动信号。在 POR 完成之前，TS1 电压不应超过 5 V。

6.2 ESD等级

			值	UNIT
$V_{(ESD)}$	静电 <code>discharge</code>	人体模型（HBM）ESD应力电压(1)	±2	kV
		带电设备模型（CDM）ESD应力电压(2)	±500	V

(1) JEDEC 文档 JEP155 指出 500-V HBM 允许使用标准 ESD 控制工艺进行安全制造。(2) JEDEC 文档 JEP157 指出 250-V CDM 允许使用标准 ESD 控制工艺进行安全制造。

6.3 推荐工作条件

超工作自由空气温度范围（如无特别说明）。参见8.1.1节了解更多关于单元配置的信息。所有电压均相对于VSS，除了“单元输入差分”。

				MIN	TYP	MAX	UNIT
V_{BAT}	电源电压范围	(BAT–VSS)	bq76920	6		25	V
		(BAT–VC5x), (VC5x–VSS)	bq76930				
		(BAT–VC10x), (VC10x–VC5x), (VC5x–VSS)	bq76940				

Recommended Operating Conditions *(continued)*

Over-operating free-air temperature range (unless otherwise noted). See 节 8.1.1 for more information on cell configurations. All voltages are relative to VSS, except "Cell input differential."

				MIN	TYP	MAX	UNIT
V_{IN}	Input voltage range	Cell input pins, differential (VCn–VCn–1) where n = 1..15/10/5 (bq76940/bq76930/bq76920, respectively), in-use cells only		2		5	V
		(VCn–VSS) where n = 1..5	bq76920	0		5 × n	V
		(VCn–VSS) where n = 1..5, (VCn–VC5x) where n = 6..10	bq76930				
		(VCn–VSS) where n = 1..5, (VCn–VC5x) where n = 6..10, (VCn–VC10x) where n = 11..15	bq76940				
		SRP		–10		10	mV
		(VC0–VSS)	bq76920				
		(VC0–VSS), (VC5b–VC5x)	bq76930				
		(VC0–VSS), (VC5b–VC5x), (VC10b–VC10x)	bq76940				
		SRN		–200		200	mV
		SCL, SDA		0		3.6	V
		(TS1–VSS)	bq76920				
		(TS1–VSS), (TS2–VC5x)	bq76930				
		(TS1–VSS), (TS2–VC5x), (TS3–VC10x)	bq76940				
		REGSRC		6		25	
V_{OUT}	Output voltage range	CHG, DSG		0		16	V
		REGOUT, ALERT		0		3.6	V
		(CAP1–VSS)	bq76920				
		(CAP1–VSS), (CAP2–VC5x)	bq76930				
		(CAP1–VSS), (CAP2–VC5x), (CAP3–VC10x)	bq76940				
I_{CB}	Cell balancing current (internal, per cell)	bq76920		0		50	mA
		bq76930, bq76940		0		5	mA
R_C	External cell input resistance	bq76920		40	100	1K	Ω
		bq76930, bq76940		500	1K	1K	Ω
C_C	External cell input capacitance			0.1	1	10	µF
R_f	External supply filter resistance			40	100	1K	Ω
C_f	External supply filter capacitance			1	10	40	µF
R_{FILT}	Sense resistor filter resistance			100	1K		Ω
R_{ALERT}	ALERT pin to VSS resistor				1M		Ω
C_L	REGOUT loading capacitance			1	4.7		µF
C_{CAP}	REGSRC, CAP1, CAP2, and CAP3 output capacitance			1			µF
R_{TS}	External thermistor nominal resistance (103AT) at 25ºC			10K			Ω
T_{OPR}	Operating free-air temperature			–40		85	°C

推荐工作条件（续）

过工作自由空气温度范围（如无特别说明）。参见第8.1.1节了解更多关于电芯配置的信息。所有电压均相对于VSS，除了“电芯输入差分”。

				MIN	TYP	MAX	UNIT
V_{IN}	输入电压范围	电芯输入引脚，差分（VCn–VCn–1），其中n = 1..15/10/5（bq76940/bq76930/bq76920，分别对应），仅限使用中的电芯		2		5	V
		(VCn–VSS) 其中 n = 1..5	bq76920	0		5 × n	V
		(VCn–VSS) 其中 n = 1..5, (VCn–VC5x) 其中 n = 6..10	bq76930				
		(VCn–VSS) 其中 n = 1..5, (VCn–VC5x) 其中 n = 6..10, (VCn–VC10x) 其中 n = 11..15	bq76940				
		SRP		–10		10	mV
		(VC0–VSS)	bq76920				
		(VC0–VSS)，(VC5b–VC5x)	bq76930				
		(VC0–VSS), (VC5b–VC5x), (VC10b–VC10x)	bq76940				
		SRN		–200		200	mV
		SCL, SDA		0		3.6	V
		(TS1–VSS)	bq76920				
		(TS1–VSS), (TS2–VC5x)	bq76930				
		(TS1–VSS), (TS2–VC5x), (TS3–VC10x)	bq76940				
		REGSRC		6		25	
V_{OUT}	输出电压范围	CHG, DSG		0		16	V
		REGOUT, ALERT		0		3.6	V
		(CAP1–VSS)	bq76920				
		(CAP1–VSS), (CAP2–VC5x)	bq76930				
		(CAP1–VSS), (CAP2–VC5x), (CAP3–VC10x)	bq76940				
I_{CB}	电池均衡电流 (内部, 每个电池)	bq76920		0		50	mA
		bq76930, bq76940		0		5	mA
R_C	外部单元输入电阻	bq76920		40	100	1K	Ω
		bq76930, bq76940		500	1K	1K	Ω
C_C	外部单元输入电容			0.1	1	10	µF
R_f	外部电源滤波器电阻			40	100	1K	Ω
C_f	外部电源滤波器电容			1	10	40	µF
R_{FILT}	检测电阻滤波器电阻			100	1K		Ω
R_{ALERT}	ALERT引脚到VSS电阻				1M		Ω
C_L	REGOUT 负载电容			1	4.7		µF
C_{CAP}	REGSRC、CAP1、CAP2 和 CAP3 输出电容			1			µF
R_{TS}	外部热敏电阻在 25°C 下的标称电阻 (103AT)			10K			Ω
T_{OPR}	工作无风温度			–40		85	°C

6.4 Thermal Information

Over-operating free-air temperature range (unless otherwise noted)

THERMAL METRIC(1)		TSSOP			UNIT
		bq76920xy 20 PINS (PW)	bq76930xy 30 PINS (DBT)	bq76940xy 44 PINS (DBT)	
$R_{\theta JA,\ High\ K}$	Junction-to-ambient thermal resistance (2)	93.7	86.5	70.1	°C/W
$R_{\theta JC(top)}$	Junction-to-case(top) thermal resistance (3)	28.7	19.4	17.5	
$R_{\theta JB}$	Junction-to-board thermal resistance (4)	44.6	41.3	33.9	
ψ_{JT}	Junction-to-top characterization parameter (5)	1.3	0.5	0.5	
ψ_{JB}	Junction-to-board characterization parameter (6)	44.1	40.6	33.4	
$R_{\theta JC(bottom)}$	Junction-to-case(bottom) thermal resistance (7)	n/a	n/a	n/a	

(1) For more information about traditional and new thermal metrics, see the *Semiconductor and IC Package Thermal Metrics* application report (SPRA953).
(2) The junction-to-ambient thermal resistance under natural convection is obtained in a simulation on a JEDEC-standard, high-K board, as specified in JESD51-7, in an environment described in JESD51-2a.
(3) The junction-to-case (top) thermal resistance is obtained by simulating a cold plate test on the package top. No specific JEDEC-standard test exists, but a close description can be found in the ANSI SEMI standard G30-88.
(4) The junction-to-board thermal resistance is obtained by simulating in an environment with a ring cold plate fixture to control the PCB temperature, as described in JESD51-8.
(5) The junction-to-top characterization parameter, ψ_{JT}, estimates the junction temperature of a device in a real system and is extracted from the simulation data for obtaining $R_{\theta JA}$, using a procedure described in JESD51-2a (sections 6 and 7).
(6) The junction-to-board characterization parameter, ψ_{JB}, estimates the junction temperature of a device in a real system and is extracted from the simulation data for obtaining $R_{\theta JA}$, using a procedure described in JESD51-2a (sections 6 and 7).
(7) The junction-to-case (bottom) thermal resistance is obtained by simulating a cold plate test on the exposed (power) pad. No specific JEDEC standard test exists, but a close description can be found in the ANSI SEMI standard G30-88.

6.5 Electrical Characteristics

Typical conditions are measured at 25ºC with nominal BAT voltages of 18 V (bq76920), 36 V (bq76930), or 48 V (bq76940) with V_{CELL} = 4 V. Min and max values include full recommended operating condition temperature range from –40ºC to +85ºC. Certain characteristics may be shown at different voltage or temperature ranges, as clarified in the Test Condition sections.

PARAMETER	DESCRIPTION	TEST CONDITION	MIN	TYP	MAX	UNIT
SUPPLY CURRENTS						
I_{DD}	NORMAL mode: ADC off, CC off	Sum of ICC_BAT and ICC_REGSRC currents		40	60	µA
	NORMAL mode: ADC on, CC off			60	90	
	NORMAL mode: ADC off, CC on			110	165	
	NORMAL mode: ADC on, CC on			130	195	
I_{CC_BAT}	NORMAL mode: ADC off	Into BAT pin		30	45	
	NORMAL mode: ADC on			50	75	
I_{CC_REGSRC}	NORMAL mode: CC off	Into REGSRC pin		10	15	
	NORMAL mode: CC on			80	120	
I_{SHIP}	SHIP/SHUTDOWN mode	Device in full shutdown, only VSTUP/BG and BOOT detector on		0.6	1.8	

6.4 热信息

过热自由空气温度范围（如另有说明）

热指标(1)		TSSOP			UNIT
		bq76920xy 20 PINS (PW)	bq76930xy 30 PINS (DBT)	bq76940xy 44 针 (DBT)	
$R_{\theta JA,\ High\ K}$	结点到环境热阻(2)	93.7	86.5	70.1	°C/W
$R_{\theta JC(top)}$	Junction-to-case(top) 热阻(3)	28.7	19.4	17.5	
$R_{\theta JB}$	Junction-to-board 热阻(4)	44.6	41.3	33.9	
ψ_{JT}	Junction-to-top 特性参数(5)	1.3	0.5	0.5	
ψ_{JB}	Junction-to-board 特性参数(6)	44.1	40.6	33.4	
$R_{\theta JC(bottom)}$	Junction-to-case(bottom) 热阻(7)	n/a	n/a	n/a	

(1) 如需了解传统和新热指标的相关信息，请参阅《半导体和IC封装热指标应用报告》（SPRA953）。
(2) 自然对流下的结到环境热阻是在JEDEC标准高K板上进行的仿真结果，具体规定见JESD51-7，环境描述见JESD51-2a。
(3) 结到壳（顶部）热阻是通过在封装顶部模拟冷板测试获得的。虽然没有特定的JEDEC标准测试，但ANSI SEMI标准G30-88中有详细的描述。
(4) 结到板热阻是通过在带有环形冷板夹具的环境中仿真获得的，该夹具用于控制PCB温度，具体描述见JESD51-8。
(5) 结到顶部特性参数 ψ_{JT}估计了器件在真实系统中的结温，并从用于获得$R_{\theta JA}$的仿真数据中提取，具体方法见JESD51-2a（第6节和第7节）。
(6) 结束到板 characterization 参数 ψ_{JB} 估计器件在真实系统中的结温，并从模拟数据中提取以获得 $R_{\theta JA}$，其使用 JESD51-2a 中描述的程序（第 6 节和第 7 节）。
(7) 结束到底部（底部）热阻通过在暴露的（功率）焊盘上模拟冷板测试获得。没有特定的 JEDEC 标准测试，但 ANSI SEMI 标准G30-88 中可以找到详细的描述。

6.5 电气特性

典型条件通常在25℃下测量，具有18 V（bq76920）、36 V（bq76930）或48 V（bq76940）的标称BAT电压，并伴有V_{CELL} = 4 V。最小值和最大值包括从–40°C到 +85°C的完整推荐工作条件温度范围。某些特性可能在不同的电压或温度范围内显示，如测试条件部分所述。

参数	描述	测试条件	MIN	TYP	MAX	UNIT
电源电流						
I_{DD}	正常模式：ADC关闭，CC关闭	ICC_BAT和ICC_REGSRC之和电流		40	60	µA
	正常模式：ADC开启，CC关闭			60	90	
	正常模式：ADC关闭，CC 开启			110	165	
	正常模式：ADC 开启，CC 开启			130	195	
I_{CC_BAT}	正常模式：ADC 关闭	连接到 BAT 引脚		30	45	
	正常模式：ADC 开启			50	75	
I_{CC_REGSRC}	正常模式：CC关闭	连接到REGSRC引脚		10	15	
	正常模式：CC开启			80	120	
I_{SHIP}	SHIP/SHUTDOWN模式	设备完全关闭，仅 VSTUP/BG 和 BOOT 检测器开启		0.6	1.8	

Electrical Characteristics *(continued)*

Typical conditions are measured at 25ºC with nominal BAT voltages of 18 V (bq76920), 36 V (bq76930), or 48 V (bq76940) with V_{CELL} = 4 V. Min and max values include full recommended operating condition temperature range from –40ºC to +85ºC. Certain characteristics may be shown at different voltage or temperature ranges, as clarified in the Test Condition sections.

PARAMETER	DESCRIPTION	TEST CONDITION	MIN	TYP	MAX	UNIT
LEAKAGE AND OFFSET CURRENTS						
dI_{NOM}	NORMAL mode supply current offset	Measured into VC5x (bq76930, bq76940) and VC10x (bq76940)	–5	±2.5	5	µA
dI_{SHIP}	SHIP mode supply current offset	Measured into VC5x (bq76930, bq76940) and VC10x (bq76940)	–1.0	±0.1	1.0	µA
dI_{ALERT}	Supply current when ALERT active	Measured into VC5x (bq76930, bq76940) or added to BAT (bq76920)		15	25	µA
dI_{CELL}	Cell measurement input current	Measured into VC0–VC15 except VC5, VC10, VC15	–0.3	±0.1	0.3	µA
		Measured into VC5, VC10, VC15			0.5	µA
I_{LKG}	Terminal input leakage				1	µA
INTERNAL POWER CONTROL (STARTUP and SHUTDOWN)						
V_{PORA}	Analog POR threshold	See Note (1)	4		5	V
V_{SHUT}	Shutdown voltage	See Note (1)			3.6	V
$t_{I2CSTARTUP}$	Time delay after boot signal on TS1 before I²C communications allowed	Delay after boot sequence when I²C communication is allowed		1		ms
$t_{BOOTREADY}$	Device boot startup delay	Delay after boot signal when device has completed full boot-up sequence			10	ms
T_{SHUTD}	Thermal shutdown voltage			100	150	°C
MEASUREMENT SCHEDULE						
t_{VCELL}	Cell voltage measurement interval	bq76920, bq76930, bq76940		250		ms
$t_{INDCELL}$	Individual cell measurement time	Per cell, balancing off		50		ms
		Per cell, balancing on		12.5		ms
t_{CB_RELAX}	Cell balancing relaxation time before cell voltage measured			12.5		ms
t_{TEMP_DEC}	Temperature measurement decimation time	Measurement duration for temperature reading		12.5		ms
t_{BAT}	Pack voltage calculation interval			250		ms
t_{TEMP}	Temperature measurement interval	Period of measurement of either TS1/TS2/TS3 or internal die temp		2		s
14-BIT ADC FOR CELL VOLTAGE AND TEMPERATURE MEASUREMENT						
ADC_{RANGE}	ADC measurement recommend operation range	V_{CELL} measurements	2		5	V
		TS/Temp measurements	0.3		3	V
ADC_{LSB}	ADC LSB value			382		µV
ADC_{25A}	ADC cell voltage accuracy	V_{CELL} = 3.6 – 4.3 V	–40	±10	40	mV
ADC_{25B}		V_{CELL} = 3.2 – 4.6 V	–40	±15	40	mV
ADC_{25C}		V_{CELL} = 2.0 – 5.0 V	–50	±25	50	mV
ADC_{60}	ADC cell voltage accuracy temperature drift adder from 25°C	T_A = 0ºC to 60ºC	–10		10	mV
ADC_{TS}	ADC thermistor measurement accuracy	Input from 0.3 to 3.0 V, BAT > 7 V	–35		35	mV

(1) Measured at BAT pin, rising.

电气特性（续）

典型条件通常在25°C下测量，具有18 V（bq76920）、36 V（bq76930）或48 V（bq76940）的标称BAT电压，并伴有V_{CELL} = 4 V。最小值和最大值包括从–40°C到 +85°C的完整推荐工作条件温度范围。某些特性可能在不同的电压或温度范围内显示，如测试条件部分所述。

参数	描述	测试条件	MIN	TYP	MAX	UNIT
漏电流和失调电流						
dI_{NOM}	正常模式供电电流偏移	测量至VC5x (bq76930, bq76940) 和 VC10x (bq76940)	–5	±2.5	5	µA
dI_{SHIP}	SHIP模式电流偏移	测量至VC5x (bq76930, bq76940) 和 VC10x (bq76940)	–1.0	±0.1	1.0	µA
dI_{ALERT}	供电电流时 ALERT激活	测量至VC5x (bq76930, bq76940) 或添加至 BAT (bq76920)		15	25	µA
dI_{CELL}	细胞测量输入 当前	测量至VC0–VC15（除VC5、VC10、VC15外）	–0.3	±0.1	0.3	µA
		测量至VC5、VC10、VC15			0.5	µA
I_{LKG}	端子输入泄漏				1	µA
内部电源控制（启动和关闭）						
V_{PORA}	模拟POR阈值	参见注释(1)	4		5	V
V_{SHUT}	关断电压	参见注释(1)			3.6	V
$t_{I2CSTARTUP}$	启动后延时 I²C前TS1上的信号 允许通信	I²C时启动序列后的延时 允许通信		1		ms
$t_{BOOTREADY}$	设备启动启动延迟	设备完成完整启动序列后，延迟信号			10	ms
T_{SHUTD}	热关机 电压			100	150	°C
测量计划						
t_{VCELL}	电池电压测量 间隔	bq76920, bq76930, bq76940		250		ms
$t_{INDCELL}$	单个电池测量时间	每个电池，平衡关闭		50		ms
		每个电池，平衡开启		12.5		ms
t_{CB_RELAX}	电池平衡弛豫 测量电池电压前的时间			12.5		ms
t_{TEMP_DEC}	温度 测量降采样时间	温度测量持续时间 读数		12.5		ms
t_{BAT}	包电压计算间隔			250		ms
t_{TEMP}	温度测量间隔	测量周期为 TS1/TS2/TS3或内部芯片温度		2		s
14位ADC用于电池电压和温度测量						
ADC_{RANGE}	ADC 测量 推荐操作 范围	V_{CELL} 测量	2		5	V
		TS/温度测量	0.3		3	V
ADC_{LSB}	ADC最低位值			382		µV
ADC_{25A}	ADC单元电压 精度	V_{CELL} = 3.6 – 4.3 V	–40	±10	40	mV
ADC_{25B}		V_{CELL} = 3.2 – 4.6 V	–40	±15	40	mV
ADC_{25C}		V_{CELL} = 2.0 – 5.0 V	–50	±25	50	mV
ADC_{60}	ADC单元电压 精度温度漂移加法器从25°C	T_A = 0oC至60oC	–10 10			mV
ADC_{TS}	ADC热敏电阻测量精度	输入0.3至3.0 V，BAT > 7 V	–35 35			mV

(1) BAT引脚测量，上升。

Electrical Characteristics *(continued)*

Typical conditions are measured at 25ºC with nominal BAT voltages of 18 V (bq76920), 36 V (bq76930), or 48 V (bq76940) with V_{CELL} = 4 V. Min and max values include full recommended operating condition temperature range from –40ºC to +85ºC. Certain characteristics may be shown at different voltage or temperature ranges, as clarified in the Test Condition sections.

PARAMETER	DESCRIPTION	TEST CONDITION	MIN	TYP	MAX	UNIT
16-BIT CC FOR PACK CURRENT MEASUREMENT						
CC_{RANGE}	CC input voltage range		–200		200	mV
CC_{FSR}	CC full scale range		–270		270	mV
CC_{LSB}	CC LSB value	CC running constantly		8.44		µV
tCC_{READ}	Conversion time	Single conversion		250		ms
CC_{INL}	Integral nonlinearity	16-bit, best fit over input voltage range ± 200 mV		± 2	± 40	LSB
CC_{OFFSET}	Offset error			± 1	± 3	LSB
CC_{GAIN}	Gain error	Over input voltage range		± 0.5%	± 1.5%	FSR
$CC_{GAINDRIFT}$	Gain error drift	Over input voltage range			150	PPM / °C
CC_{RIN}	Effective input resistance			2.5		MΩ
THERMISTOR BIAS						
R_{TS}	Pull-up resistance	T_A = 25°C	9.85	10	10.15	kΩ
$R_{TSDRIFT}$	Pull-up resistance across temp	T_A = –40°C to 85°C	9.7		10.3	kΩ
DIETEMP						
$V_{DIETEMP25}$	Die temperature voltage	T_A = 25°C		1.20		V
$V_{DIETEMPDRIFT}$	Die temperature voltage drift			–4.2		mV/°C
INTEGRATED HARDWARE PROTECTIONS						
OV_{RANGE}	OV threshold range		0x2008		0x2FF8	ADC
UV_{RANGE}	UV threshold range		0x1000		0x1FF0	ADC
OV_{UVSTEP}	OV and UV threshold step size			16		LSB
$UV_{MINQUAL}$	UV minimum value to qualify	Below $UV_{MINQUAL}$, cell is shorted (unused)		0x0518		ADC
OV_{DELAY}	OV delay timer options	OV delay = 1 s	0.7	1	1.75	s
		OV delay = 2 s	1.6	2	2.75	
		OV delay = 4 s	3.5	4	5	
		OV delay = 8 s	7	8	10	
UV_{DELAY}	UV delay timer options	UV delay = 1 s	0.7	1	1.75	
		UV delay = 4 s	3.5	4	5	
		UV delay = 8 s	7	8	10	
		UV delay = 16 s	14	16	20	
OCD_{RANGE}	OCD threshold options	Measured across (SRP–SRN) (2)	8		100	mV
OCD_{STEP}	OCD threshold step size	RSNS = 0		2.78		mV
		RSNS = 1		5.56		mV
OCD_{DELAY}	OCD delay options	See Note (3)	8		1280	ms
SCD_{RANGE}	SCD threshold options	Measured across (SRP–SRN) (2)	22		200	mV
SCD_{STEP}	SCD threshold step size	RSNS = 0		11.1		mV
		RSNS = 1		22.2		mV
SCD_{DELAY}	SCD delay options		35	70	105	µs
			50	100	150	µs
			140	200	260	µs
			280	400	520	µs

(2) Values indicate nominal thresholds only. For Min and Max variation, apply OC_{OFFSET} and $OC_{SCALERR}$.
(3) Values indicate nominal thresholds only. For Min and Max variation, apply $t_{PROTACC}$.

电气特性（续）

典型条件在25°C下测量，使用标称BAT电压18 V（bq76920）、36 V（bq76930）或48 V（bq76940），并配合V_{CELL} = 4 V。最小值和最大值包括从–40°C到 +85°C的完整推荐工作温度范围。
某些特性可能在不同的电压或温度范围内显示，如测试条件部分所述。

参数	描述	测试条件	MIN	TYP	MAX	UNIT
16-BIT CC FOR PACK CURRENT MEASUREMENT						
CC_{RANGE}	CC输入电压范围		–200		200	mV
CC_{FSR}	CC满量程范围		–270		270	mV
CC_{LSB}	CC LSB值	CC持续运行		8.44		µV
tCC_{READ}	转换时间	单次转换		250		ms
CC_{INL}	积分非线性度	16位，输入电压范围± 200 mV最佳拟合		± 2	± 40	LSB
CC_{OFFSET}	偏移误差			± 1	± 3	LSB
CC_{GAIN}	误差增益	输入电压范围超限		± 0.5%	± 1.5%	FSR
$CC_{GAINDRIFT}$	误差增益漂移	输入电压范围			150	PPM / °C
CC_{RIN}	有效输入电阻			2.5		MΩ
热敏电阻偏置						
R_{TS}	上拉电阻	T_A = 25°C	9.85	10	10.15	kΩ
$R_{TSDRIFT}$	上拉电阻 temp	T_A =–40°C至85°C	9.7		10.3	kΩ
DIETEMP						
$V_{DIETEMP25}$	芯片温度电压	T_A = 25°C		1.20		V
$V_{DIETEMPDRIFT}$	芯片温度电压漂移			–4.2		mV/°C
集成硬件保护						
OV_{RANGE}	OV阈值范围		0x2008		0x2FF8	ADC
UV_{RANGE}	UV阈值范围		0x1000		0x1FF0	ADC
OV_{UVSTEP}	OV 和 UV 阈值步长			16		LSB
$UV_{MINQUAL}$	UV 最小值以符合资格	在$UV_{MINQUAL}$以下，单元格短路（未使用）		0x0518		ADC
OV_{DELAY}	OV延迟定时器选项	OV延迟 = 1 秒	0.7	1	1.75	s
		OV延迟 = 2 秒	1.6	2	2.75	
		OV延迟 = 4 秒	3.5	4	5	
		OV延迟 = 8 秒	7	8	10	
UV_{DELAY}	UV延迟定时器选项	UV延迟 = 1 秒	0.7	1	1.75	
		UV延迟 = 4 秒	3.5	4	5	
		UV延迟 = 8 秒	7	8	10	
		UV延迟 = 16 秒	14	16	20	
OCD_{RANGE}	OCD阈值选项	在(SRP–SRN) (2)范围内测量	8		100	mV
OCD_{STEP}	OCD阈值步长	RSNS = 0		2.78		mV
		RSNS = 1		5.56		mV
OCD_{DELAY}	OCD延迟选项	参见注释(3)	8		1280	ms
SCD_{RANGE}	SCD阈值选项	在(SRP–SRN) (2)上测量	22		200	mV
SCD_{STEP}	SCD阈值步长	RSNS = 0		11.1		mV
		RSNS = 1		22.2		mV
SCD_{DELAY}	SCD延迟选项		35	70	105	µs
			50	100	150	µs
			140	200	260	µs
			280	400	520	µs

(2) 值仅表示标称阈值。对于最小值和最大值变化，请应用OC_{OFFSET}和$OC_{SCALERR}$。
(3) 数值仅表示标称阈值。对于Min和Max变化，应用$t_{PROTACC}$。

Electrical Characteristics *(continued)*

Typical conditions are measured at 25ºC with nominal BAT voltages of 18 V (bq76920), 36 V (bq76930), or 48 V (bq76940) with V_{CELL} = 4 V. Min and max values include full recommended operating condition temperature range from –40ºC to +85ºC. Certain characteristics may be shown at different voltage or temperature ranges, as clarified in the Test Condition sections.

PARAMETER	DESCRIPTION	TEST CONDITION	MIN	TYP	MAX	UNIT
$t_{PROTACC}$	Delay accuracy for OCD		–20%		20%	
OC_{OFFSET}	OCD and SCD voltage offset		–2.5		2.5	mV
$OC_{SCALEERR}$	OCD and SCD scale accuracy		–10%		10%	
CHARGE AND DISCHARGE DRIVERS						
V_{FETON}	CHG and DSG on	REGSRC ≥ 12 V with load resistance of 10 MΩ	10	12	14	V
		REGSRC < 12 V with load resistance of 10 MΩ	REGSRC –2	REGSRC –1	REGSRC	V
t_{FET_ON}	CHG and DSG ON rise time	CHG/DSG driving an equivalent load capacitance of 10 nF, measured from 10% to 90% of V_{FETON}		200	250	µs
t_{DSG_OFF}	DSG pull-down OFF fall time	DSG driving an equivalent load capacitance of 10 nF, measured from 90% to 10%		60	90	µs
R_{CHG_OFF}	CHG pull-down OFF resistance to VSS	When CHG disabled, CHG held at 12 V	750	1000	1250	kΩ
R_{DSG_OFF}	DSG pull-down OFF resistance to VSS	When DSG disabled, DSG held at 12 V	1.75	2.50	4.25	kΩ
V_{LOAD_DETECT}	Load detection threshold		0.4	0.7	1.0	V
V_{CHG_CLAMP}	CHG clamp voltage	If the CHG pin externally pulled high (through PACK–, if load applied), 500 µA max sink current into CHG pin. With CHG_ON bit cleared.	18	20	22	V
ALERT PIN						
V_{ALERT_OH}	ALERT output voltage high	I_{OL} = 1 mA	REGOUT x 0.75			V
V_{ALERT_OL}	ALERT output voltage low	Unloaded			REGOUT x 0.25	V
V_{ALERT_IH}	ALERT input high	ALERT externally forced high when internally driven low. See note (4).	1		1.5	V
R_{ALERT_PD}	ALERT pin weak pulldown resistance when driven low	Measured into ALERT pin with ALERT = REGOUT	0.8	2.5	8	MΩ
CELL BALANCING DRIVER						
R_{DSFET}	Internal cell balancing driver resistance	V_{CELL} = 3.6 V	1	5	10	Ω
X_{BAL}	Cell balancing duty cycle when enabled	Every 250 ms		70%		
EXTERNAL REGULATOR						
V_{EXTLDO}	External LDO voltage options	Nominal values, TI factory programmed, unloaded, across temp	2.45	2.50	2.55	V
			3.20	3.30	3.40	V
V_{EXTLDO_LN}	Line regulation	REGSRC pin stepped from 6 to 25 V, with 10 mA load, in 100 µs			100	mV
V_{EXTLDO_LD}	Load regulation	I_{REGOUT} = 0 mA to 10 mA	–4%		4%	
V_{EXTLDO_DC}	External LDO minimum voltage under DC load	REGOUT = 10 mA DC, 2.5-V version	2.4			V
		REGOUT = 20 mA DC, 2.5-V version	2.3			V
		REGOUT = 10 mA DC, 3.3-V version	3.15			V
		REGOUT = 20 mA DC, 3.3-V version	3.05			V

(4) MIN specifies the threshold below which the device will never register that an external alert has occurred. MAX specifies the minimum threshold above which the device will always register that an external alert has occurred.

电气特性（续）

典型条件在25°C下测量，使用标称BAT电压18 V（bq76920）、36 V（bq76930）或48 V（bq76940）并配合V_{CELL} = 4 V。最小值和最大值包括从–40°C到 +85°C的完整推荐工作条件温度范围。某些特性可能在不同的电压或温度范围内显示，如测试条件部分所述。

参数	描述	测试条件	MIN	TYP	MAX	UNIT
$t_{PROTACC}$	OCD延迟精度		–20%		20%	
OC_{OFFSET}	OCD和SCD电压偏移		–2.5		2.5	mV
$OC_{SCALEERR}$	OCD和SCD比例精度		–10%		10%	
充放电驱动						
V_{FETON}	CHG和DSG开启	REGSRC ≥ 12 V负载电阻为 10 MΩ	10	12	14	V
		REGSRC < 12 V负载电阻为 10 MΩ	REGSRC –2	REGSRC –1	REGSRC	V
t_{FET_ON}	CHG和DSG上升 time	CHG/DSG 驱动等效负载电容 10 nF，测量范围从 10% 到 90% 的 V_{FETON}		200	250	µs
t_{DSG_OFF}	DSG 拉低 OFF 下降 time	DSG 驱动等效负载电容 10 nF，测量范围从 90% 到 10%		60	90	µs
R_{CHG_OFF}	CHG 拉低 OFF 抗VSS	当CHG禁用时，CHG保持在12 V	750	1000	1250	kΩ
R_{DSG_OFF}	DSG下拉OFF抗VSS	当DSG禁用时，DSG保持在12 V	1.75	2.50	4.25	kΩ
V_{LOAD_DETECT}	负载检测阈值		0.4	0.7	1.0	V
V_{CHG_CLAMP}	CHG 端钳位电压	如果 CHG 引脚外部被拉高（通过 PACK–，如果施加负载），则 CHG 引脚最大 500 µA 源电流。清除 CHG_ON 位。	18	20	22	V
ALERT 引脚						
V_{ALERT_OH}	警报输出电压 high	I_{OL} = 1 mA	REGOUT x 0.75			V
V_{ALERT_OL}	警报输出电压低	未加载			REGOUT x 0.25	V
V_{ALERT_IH}	ALERT 输入高	ALERT 外部强制高时内部驱动低。参见注释 (4)。	1		1.5	V
R_{ALERT_PD}	ALERT引脚弱低电平时下拉电阻	通过ALERT= REGOUT测量到ALERT引脚	0.8	2.5	8	MΩ
单元均衡驱动器						
R_{DSFET}	内部单元均衡驱动器电阻	V_{CELL} = 3.6 V	1	5	10	Ω
X_{BAL}	启用时的电池均衡占空比	每 250 ms		70%		
外部稳压器						
V_{EXTLDO}	外部 LDO 电压选项	标称值，TI工厂编程，未加载，跨温度	2.45	2.50	2.55	V
			3.20	3.30	3.40	V
V_{EXTLDO_LN}	线路调节	REGSRC引脚从6V stepped到25V，在100 µs内施加10 mA负载			100	mV
V_{EXTLDO_LD}	负载调节	I_{REGOUT} = 0 mA 至 10 mA	–4%		4%	
V_{EXTLDO_DC}	外部LDO最小值直流负载电压	REGOUT = 10 mA 直流，2.5-V版本	2.4			V
		REGOUT = 20 mA DC, 2.5-V 版本	2.3			V
		REGOUT = 10 mA DC, 3.3-V 版本	3.15			V
		REGOUT = 20 mA DC, 3.3-V 版本	3.05			V

(4) MIN表示设备不会检测到外部警报发生的阈值以下值。MAX表示设备始终会检测到外部警报发生的阈值以上最小值。

Electrical Characteristics *(continued)*

Typical conditions are measured at 25ºC with nominal BAT voltages of 18 V (bq76920), 36 V (bq76930), or 48 V (bq76940) with V_{CELL} = 4 V. Min and max values include full recommended operating condition temperature range from –40ºC to +85ºC. Certain characteristics may be shown at different voltage or temperature ranges, as clarified in the Test Condition sections.

PARAMETER	DESCRIPTION	TEST CONDITION	MIN	TYP	MAX	UNIT
I_{EXTLDO_LIMIT}	External LDO current limit	REGOUT = 0 V	30	38	45	mA
BOOT DETECTOR						
V_{BOOT}	Boot threshold voltage	Measured at TS1 pin with device in SHIP mode. Below MIN, device will not boot up. Above MAX, device will be guaranteed to boot up.	300		1000	mV
t_{BOOT_max}	Boot threshold application time	Measured at TS1 pin. Below MIN, device will not boot up. Above MAX, device will be guaranteed to boot up.	10		2000	µs

6.6 Timing Requirements

I²C COMPATIBLE INTERFACE		MIN	TYP	MAX	UNIT
V_{IL}	Input Low Logic Threshold			REGOUT x 0.25	V
V_{IH}	Input High Logic Threshold	REGOUT x 0.75			V
V_{OL}	Output Low Logic Drive			0.20	V
t_f	SCL, SDA Fall Time			0.40	
V_{OH}	Output High Logic Drive (Not applicable due to open-drain outputs)	N/A		N/A	V
t_{HIGH}	SCL Pulse Width High	4.0			µs
t_{LOW}	SCL Pulse Width Low	4.7			µs
$t_{SU;STA}$	Setup time for START condition	4.7			µs
$t_{HD;STA}$	START condition hold time after which first clock pulse is generated	4.0			µs
$t_{SU;DAT}$	Data setup time	250			ns
$t_{HD;DAT}$	Data hold time	0			µs
$t_{SU;STO}$	Setup time for STOP condition	4.0			µs
t_{BUF}	Time the bus must be free before new transmission can start	4.7			µs
$t_{VD;DAT}$	Clock Low to Data Out Valid			900	ns
$t_{HD;DAT}$	Data Out Hold Time After Clock Low	0			ns
f_{SCL}	Clock Frequency	0		100	kHz

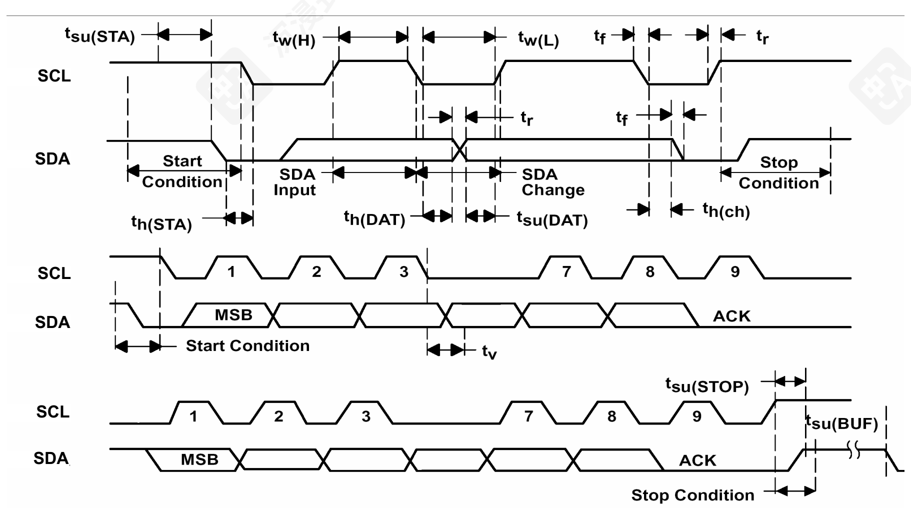


图 6-1. I²C Timing

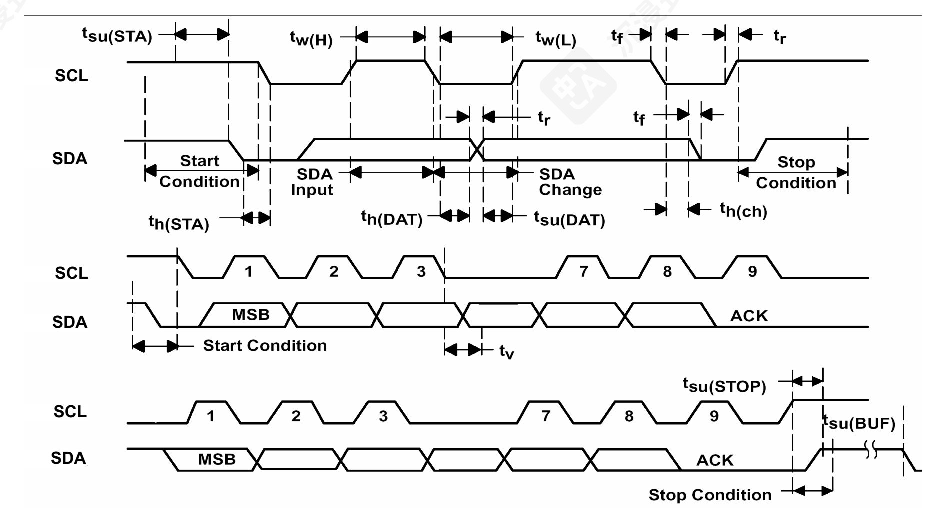
电气特性（续）

典型条件在25°C下测量，使用标称BAT电压18 V（bq76920）、36 V（bq76930）或48 V（bq76940）及V_{CELL} = 4 V。最小值和最大值包括从–40°C到 +85°C的完整推荐工作温度范围。某些特性可能在不同电压或温度范围内显示，如测试条件部分所述。

参数	描述	测试条件	MIN	TYP	MAX	UNIT
I_{EXTLDO_LIMIT}	外部LDO电流限制	REGOUT = 0 V	30	38	45	mA
启动检测器						
V_{BOOT}	启动阈值电压	在TS1引脚上测量，设备处于SHIP模式。低于MIN值，设备将无法启动。高于MAX值，设备将保证能启动。	300		1000	mV
t_{BOOT_max}	启动阈值应用时间	在TS1引脚处测量。低于MIN值，设备将无法启动。高于MAX值，设备将保证启动。	10		2000	µs

6.6 定时要求

I²C 兼容接口		MIN	TYP	MAX	UNIT
V_{IL}	输入低电平逻辑阈值			REGOUT x 0.25	V
V_{IH}	输入高电平阈值	REGOUT x 0.75			V
V_{OL}	输出低电平驱动			0.20	V
t_f	SCL、SDA 下降时间			0.40	
V_{OH}	输出高电平驱动（由于开漏输出，不适用）	N/A		N/A	V
t_{HIGH}	SCL脉冲宽度高	4.0			µs
t_{LOW}	SCL脉冲宽度低	4.7			µs
$t_{SU;STA}$	START条件建立时间	4.7			µs
$t_{HD;STA}$	起始条件保持时间，此后生成第一个时钟脉冲	4.0			µs
$t_{SU;DAT}$	数据建立时间	250			ns
$t_{HD;DAT}$	数据保持时间	0			µs
$t_{SU;STO}$	STOP条件设置时间	4.0			µs
t_{BUF}	新传输开始前，巴士必须空闲的时间	4.7			µs
$t_{VD;DAT}$	时钟低电平至数据输出有效			900	ns
$t_{HD;DAT}$	时钟低电平后数据输出保持时间	0			ns
f_{SCL}	时钟频率	0		100	kHz



图图 6-1. I²C 定时

6.7 Typical Characteristics

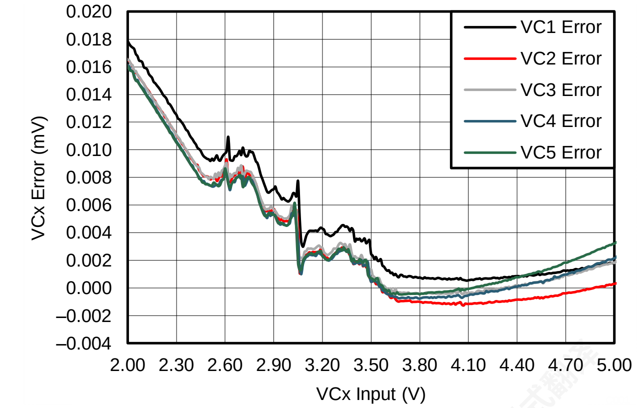


图 6-2. bq76930 VCx Error Across Input Range at 25°C with VIN at 3.6 V

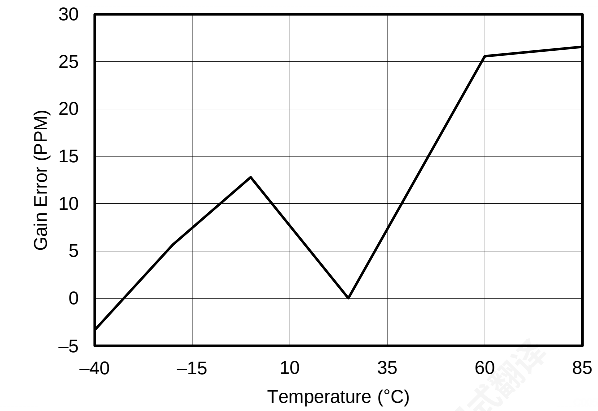


图 6-3. Coulomb Counter Gain Error Temperature Drift (from –0.2 V to 0.2 V)

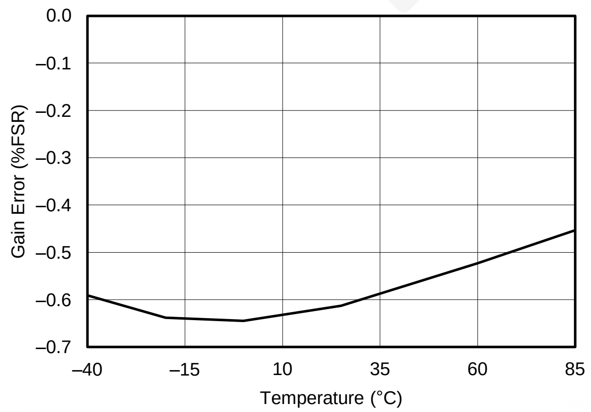


图 6-4. Coulomb Counter Gain Error (from –0.2 V to 0.2 V)

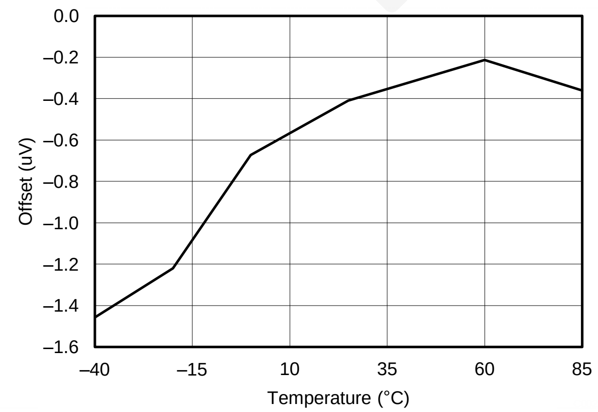


图 6-5. Coulomb Counter Offset

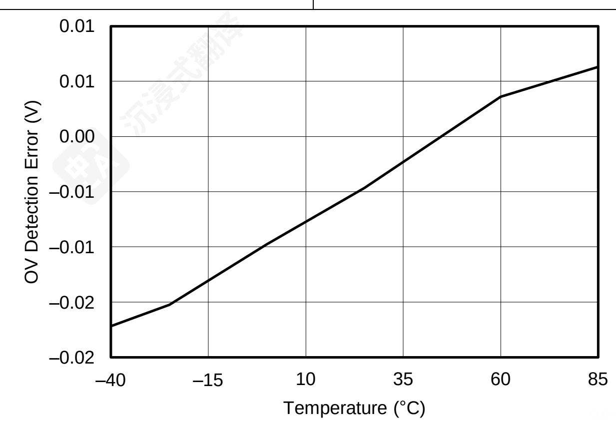
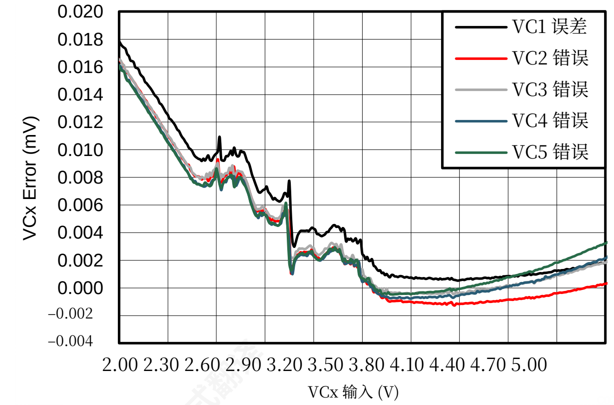
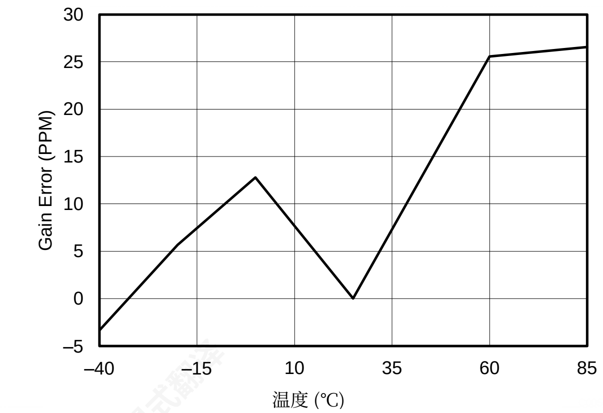


图 6-6. OV Protection Detection Error (0xFF Setting)

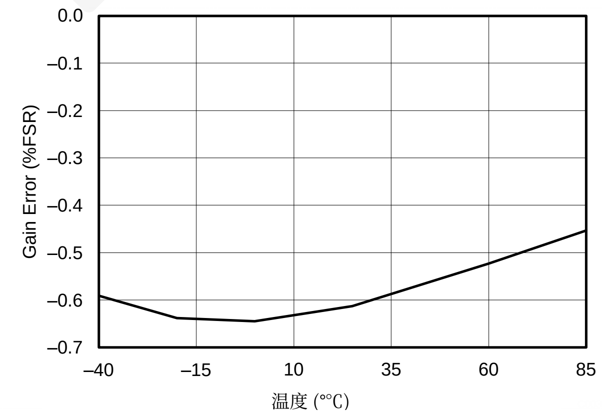
6.7 典型特性



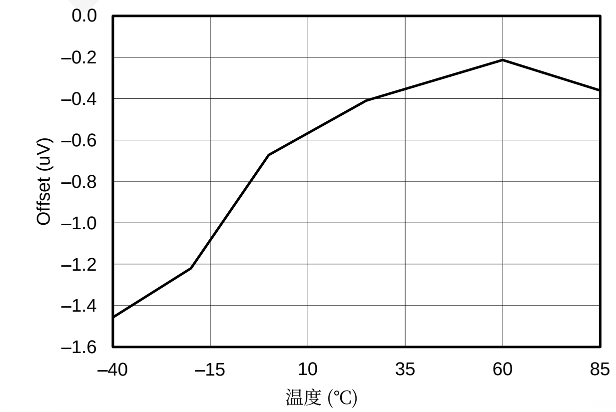
图图 6-2. bq76930 VCx 误差跨输入范围（25°C，VIN 为 3.6 V）



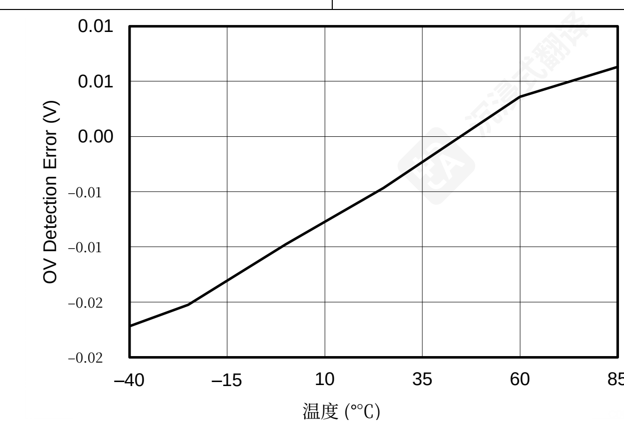
图图 6-3. 库仑计数器增益误差温度漂移（-0.2 V 至 0.2 V）



图图 6-4. 库仑计数器增益误差（-0.2 V 至 0.2 V）



图图 6-5. 库仑计数器失调



图图 6-6. 过压保护检测误差（0xFF 设置）

7 Detailed Description

7.1 Overview

In the bq769x0 family of analog front-end (AFE) devices, the bq76920 device supports up to 5-series cells, the bq76930 device supports up to 10-series cells, and the bq76940 device supports up to 15-series cells. Via I^2C, a host controller can use the bq769x0 to implement battery pack management functions, such as monitoring (cell voltages, pack current, pack temperatures), protection (controlling charge/discharge FETs), and balancing. Integrated A/D converters enable a purely digital readout of critical system parameters including cell voltages and internal or external temperature, with calibration handled in TI's manufacturing process. For an additional degree of pack reliability, the bq769x0 includes hardware protections for voltage (OV, UV) and current (OCD, SCD).

The bq769x0 provides two low-side FET drivers, charge (CHG) and discharge (DSG), which may be used to directly manipulate low-side power NCH FETs, or as signals that control an external circuit that enables high-side PCH or NCH FETs. A dedicated ALERT input/output pin serves as an interrupt signal to the host microcontroller, quickly informing the microcontroller of an updated status in the AFE. This may include a fault event or that a coulomb counter sample is available for reading. An available ALERT pin may also be driven externally by a secondary protector to provide a redundant means of disabling the CHG and DSG signals and higher system visibility.

7.2 Functional Block Diagram

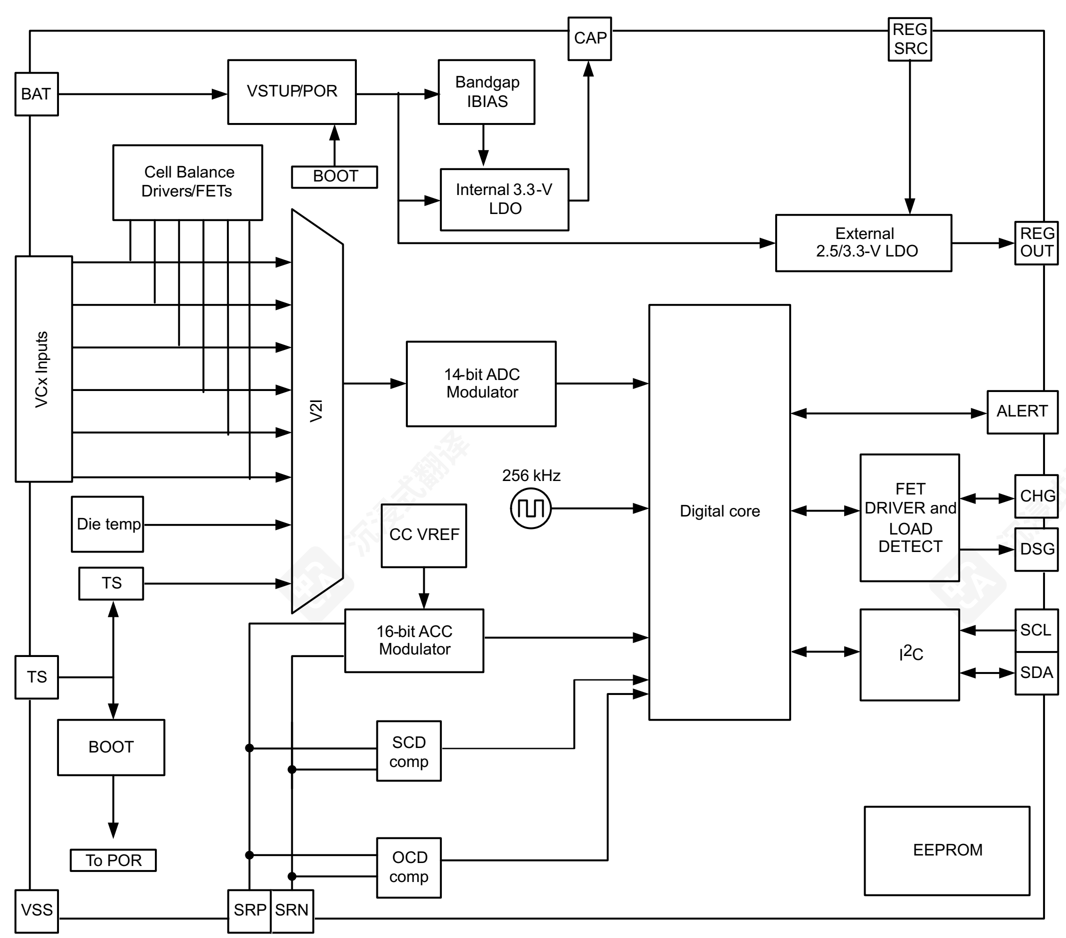


图 7-1. Functional Block Diagram

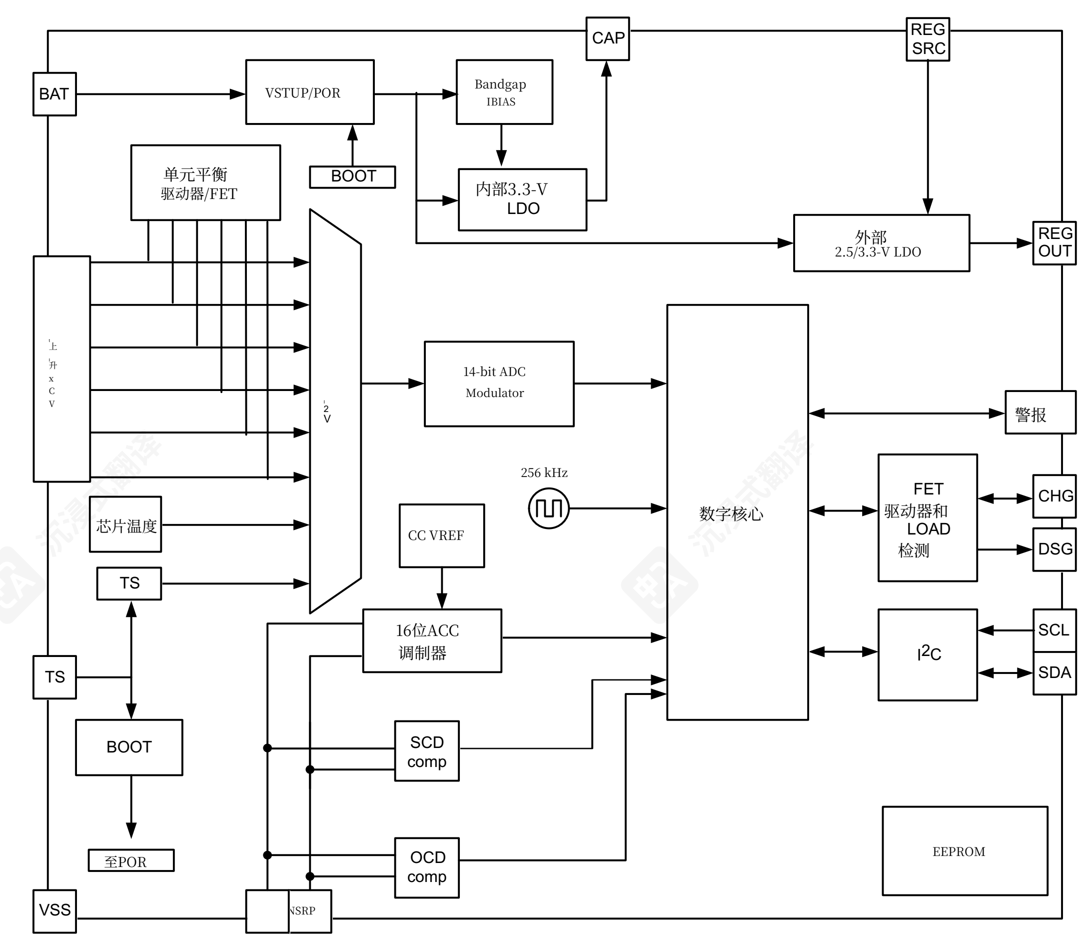
7 详细描述

7.1 概述

在 bq769x0 系列模拟前端（AFE）器件中，bq76920 器件支持高达 5 系列电池，bq76930 器件支持高达 10 系列电池，bq76940 器件支持高达 15 系列电池。通过 I^2C，主机控制器可以使用 bq769x0 实现电池组管理功能，例如监控（电池电压、电池组电流、电池组温度）、保护（控制充放电 FET）和均衡。集成的 A/D 转换器可对关键系统参数（包括电池电压和内部或外部温度）进行纯数字读取，校准由 TI 的制造工艺处理。为了进一步提高电池组的可靠性，bq769x0 包括针对电压（过压、欠压）和电流（过流、短路）的硬件保护。

bq769x0 提供两个低边 FET 驱动器，充电 (CHG) 和放电 (DSG)，可用于直接控制低边功率 NCH FET，或作为控制外部电路的信号，该电路可启用高边 PCH 或 NCH FET。一个专用的 ALERT 输入/输出引脚作为主机微控制器的中断信号，快速通知微控制器 AFE 状态已更新。这可能包括故障事件或库仑计数器样本可供读取。可用的 ALERT 引脚也可能由二级保护器外部驱动，以提供禁用 CHG 和 DSG 信号及更高系统可见性的冗余方式。

7.2 功能框图



图图 7-1. 功能框图

7.3 Feature Description

7.3.1 *Subsystems*

bq769x0 consists of three major subsystems: Measurement, Protection, and Control. These work together to ensure that the fundamental battery pack parameters—voltage, current and temperature—are accurately captured and easily available to a host controller, while ensuring a baseline or secondary level of hardware protection in the event that a host controller is unable or unavailable to manage certain fault conditions.

注

The bq769x0 is intended to serve as an analog front-end (AFE) as part of a chipset system solution: A companion microcontroller is required to oversee and control this AFE.

- The Measurement subsystem's core responsibility is to digitize the cell voltages, pack current (integrated into a passed charge calculation), external thermistor temperature, and internal die temperature. It also performs an automatic calculation of the total battery stack voltage, by simply adding up all measured cell voltages.
- The Protection subsystem provides a baseline or secondary level of hardware protections to better support a battery pack's FMEA requirements in the event of a loss of host control or simply if a host is unable to respond to a certain fault event in time. Integrated protections include pack-level faults such as OV, UV, OCD, SCD, detection of an external secondary protector fault, and internal logic "watchdog"-style device fault (XREADY). Protection events will trigger toggling of the ALERT pin, as well as automatic disabling of the DSG and/or CHG FET driver (depending on the fault). Recovery from a fault event must be handled by the host microcontroller.
- The Control subsystem implements a suite of useful pack features, including direct low-side NCH FET drivers, cell balancing drivers, the ALERT digital output, an external LDO and more.

The following sections describe each subsystem in greater detail, as well as explaining the various power states that are available.

7.3.1.1 Measurement Subsystem Overview

The monitoring subsystem ensures that all cell voltages, temperatures, and pack current may be easily measured by the host. All ADCs are trimmed by TI.

ADC and CC data are always returned as atomic values if both high and low registers are read in the same transaction (using address auto-increment).

7.3.1.1.1 Data Transfer to the Host Controller

The bq769x0 has a fully digital interface: All information is transferred through I^2C, simply by reading and/or writing to the appropriate register(s) storing the relevant data. Block reads and writes, buffered by an 8-bit CRC code per byte, ensure a fast and robust transmission of data.

7.3.1.1.2 14-Bit ADC

Each bq769x0 device measures cell voltages and temperatures using a 14-bit ADC. This ADC measures all differential cell voltages, thermistors and/or die temperature with a nominal full-scale unsigned range of 0–6.275 V and LSB of 382 µV.

To enable the ADC, the *[ADC_EN]* bit in the SYS_CTRL1 register must be set. This bit is set automatically whenever the device enters NORMAL mode. When enabled, the ADC ensures that the integrated OV and UV protections are functional.

7.3 功能描述

7.3.1 子系统

bq769x0 由三大子系统组成：测量、保护和控制。这些子系统协同工作，确保电池包的基本参数——电压、电流和温度——被准确捕获并易于主机控制器访问，同时在主机控制器无法或不可用管理某些故障条件时，提供基线或二级硬件保护。

注

bq769x0 芯片旨在作为芯片组系统解决方案中的模拟前端 (AFE)：需要一位微控制器来监督和控制这个 AFE。

- 测量子系统的主要职责是将电池电压数字化、打包电流（集成到通过电荷计算中）、外部热敏电阻温度和内部芯片温度。它还通过简单地将所有测量的电池电压相加来执行总电池堆栈电压的自动计算。

- 保护子系统提供基础或次要级别的硬件保护，以便在主机控制丢失或主机无法及时响应特定故障事件的情况下，更好地支持电池包的 FMEA 要求。集成的保护功能包括电池包级别的故障，如过压 (OV)、欠压 (UV)、过流 (OCD)、短路 (SCD)、检测外部二级保护器故障，以及内部逻辑“看门狗”式设备故障 (XREADY)。保护事件将触发 ALERT 引脚的切换，以及根据故障情况自动禁用 DSG 和/或 CHG FET 驱动器。从故障事件中恢复必须由主机微控制器处理。

- 控制子系统实现了一套实用的包功能，包括直接低侧NCH FET驱动器、电池均衡驱动器、ALERT数字输出、外部LDO等。

以下各节将更详细地描述每个子系统，并解释可用的各种电源状态。

7.3.1.1 测量子系统概述

监控子系统确保所有电池电压、温度和包电流均可被主机轻松测量。所有ADC均由TI校准。

如果同一事务中读取高低寄存器（使用地址自动递增），ADC和CC数据始终作为原子值返回。

7.3.1.1.1 数据传输至主机控制器

bq769x0 拥有全数字接口：所有信息通过 I^2C 传输，只需读取和/或写入存储相关数据的适当寄存器即可。带每字节 8 位 CRC 码缓冲的块读和写，确保数据快速且可靠的传输。

7.3.1.1.2 14 位 ADC

每个 bq769x0 设备使用 14 位 ADC 测量电池电压和温度。该 ADC 测量所有差分电池电压、热敏电阻和/或芯片温度，其标称满量程无符号范围为 0–6.275V，且 LSB 为 382 µV。

要启用ADC，必须设置SYS_CTRL1寄存器中的 *[ADC_EN]* 位。每当设备进入NORMAL模式时，该位会自动设置。启用后，ADC确保集成的OV和UV保护功能正常。

For each contiguous set of five cells (VC1 to VC5, VC6 to VC10), when no cells in that particular set are being balanced, each cell is measured over a 50-ms decimation window and a complete update is available every 250 ms. In the bq76930 and bq76940, every set of five cells above the primary five cells is measured in parallel. The 50-ms decimation greatly assists with removing the aliasing effects present in a noisy motor environment.

When any cells in a contiguous set of 5 cells are being balanced, those affected cells are measured in a reduced 12.5-ms decimation period, to allow the cell balancing to function properly without affecting the integrated OV and UV protections. Since cell balancing is typically only performed during pack charge or idle periods, the shortened decimation periods should not impact accuracy as the system noise during these times is greatly reduced. This reduced decimation period is only applied to sets where one of the cells is being balanced. The following summarizes this for the bq76920–bq76940 devices:

- VC1 to VC5 measurements are each taken in a 50-ms decimation period when all bits in CELLBAL1 register are 0, and a 12.5-ms decimation period when any bits in CELLBAL1 register are 1.
- VC6 to VC10 measurements are each taken in a 50-ms decimation period when all bits in CELLBAL2 register are 0, and a 12.5-ms decimation period when any bits in CELLBAL2 register are 1.
- VC11 to VC15 measurements are each taken in a 50-ms decimation period when all bits in CELLBAL3 register are 0, and a 12.5-ms decimation period when any bits in CELLBAL3 register are 1.
- Total update interval is 250 ms.

Each differential cell input is factory-trimmed for gain and/or offset, such that the resulting reading via I^2C is always consistent from part-to-part and requires no additional calibration or correction factor application.

The ADC is required to be enabled in order for the integrated OV and UV protections to be operating.

The following shows how to convert the 14-bit ADC reading into an analog voltage. Each device is factory calibrated, with a GAIN and OFFSET stored into EEPROM.

The ADC transfer function is a linear equation defined as follows:

V(cell) = GAIN x ADC(cell) + OFFSET (1)

GAIN is stored in units of µV/LSB, while OFFSET is stored in mV units.

Some example cell voltage calculations are provided in the table below. For illustration purposes, the example uses a hypothetical GAIN of 380 µV/LSB (ADCGAIN<4:0> = 0x0F) and OFFSET of 30 mV (ADCOFFSET<7:0> = 0x1E).

14-Bit ADC Result	ADC Result in Decimal	GAIN (µV/LSB)	OFFSET (mV)	Cell Voltage (mV)
0x1800	6144	380	30	2365
0x1F10	7952	380	30	3052

注

When entering NORMAL mode from SHIP mode, please allow for the following times before reading out initial cell voltage data:

bq76920: 250 ms

bq76930: 400 ms

bq76940: 800 ms

对于每五个连续的电池单元（VC1至VC5、VC6至VC10），当该特定组中没有任何电池单元进行均衡时，每个电池单元会在50毫秒的降采样窗口内进行测量，并且每250毫秒提供一次完整更新。在bq76930和bq76940中，主五个电池单元以上的每五个电池单元组会并行测量。50毫秒的降采样有助于消除噪声电机环境中存在的混叠效应。

当连续5个单元中的任意单元正在进行均衡时，受影响的单元将在一个缩短的12.5毫秒抽取周期内进行测量，以允许均衡功能正常工作而不影响集成的OV和UV保护。由于均衡通常仅在电池组充电或空闲期间执行，因此缩短的抽取周期不会影响精度，因为这些时段的系统噪声已大大降低。此缩短的抽取周期仅应用于其中有一个单元正在均衡的单元组。以下总结了bq76920–bq76940设备的相关内容：

- VC1至VC5的测量在CELLBAL1寄存器所有位均为0时，每个测量均在50毫秒抽取周期内进行，而在CELLBAL1寄存器任何位为1时，每个测量均在12.5毫秒抽取周期内进行。
- VC6至VC10的测量在CELLBAL2寄存器所有位均为0时，每个测量均在50毫秒抽取周期内进行，而在CELLBAL2寄存器任何位为1时，每个测量均在12.5毫秒抽取周期内进行。
- VC11至VC15的测量值均在50 ms的降采样周期内进行，当CELLBAL3寄存器中所有位均为0时；而在任何位为1时，降采样周期为12.5 ms。
- 总更新间隔为250 ms。

每个差分单元输入均经过工厂校准增益和/或偏移，因此通过I^2C读取的值始终在不同器件间保持一致，无需额外的校准或修正因子。

ADC必须启用，以便集成过压(OV)和欠压(UV)保护功能正常工作。

以下展示了如何将14位ADC读数转换为模拟电压。每个设备均经过工厂校准，其增益(GAIN)和偏移(OFFSET)存储在EEPROM中。

ADC传输函数是一个线性方程，定义如下：

V(cell) = GAIN x ADC(cell) + OFFSET (1)

GAIN 以 µV/LSB 单位存储，而 OFFSET 以 mV 单位存储。

表中提供了示例单元电压计算。为说明目的，示例使用假设的 GAIN 为 380 µV/LSB (ADCGAIN<4:0> = 0 x0F) 和 OFFSET 为 30 mV (ADCOFFSET<7:0> = 0x1E)。

14位ADC结果	ADC结果的十进制表示	GAIN (µV/LSB)	OFFSET (mV)	单元电压 (mV)
0x1800	6144	380	30	2365
0x1F10	7952	380	30	3052

注

从SHIP模式进入NORMAL模式时，在读取初始电池电压数据前，请允许以下时间：

bq76920: 250 ms

bq76930: 400 ms

bq76940: 800 ms

7.3.1.1.2.1 Optional Real-time Calibration Using the Host Microcontroller

The performance of the cell voltage values measured by the 14-bit ADC has a factory-calibrated accuracy, as follows:

- +/– 10 mV TYP, +/– 40 mV MIN and MAX from 3.6 to 4.3 V,
- +/– 15 mV TYP, +/– 40 mV MIN and MAX from 3.2 to 4.6 V, and
- +/– 50 mV MIN and MAX from 2.0 to 5.0 V

While this is suitable for the majority of pack protection and basic monitoring applications the bq769x0 AFE family is intended to support, certain systems may require a higher accuracy performance.

To achieve this, use an available ADC channel and general purpose output terminal on the host microcontroller paired with the bq769x0. A simple external circuit consisting of two precision resistors and a small-signal FET is activated by the host microcontroller to determine the total stack voltage, V_{STACK}. This is then compared against the sum of the individual cell voltages as measured by the internal ADC of the bq769x0. The resulting transfer function coefficient, $GAIN_2$, is simply applied to each cell voltage ADC value for improved accuracy.

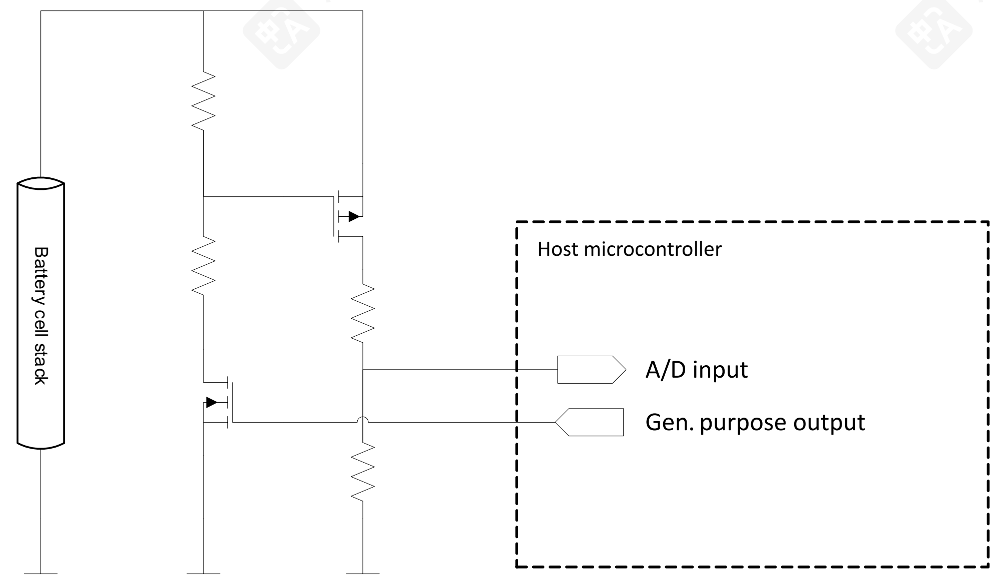


图 7-2. External Real-Time Calibration Circuit to Host Microcontroller

The process is as follows:

1. Periodically measure V_{STACK}.
   (a) $V_{STACK} = V_{AD} \times (R1 + R2) / R1$
2. Read out all V_{CELL} ADC readings from the bq769x0 and apply the standard GAIN and OFFSET values stored in the bq769x0.
   (a) $V(1) = GAIN \times ADC_1 + OFFSET$, $V(2) = GAIN \times ADC_2 + OFFSET$, and so on
3. Sum up all V_{CELL} values, V_{SUM}.
   (a) $V_{SUM} = V(1) + V(2) + V(3) \ldots$
4. Calculate $GAIN_2$.
   (a) $GAIN_2 = V_{STACK} / V_{SUM}$

As a general recommendation, a new $GAIN_2$ function should be generated when the cell voltages increase or decrease by more than 100 mV. With $GAIN_2$, each cell voltage calculation becomes:

$$V(cell) = GAIN_2 \times (GAIN \times ADC(cell) + OFFSET) \quad (2)$$

For systems that do not require this additional in-use calibration function, $GAIN_2$ is simply “1”.

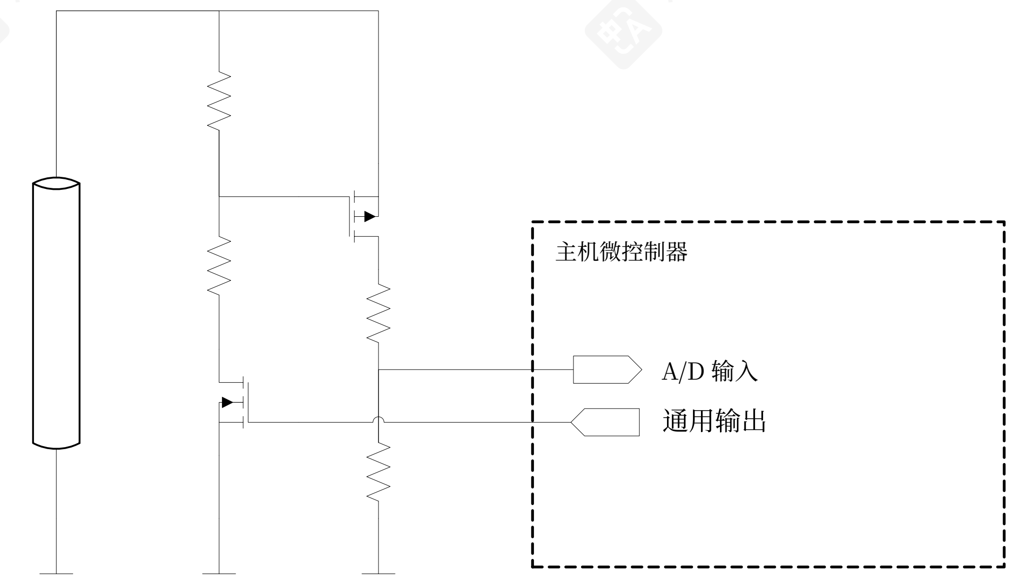
7.3.1.1.2.1 可选实时校准使用主机微控制器

由14位ADC测量的电池电压值的性能具有工厂校准的精度，如下：

- +/– 10 mV 典型值，+/– 40 mV 最小值和最大值从 3.6 到 4.3 V，
- +/– 15 mV 典型值， +/– 40 mV 最小值和最大值，范围从 3.2 至 4.6 V，和
- +/– 50 mV 最小值和最大值，范围从 2.0 至 5.0 V

虽然这对于大多数包保护和基本监控应用来说是合适的，但 bq769x0 AFE 系列旨在支持，某些系统可能需要更高的精度性能。

为此，请使用主机微控制器上的可用 ADC 通道和通用输出端，与 bq769x0 配合使用。一个由两个精密电阻和小信号 FET 组成的简单外部电路由主机微控制器激活，以确定总堆叠电压，V_{STACK}。然后，将其与由 bq769x0 内部 ADC 测量的单个电池电压之和进行比较。得到的传输函数系数，$GAIN_2$，，简单地应用于每个电池电压 ADC 值，以实现更高的精度。



图图 7-2. 外部实时校准电路至主机微控制器

流程如下：

1. 定期测量 VSTACK STACK 。
   (a) $V_{STACK} = V_{AD} \times (R1 + R2) / R1$
2. 从 bq769x0 读取所有 V_{CELL} ADC 读数，并应用 bq769x0 中存储的标准 GAIN 和 OFFSET 值。
   (a) V(1) ＝增益 x ADC_1 + 偏移，V(2) ＝增益 x ADC_2 + 偏移，等等
3. 将所有 V_{CELL}值、V_{SUM}相加。
   (a) $V_{SUM} = V(1) + V(2) + V(3)\ldots$
4. 计算 $GAIN_2$。
   (a) $GAIN_2 = V_{STACK} / V_{SUM}$一般建议，当电池电压变化超过 100 mV 时，应生成新的 $GAIN_2$函数。使用 GAIN 2，每个电池电压计算变为：

$$V(cell) = GAIN_2 \times (GAIN \times ADC(cell) + OFFSET) \quad (2)$$

对于不需要此附加使用中校准功能的系统，$GAIN_2$只需为“1”。

7.3.1.1.3 16-Bit CC

A 16-bit integrating ADC, commonly referred to as the coulomb counter (CC), provides measurements of accumulated charge across the current sense resistor. The integration period for this reading is 250 ms.

The CC may be operated in one of two modes: ALWAYS ON and 1-SHOT.

- In ALWAYS ON mode, the CC runs at 100%, gathering a fresh reading every 250 ms. The conclusion of each reading sets the CC_READY bit, which toggles the ALERT pin high to inform the microcontroller that a new reading is available. To enable Always On mode, set *[CC_EN]* = 1.
- In 1-SHOT mode, the CC performs a single 250-ms reading, and similarly sets the CC_READY bit when completed. This mode is intended for non-gauging usages, where the host simply desires to check the pack current.

  To enable a 1-SHOT reading, ensure *[CC_EN]* = 0 and set *[CC_ONESHOT]* = 1.

The full scale range of the CC is ± 270 mV, with a max recommended input range of ± 200 mV, thus yielding an LSB of approximately 8.44 µV.

The following equation shows how to convert the 16-bit CC reading into an analog voltage if no board-level calibration is performed:

CC Reading (in µV) = ***[16-bit 2's Complement Value] × (8.44 µV/LSB)*** (3)

16-Bit CC Result	ADC Result in Decimal	CC Reading (in µV)
0x0001	1	8.44
0x2710	10000	84,400
0x7D00	32000	270,080
0x8300	–32000	–270,080
0xC350	–15536	–131,123.84
0xFFFF	–1	–8.44

7.3.1.1.4 External Thermistor

One (bq76920), two (bq76930), or three (bq76940) 10 kΩ NTC 103AT thermistors may be measured by the device. These are measured by applying a factory-trimmed internal 10k pull-up resistance to an internal regulator value of nominally 3.3 V, the result of which can be read out from the TSx (TS1, TS2, TS3) registers.

To select thermistor measurement mode, set *[TEMP_SEL]* = 1.

Thermistor TS1 is connected between TS1 and VSS; TS2 is connected between TS2 and VC5x (bq76930 and bq76940 only); and TS3 is connected between TS3 and VC10x (bq76940 only). These thermistors may be placed in various areas in the battery pack to measure such things as localized cell temperature, FET heating, etc.

The thermistor impedance may be calculated using the 14-bit ADC reading in the TS1/TS2/TS3 registers and 10k internal pull-up resistance as follows:

The following equations show how to use the 14-bit ADC readings in TS1, TS2, and TS3 to determine the resistance of the external 103AT thermistor:

V_{TSX} = ***(ADC in Decimal) x 382 µV/LSB*** (4)

R_{TS} = ***(10,000 × VTSX) ÷ (3.3 – VTSX)*** (5)

To convert the thermistor resistance into temperature, please refer to the thermistor component manufacturer's datasheet.

7.3.1.1.3 16位CC

一个16位积分式ADC，通常称为库仑计（CC），提供对电流检测电阻上累积电荷的测量。此读数的积分周期为250 ms。

CC可以在两种模式下操作：ALWAYS ON和1-SHOT。

- 在ALWAYS ON模式下，CC以100%运行，每250毫秒采集一次新读数。每次读数结束时都会设置CC_READY位，该位切换ALERT引脚为高电平，以通知微控制器有新的读数可用。要启用Always On模式，请设置 *[CC_EN]* = 1。
- 在1-SHOT模式下，CC执行一次250毫秒的读数，完成后同样设置CC_READY位。此模式用于非测量用途，主机只需检查包电流。

  要启用1-SHOT读数，请确保 [CC_EN*]* = 0 并设置 *[*CC_ONESHOT*]* = 1。

CC的完整量程范围是 ± 270 mV，最大推荐输入范围为 ± 200 mV，因此产生约8.44 µV的LSB。

以下公式展示了若未进行板级校准，如何将16位CC读数转换为模拟电压：

CC读数（单位：µV） = ***[16**-位2的补码值**] ×**(8.44 µV/LSB) (3)

16位CC结果	ADC结果（十进制）	CC读数（单位：µV）
0x0001	1	8.44
0x2710	10000	84,400
0x7D00	32000	270,080
0x8300	–32000	–270,080
0xC350	–15536	–131,123.84
0xFFFF	–1	–8.44

7.3.1.1.4 外部热敏电阻

设备可以测量一个（bq76920）、两个（bq76930）或三个（bq76940）10 kΩ NTC103AT热敏电阻。这些热敏电阻通过向内部稳压器施加标称3.3 V的内部工厂校准上拉电阻进行测量，测量结果可以从TSx（TS1、TS2、TS3）寄存器中读出。

要选择热敏电阻测量模式，请设置 *[*TEMP_SEL*]* = 1。

热敏电阻TS1连接在TS1和VSS之间；TS2连接在TS2和VC5x（仅限bq76930和bq76940）；TS3连接在TS3和VC10x（仅限bq76940）。这些热敏电阻可以放置在电池包的各个位置，用于测量局部电池温度、FET加热等。

可通过TS1/TS2/TS3寄存器中的14位ADC读数和10k内部上拉电阻来计算热敏电阻阻抗，具体方法如下：

以下公式展示了如何使用TS1、TS2和TS3中的14位ADC读数来确定外部103AT热敏电阻的电阻值：

V_{TSX} = ***(ADC in Decimal) x 382 µV/LSB*** (4)

R_{TS} = ***(10,000 × VTSX) ÷ (3.3 – VTSX)*** (5)

将热敏电阻阻值转换为温度时，请参考热敏电阻组件制造商的数据手册。

7.3.1.1.5 Die Temperature Monitor

注

When switching between external and internal temperature monitoring, a 2-s latency may be incurred due to the natural scheduler update interval.

A die temperature block generates a voltage that is proportional to the die temperature, and provides a way of reducing component count if pack thermistors are not used or ensuring that the die power dissipation requirements are observed. The die is measured using the same on-board 14-bit ADC as the cell voltages.

To select internal die temperature measurement mode, set *[TEMP_SEL]* = 0.

For bq76930 and bq76940, multiple die temperature measurements are available. These are stored in TS2 and TS3.

To convert a DIETEMP reading into temperature, refer to the following equation box. If more accurate temperature readings are needed from DIETEMP, the DIETEMP at room temperature value should be stored during production calibration.

The following equation shows how to use the 14-bit ADC readings in TS1, TS2, and TS3 when [TEMPSEL] = 0 to determine the internal die temperature:

$$V_{25} = 1.200\text{ V (nominal)} \quad (6)$$

$$V_{TSX} = (\text{ADC in Decimal}) \times 382\ \mu\text{V/LSB} \quad (7)$$

$$TEMP_{DIE} = 25° – ((V_{TSX} – V_{25}) \div 0.0042) \quad (8)$$

7.3.1.1.6 16-Bit Pack Voltage

Once converted to digital form, each cell voltage is added up and the summation result stored in the BAT registers. This 16-bit value has a nominal LSB of 1.532 mV.

The following shows how to convert the 16-bit pack voltage ADC reading into an analog voltage. This value also uses the GAIN and OFFSET stored into EEPROM.

The ADC transfer function is a linear equation defined as follows:

$$V_{(BAT)} = 4 \times \text{GAIN} \times \text{ADC(cell)} + (\#\text{Cells} \times \text{OFFSET}) \quad (9)$$

GAIN is stored in units of µV/LSB, while OFFSET is stored in mV units.

7.3.1.1.7 System Scheduler

A master scheduler oversees the monitoring intervals, creating a full update every 250 ms. Temperature measurements are taken every 2 seconds. Pack voltage is calculated every 250 ms.

7.3.1.2 Protection Subsystem

7.3.1.2.1 Integrated Hardware Protections

Integrated hardware protections are provided as an extra degree of safety and are meant to supplement the standard protection feature set that would be incorporated into the host controller firmware. They should not be used as the sole means of protecting a battery pack, but are useful for FMEA purposes; for example, in the event that a host microcontroller is unable to react to any of the below protection situations. All hardware protection thresholds and delays should be loaded into the AFE by the host microcontroller during system startup. The AFE will also default to pre-defined threshold and delay settings, in case the host microcontroller is unable to or does not wish to program the protection settings.

7.3.1.1.5 芯片温度监控器

注

在外部和内部温度监控之间切换时，由于 naturalscheduler 更新间隔，可能会产生 2 秒的延迟。

芯片温度模块会生成与芯片温度成正比的电压，并在不使用封装热敏电阻时减少元件数量，或确保满足芯片功耗要求。芯片的测量使用与电池电压相同的板载14位ADC。

要选择内部芯片温度测量模式，请设置 *[TEMP_SEL]* = 0。

对于bq76930和bq76940，支持多次芯片温度测量。这些测量值存储在TS2和TS3中。

要将DIETEMP读数转换为温度值，请参考以下公式框。若需要从DIETEMP获取更精确的温度读数，应在生产校准时存储室温下的DIETEMP值。

以下公式展示了当[TEMPSEL] = 0 选择时，如何使用TS1、TS2和TS3的14位ADC读数来确定内部芯片温度：

$$V_{25} = 1.200\text{ V（标称值）} \quad (6)$$

$$V_{TSX} = (\text{ADC in Decimal}) \times 382\ \mu\text{V/LSB} \quad (7)$$

$$TEMP_{DIE} = 25° – ((V_{TSX} – V_{25}) \div 0.0042) \quad (8)$$

7.3.1.1.6 16位包电压

一旦转换为数字形式，每个单元电压会被累加，并将累加结果存储在BAT寄存器中。该16位值具有1.532 mV的标称LSB值。

以下展示了如何将16位包电压ADC读数转换为模拟电压。此值也使用存储在EEPROM中的GAIN和OFFSET。

ADC转换函数是一个线性方程，定义如下：

$$V_{(BAT)} = 4 \times \text{GAIN} \times \text{ADC(cell)} + (\#\text{Cells} \times \text{OFFSET}) \quad (9)$$

GAIN 以 µV/LSB 单位存储，而 OFFSET 以 mV 单位存储。

7.3.1.1.7 系统调度器

主调度器负责监控间隔，每 250 ms 进行一次完整更新。温度测量每 2 秒进行一次。包电压每 250 ms 计算。

7.3.1.2 保护子系统

7.3.1.2.1 集成硬件保护

集成硬件保护作为额外的安全措施提供，旨在补充主机控制器固件中集成的标准保护功能集。它们不应作为保护电池包的唯一手段，但对于FMEA目的很有用；例如，在主机微控制器无法对以下任何保护情况做出反应的情况下。所有硬件保护阈值和延迟应在系统启动时由主机微控制器加载到AFE中。如果主机微控制器无法或不愿编程保护设置，AFE将默认使用预定义的阈值和延迟设置。

Overcurrent in Discharge (OCD) and Short Circuit in Discharge (SCD) are implemented using sampled analog comparators that run at 32 kHz, and that continuously monitor the voltage across (SRP–SRN) while the device is in NORMAL mode. Upon detection of a voltage that exceeds the programmed OCD or SCD threshold, a counter begins to count up to a programmed delay setting. If the counter reaches its target value, the SYS_STAT register is updated to indicate the fault condition, the FET state(s) are updated as shown in
表 7-1, and the ALERT pin is driven high to interrupt the host.

The protection fault threshold and delay settings for OCD and SCD protections are configured via the PROTECT1 and PROTECT2 registers. See 节 7.5 for details about supported values.

Overvoltage (OV) and Undervoltage (UV) protections are handled digitally, by comparing the cell voltage readings against the 8-bit programmed thresholds in the OV and UV registers.

The OV threshold is stored in the OV_TRIP register and is a direct mapping of 8 bits of the 14-bit ADC reading, with the upper 2 MSB preset to “10” and lower 4 LSB preset to “1000”. In other words, the corresponding OV trip level is mapped to “10-XXXX-XXXX–1000”. The programmable range of OV thresholds is approximately 3.15 to 4.7 V, but this is subject to variation due to the (GAIN, OFFSET) linear equation used to map the ADC values.

The UV threshold is stored in the UV_TRIP register and is a direct mapping of 8 bits of the 14-bit ADC reading, with the upper 2 MSB preset to “01” and lower 4 LSB preset to “0000”. In other words, the corresponding OV trip level is mapped to “01-XXXX-XXXX–0000”. The programmable range of UV thresholds is approximately 1.58 to 3.1 V, but this is subject to variation due to the (GAIN, OFFSET) linear equation used to map the ADC values.

Protection	Upper 2 MSB	Middle 8 Bits	Lower 4 LSB
OV	10	Set in OV_TRIP Register	1000
UV	01	Set in UV_TRIP Register	0000

注

To support flexible cell configurations within bq76920, bq76930, and bq76940, UV is ignored on any cells that have a reading under $UV_{MINQUAL}$. This allows cell pins to be shorted in implementations where not all cells are needed (for example, 6-series cells using the bq76930).

Default protection thresholds and delays are shown in the register description at the end of this datasheet. These are loaded into the digital register (RAM) of the device when the device enters NORMAL mode. These RAM values may then be overwritten by the host controller to any other values, which they will retain until a POR event. It is recommended that the host controller reload these values during its standard power-up and/or re-initialization sequence.

To calculate the correct OV_TRIP and UV_TRIP register values for a device, use the following procedure:

1. Determine desired OV.
2. Read out *[ADCGAIN]* and *[ADCOFFSET]* from their corresponding registers. Note that ADCGAIN is stored in units of µV/LSB, while ADCOFFSET is stored in mV.
3. Calculate the full 14-bit ADC value needed to meet the desired OV and UV trip thresholds as follows:
   (a) OV_TRIP_FULL = (OV – ADCOFFSET) ÷ ADCGAIN
   (b) UV_TRIP_FULL = (UV – ADCOFFSET) ÷ ADCGAIN
4. Remove the upper 2 MSB and lower 4 LSB from the full 14-bit value, retaining only the remaining middle 8 bits. This can be done by shifting the OV_TRIP_FULL and UV_TRIP_FULL binary values 4 bits to the right and removing the upper 2 MSB.
5. Write OV_TRIP and UV_TRIP to their corresponding registers.

放电过流（OCD）和放电短路（SCD）通过运行在32 kHz的采样模拟比较器实现，这些比较器在设备处于NORMAL模式时持续监测（SRP–SRN）之间的电压。一旦检测到超过编程的OCD或SCD阈值的电压，计数器开始向上计数至编程的延迟设置。如果计数器达到目标值，SYS_STAT寄存器将更新以指示故障状态，FET状态将按所示更新。

表 7-1，以及警报引脚被驱动为高电平以中断主机。

OCD 和 SCD 保护的保护故障阈值和延迟设置通过 PROTECT1 和 PROTECT2 寄存器进行配置。有关支持值的详细信息，请参阅 7.5 节。

过压 (OV) 和欠压 (UV) 保护由数字方式处理，通过将电池电压读数与 OV 和 UV 寄存器中的 8 位编程阈值进行比较。

OV 阈值存储在 OV_TRIP 寄存器中，它是 14 位 ADC 读数的 8 位直接映射，其中最高 2 位 MSB 预设为“10”，最低 4 位 LSB 预设为“1000”。换句话说，相应的 OV 跳转电平映射为“10-XXXX-XXXX–1000”。OV 阈值的可编程范围约为 3.15 至 4.7 V，但由于用于映射 ADC 值的 (GAIN, OFFSET) 线性方程，此范围可能会有所变化。

UV阈值存储在UV_TRIP寄存器中，是14位ADC读数的8位直接映射，其中高2位MSB预设为“01”，低4位LSB预设为“0000”。换句话说，相应的OV触发电平映射为“01-XXXX-XXXX–0000”。UV阈值的可编程范围约为1.58至3.1 V，但由于用于映射ADC值的(GAIN, OFFSET)线性方程，此范围可能有所变化。

保护	高2位MSB	中8位	低4位LSB
OV	10	设置在OV_TRIP寄存器中	1000
UV	01	设置在UV_TRIP寄存器中	0000

注

为了支持bq76920、bq76930和bq76940内的灵活单元配置，当任何单元的读数低于$UV_{MINQUAL}$时，UV将被忽略。这允许在不需要所有单元的实现中（例如，使用bq76930的6系列单元）将单元引脚短路。

bq76930）。

默认的保护阈值和延迟显示在本数据表末尾的寄存器描述中。当设备进入NORMAL模式时，这些值会加载到设备的数字寄存器（RAM）中。然后，这些RAM值可能会被主机控制器覆盖为任何其他值，并且它们会保留到POR事件发生。建议主机控制器在其标准上电和/或重新初始化序列期间重新加载这些值。

要计算设备的正确OV_TRIP和UV_TRIP寄存器值，请使用以下步骤：

1. 确定所需的OV。
2. 从其对应的寄存器中读出 *[ADCGAIN]* 和 *[ADCOFFSET]*。注意，ADCGAIN以µV/LSB为单位存储，而ADCOFFSET以mV为单位存储。
3. 计算满足所需 OV 和 UV 跳变阈值所需的完整 14 位 ADC 值，方法如下：(a) OV_TRIP_FULL = (OV – ADCOFFSET) ÷ADCGAIN
   (b) UV_TRIP_FULL = (UV – ADCOFFSET) ÷ ADCGAIN
4. 从完整 14 位值中移除最高 2 MSB 和最低 4 LSB，仅保留中间的 8 位。这可以通过将 OV_TRIP_FULL 和 UV_TRIP_FULL 二进制值右移 4 位并移除最高 2 MSB 来完成。
5. 将 OV_TRIP 和 UV_TRIP 写入它们对应的寄存器。

Both OV and UV protections require the ADC to be enabled. Ensure that the *[ADC_EN]* bit is set to 1 if OV and UV protections are needed.

7.3.1.2.2 Reduced Test Time

A special debug and test configuration bit is provided in the SYS_CTRL2 register, called *[DELAY_DIS]*. Setting *[DELAY_DIS]* bypasses the OV/UV protection fault timers and allows a fault condition to be registered within 200 ms after application of such a fault condition.

7.3.1.3 Control Subsystem

7.3.1.3.1 FET Driving (CHG AND DSG)

Each bq769x0 device provides two low-side FET drivers, CHG and DSG, which control NCH power FETs or may be used as a signal to enable various other circuits such as a high-side NCH charge pump circuit.

Both DSG and CHG drivers have a fast pull-up to nominally 12 V when enabled. DSG uses a fast pull-down to VSS when disabled, while CHG utilizes a high impedance (nominally 1 MΩ) pull-down path when disabled.

An additional internal clamp circuit ensures that the CHG pin does not exceed a maximum of 20 V.

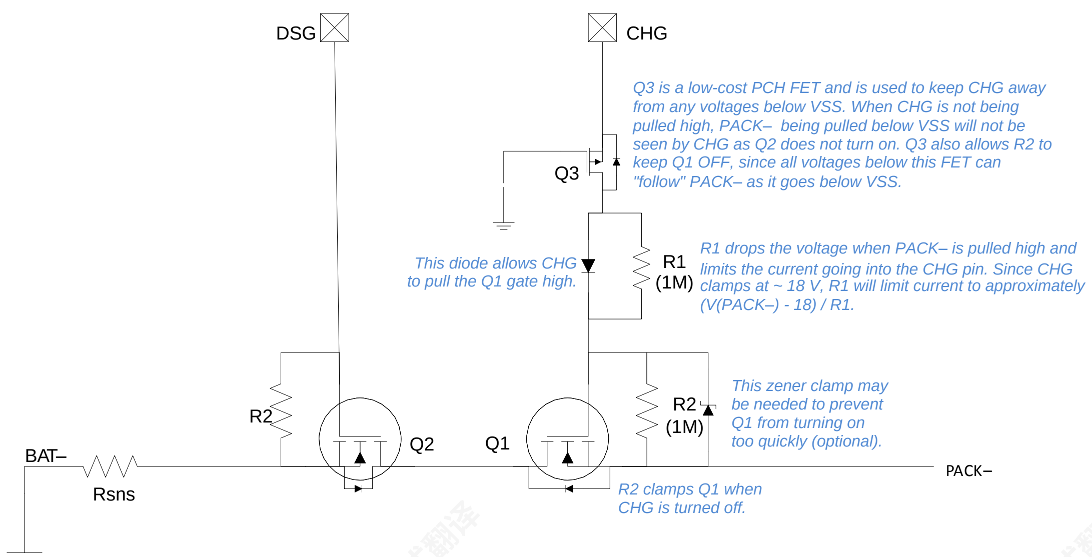


图 7-3. CHG and DSG FET Circuit

The power path for the CHG and DSG pull-up circuit originates from the REGSRC pin, instead of BAT.

To enable the CHG fet, set the [CHG_ON] register bit to 1; to disable, set [CHG_ON] = 0. The discharge FET may be similarly controlled via the [DSG_ON] register bit.

Certain fault conditions or power state transitions will clear the state of the CHG/DSG FET controls. 表 7-1 shows what action, if any, to take to [CHG_ON] and [DSG_ON] in response to various system events:

表 7-1. CHG, DSG Response Under Various System Events

EVENT	[CHG_ON]	[DSG_ON]
OV Fault	Set to 0	—
UV Fault	—	Set to 0
OCD Fault	—	Set to 0
SCD Fault	—	Set to 0

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OV和UV保护都需要启用ADC。如果需要OV和UV保护，请确保 *[ADC_EN]* 位被设置为1。

7.3.1.2.2 缩短测试时间

SYS_CTRL2 寄存器中提供了一个特殊的调试和测试配置位，称为 *[DELAY_DIS]*。设置 *[DELAY_DIS]* 会绕过 OV/UV 保护故障定时器，并允许在应用此类故障条件后 200 msafter 注册故障状态。

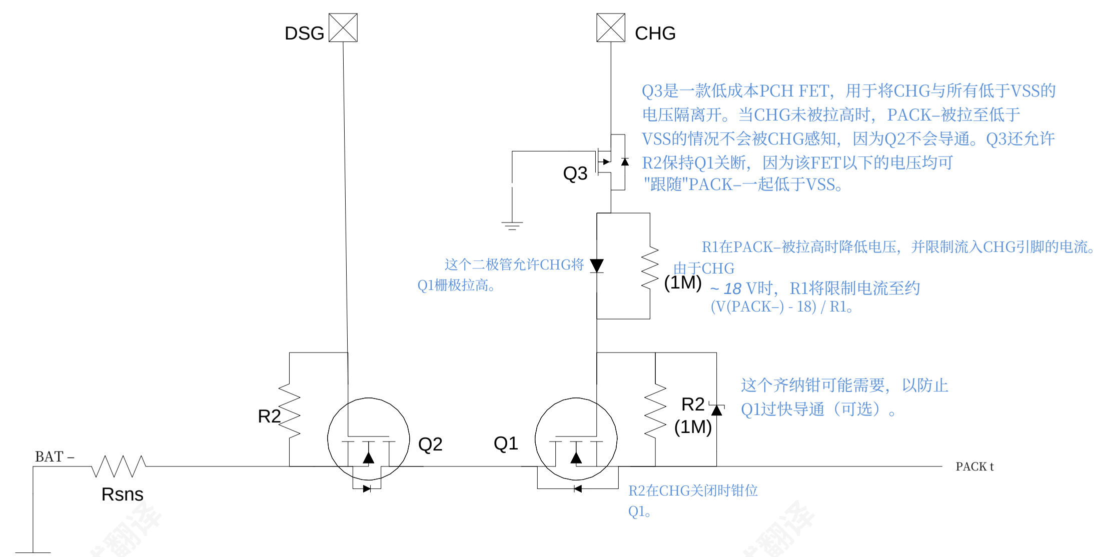
7.3.1.3 控制子系统

7.3.1.3.1 FET 驱动 (CHGAND DSG)

每个 bq769x0 设备提供两个低边 FET 驱动器，CHG 和 DSG，它们控制 NCH 功率 FET，或者可以用来作为启用其他各种电路（如高边 NCH 充电泵电路）的信号。

DSG 和 CHG 驱动器在启用时具有快速上拉至标称 12 V。DSG 在禁用时使用快速下拉至 VSS，而 CHG 在禁用时采用高阻抗（标称 1MΩ）下拉路径。

一个额外的内部钳位电路确保 CHG 引脚电压不超过 20 V 的最大值。



图图 7-3. CHG 和 DSG FET 电路

CHG 和 DSG 上拉电路的电源路径源自 REGSRC 引脚，而非 BAT。要启用 CHG FET，请将 [CHG_ON] 寄存器位设置为 1；要禁用，请设置 [CHG_ON] = 0。放电 FET 也可以通过 [DSG_ON] 寄存器位进行类似控制。

某些故障条件或电源状态转换将清除 CHG/DSG FET 控制的状态。表 7-1 显示了在响应各种系统事件时，针对 [CHG_ON] 和 [DSG_ON] 应采取何种操作（如有）：

表表 7-1. CHG、DSG 在各种系统事件下的响应

事件	[CHG_ON]	[DSG_ON]
OV故障	设为0	—
UV故障	—	设为0
OCD故障	—	设为0
SCD故障	—	设为0

表 7-1. CHG, DSG Response Under Various System Events (continued)

EVENT	[CHG_ON]	[DSG_ON]
ALERT Override	Set to 0	Set to 0
DEVICE_XREADY is set	Set to 0	Set to 0
Enter SHIP mode from NORMAL	Set to 0	Set to 0

注

All protection recovery must be initiated by the host microcontroller. In order to resume FET operation after a fault condition has occurred, the host microcontroller must first clear the corresponding status bit in the SYS_STAT register, which will clear the ALERT pin, and then manually re-enable the CHG and/or DSG bit. Certain faults, such as OV or UV, may immediately re-toggle if such a condition still persists. Refer to 表 7-3 for details on clearing status bits.

There are no conditions under which the bq769x0 automatically sets either *[CHG_ON]* or *[DSG_ON]* to 1.

7.3.1.3.2 Load Detection

A load detection circuit is present on the CHG pin and activated whenever the CHG FET is disabled ([CHG_ON] = 0). This circuit detects if the CHG pin is externally pulled high when the high impedance (approximately 1 MΩ) pull-down path should actually be holding the CHG pin to VSS, and is useful for determining if the PACK– pin (outside of the AFE) is being held at a high voltage—for example, if the load is present while the power FETs are off. The state of the load detection circuit is read from the *[LOAD_PRESENT]* bit of the SYS_CTRL1 register.

After an OCD or SCD fault has occurred, the DSG FET will be disabled ([DSG_ON] cleared), and the CHG FET must similarly be explicitly disabled to activate the load detection circuit. The host microcontroller may periodically poll the *[LOAD_PRESENT]* bit to determine the state of the PACK– pin and determine when the load is removed (*[LOAD_PRESENT]* = 0).

7.3.1.3.3 Cell Balancing

Both internal and external passive cell balancing options are fully supported by the bq76920, while external cell balancing is recommended for bq76930 and bq76940. It is left to the host controller to determine the exact balancing algorithm to be used in any given system. Each bq769x0 device provides the cell voltages and balancing drivers to enable this. If using the internal cell balance drivers, up to 50 mA may be balanced per cell. If using external cell balancing, much higher balancing currents may be employed.

To activate a particular cell balancing channel, simply set the corresponding bit for that cell in the CELLBAL1, CELLBAL2, or CELLBAL3 register. For example, VC1–VC0 is enabled by setting *[CB1]*, while VC12–VC11 is set via *[CB12]*.

Multiple cells may be simultaneously balanced. It is left to the user's discretion to determine the ideal number of cells to concurrently balance. Adjacent cells should not be balanced simultaneously. This may cause cell pins to exceed their absolute maximum conditions and is also not recommended for external balancing implementations. Additionally, if internal balancing is used, care should be taken to avoid exceeding package power dissipation ratings.

注

The host controller must ensure that no two adjacent cells are balanced simultaneously within each set of the following:

- VC1–VC5
- VC6–VC10
- VC11–VC15

表7-1. 在各种系统事件下的CHG、DSG响应（续）

事件	[CHG_ON]	[DSG_ON]
ALERT覆盖	设置为 0	设置为 0
DEVICE_XREADY 已设置	设置为 0	设置为 0
从 NORMAL 进入 SHIP 模式	设置为 0	设置为 0

注

所有保护恢复必须由主机微控制器启动。为了在发生故障条件后恢复 FET 操作，主机微控制器必须首先清除 SYS_STAT 寄存器中相应的状态位，这将清除 ALERT 引脚，然后手动重新启用 CHG 和/或 DSG 位。某些故障，如 OV 或 UV，如果该条件仍然存在，可能会立即重新触发。有关清除状态位的详细信息，请参阅表 7-3。

在没有任何条件下，bq769x0 都不会自动将 *[CHG_ON]* 或 *[DSG_ON]* 设置为 1。

7.3.1.3.2 负载检测

CHG 引脚上有一个负载检测电路，当 CHG FET 被禁用（[CHG_ON] = 0）时会被激活。该电路检测 CHG 引脚是否在外部被拉高，而此时高阻抗（约 1 MΩ）下拉路径实际上应该将 CHG 引脚拉至 VSS，这对于确定 PACK– 引脚（AFE 外部）是否被维持在高电压很有用——例如，如果负载存在而电源 FET 关闭时。负载检测电路的状态从 SYS_CTRL1 寄存器的 *[LOAD_PRESENT]* 位读取。

在发生 OCD 或 SCD 故障后，DSG FET 将被禁用（[DSG_ON] 清除），并且 CHG FET 也必须同样被显式禁用以激活负载检测电路。主机微控制器可以定期轮询 *[LOAD_PRESENT]* 位以确定 PACK– 引脚的状态并确定何时移除负载（*[LOAD_PRESENT]* = 0）。

7.3.1.3.3 电池均衡

bq76920 完全支持内部和外部无源电池均衡选项，而 bq76930 和 bq76940 建议使用外部电池均衡。具体系统中使用的确切均衡算法由主机控制器决定。每个 bq769x0 设备提供电池电压和均衡驱动器以启用此功能。如果使用内部电池均衡驱动器，每个电池最多可均衡高达 50 mA。如果使用外部电池均衡，可以采用更高的均衡电流。

要激活特定的电芯平衡通道，只需在 CELLBAL1、CELLBAL2 或 CELLBAL3 寄存器中为该电芯设置相应的位即可。例如，VC1–VC0 通过设置 *[CB1]*，启用，而 VC12–VC11 则通过 *[CB12]*设置。

多个电芯可以同时进行平衡。理想的同时平衡电芯数量由用户自行决定。相邻电芯不应同时进行平衡。这可能导致电芯引脚超出其绝对最大条件，并且也不建议用于外部平衡实现。此外，如果使用内部平衡，应小心避免超过封装功耗散逸额定值。

注

主机控制器必须确保在以下每一组中，没有两个相邻的单元同时被平衡：

- VC1–VC5
- VC6–VC10
- VC11–VC15

The total duty cycle devoted to balancing is approximately 70% per 250 ms. This is because a portion of the 250 ms is allotted for normal cell voltage measurements via the ADC.

If *[ADC_EN]* =1, OV and UV protections are not affected by cell balancing, since the cell balancing is temporarily suspended for a small slice of time every 250 ms during which the cell voltage readings are taken. This ensures that the OV and UV protections do not accidentally trigger, or miss an actual OV/UV condition on the cells while balancing is enabled.

注

All cell balancing control bits in CELLBAL1, CELLBAL2, and CELLBAL3 are automatically cleared under the following events, and must be explicitly re-written by the host microcontroller following clearing of the event:

- DEVICE_XREADY is set
- Enters NORMAL mode from SHIP mode

7.3.1.3.4 Alert

The ALERT pin serves as an active high digital interrupt signal that can be connected to a GPIO port of the host microcontroller. This signal is an OR of all bits in the SYS_STAT register.

In order to clear the ALERT signal, the source bit in the SYS_STAT register must first be cleared by writing a “1” to that bit. This will cause an automatic clear of the ALERT pin once all bits are cleared.

The ALERT pin may also be driven by an external source; for example, the pack may include a secondary overvoltage protector IC. When the ALERT pin is forced high externally while low, the device will recognize this as an OVRD_ALERT fault and set the *[OVRD_ALERT]* bit. This triggers automatic disabling of both CHG and DSG FET drivers. The device cannot recognize the ALERT signal input high when it is already forcing the ALERT signal high from another condition.

The ALERT pin has no internal debounce support so care should be taken to protect the pin from noise or other parasitic transients.

注

It is highly recommended to place an external 500 kΩ–1 MΩ pull-down resistor from ALERT to VSS as close to the IC as possible. Additional recommendations are:

a) To keep all traces between the IC and components connected to the ALERT pin very short.

b) To include a guard ring around the components connected to the ALERT pin and the pin itself.

7.3.1.3.5 Output LDO

An adjustable output voltage regulator LDO is provided as a simple way to provide power to additional components in the battery pack, such as the host microcontroller or LEDs. The LDO is configured in EEPROM by TI during the production test process, and can support 2.5-V or 3.3-V options.

A cascode small-signal FET must be added in the external path between BAT and REGSRC with the bq76930 and bq76940. This helps drop most of the power dissipation outside of the package and cuts down on package power dissipation.

7.3.1.4 Communications Subsystem

The AFE implements a standard 100-kHz I^2C interface and acts as a slave device. The I^2C device address is 7-bits and is factory programmed. Consult the Device Comparison Table (Section 4) of this datasheet for more information.

用于平衡的总占空比约为每250毫秒70%。这是因为250毫秒的一部分时间被分配用于通过ADC进行正常单元电压测量。

如果 *[ADC_EN]* =1，过压和欠压保护不会受到电池平衡的影响，因为在每250毫秒内采集电池电压读数时，电池平衡会暂时暂停一小段时间。这确保了在电池平衡启用时，过压和欠压保护不会意外触发，或者在平衡过程中错过电池的实际过压/欠压条件。

注

在以下事件中，CELLBAL1、CELLBAL2和CELLBAL3中的所有电池平衡控制位都会自动清除，并且必须在事件清除后由主机微控制器显式重写：

- DEVICE_XREADY被设置
- 从SHIP模式进入NORMAL模式

7.3.1.3.4 警报

ALERT引脚用作一个高电平数字中断信号，可以连接到主机微控制器的GPIO端口。该信号是SYS_STAT寄存器中所有位的或运算结果。

要清除ALERT信号，必须先通过向SYS_STAT寄存器中的该位写入“1”来清除源位。一旦所有位都被清除，ALERT引脚将自动清除。

ALERT引脚也可能由外部源驱动；例如，包中可能包含一个二级过压保护IC。当ALERT引脚在外部强制置高而原本为低时，设备会将其识别为OVRD_ALERT故障并设置 *[OVRD_ALERT]* 位。这会触发CHG和DSG FET驱动器的自动禁用。当设备已经从其他条件强制将ALERT信号置高时，它无法识别ALERT引脚输入的高电平。

ALERT引脚没有内部消抖支持，因此应小心保护该引脚免受噪声或其他寄生瞬态的影响。

注

强烈建议从ALERT引脚处放置一个外部500 kΩ–1 MΩ下拉电阻，并将其尽可能靠近IC连接到VSS。其他建议如下：
将IC和连接到ALERT引脚的组件之间的所有走线保持尽可能短。其他建议包括：

a) 保持IC和ALERT引脚连接的组件之间的所有走线非常短。

b) 在连接到ALERT引脚的组件和引脚本身周围设置一个保护环。

7.3.1.3.5 输出LDO

提供一个可调输出电压的LDO，作为向电池组中的其他组件（如主机微控制器或LED）供电的一种简单方式。LDO由TI在生产测试过程中通过EEPROM配置，并支持2.5-V或3.3-V选项。

在BAT和REGSRC之间的外部路径中，必须添加一个级联小信号FET，与bq76930和bq76940配合使用。这有助于将大部分功耗分散到封装外部，从而降低封装功耗。

7.3.1.4 通信子系统

AFE 实现了标准的 100-kHz I^2C 接口，并作为从设备工作。I^2C 设备地址为 7 位，由工厂预编程。请参阅本数据表中的设备比较表（第 4 节）以获取更多信息。

A write transaction is shown in 图 7-4. Block writes are allowed by sending additional data bytes before the Stop. The I²C block will auto-increment the register address after each data byte.

When enabled, the CRC is calculated as follows:

- In a single-byte write transaction, the CRC is calculated over the slave address, register address, and data.
- In a block write transaction, the CRC for the first data byte is calculated over the slave address, register address, and data. The CRC for subsequent data bytes is calculated over the data byte only.

The CRC polynomial is x8 + x2 + x + 1, and the initial value is 0.

When the slave detects a bad CRC, the I²C slave will NACK the CRC, which causes the I²C slave to go to an idle state.

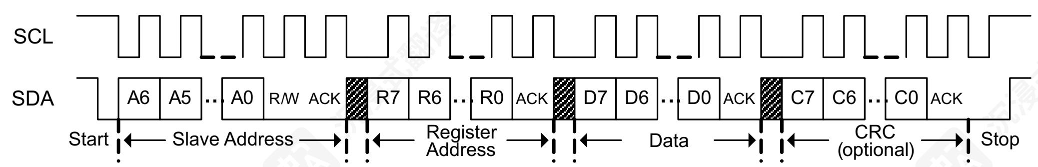


图 7-4. I²C Write

图 7-5 shows a read transaction using a Repeated Start.

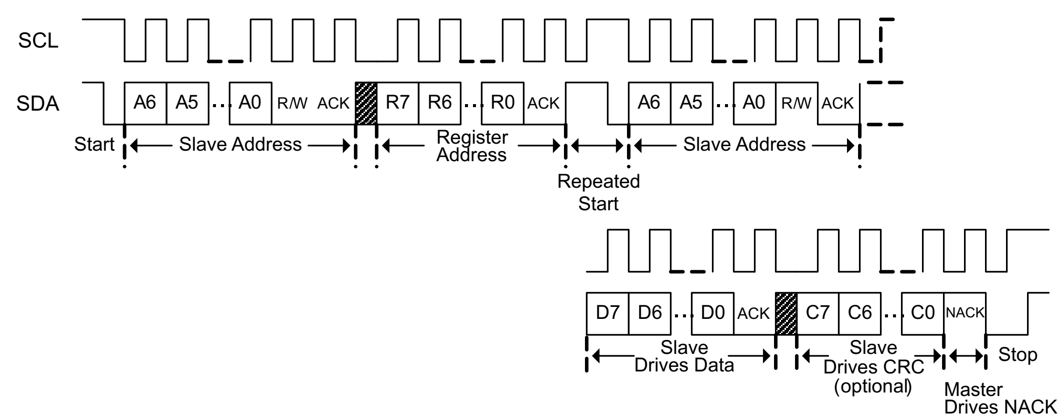


图 7-5. I²C Read with Repeated Start

图 7-6 shows a read transaction where a Repeated Start is not used, for example if not available in hardware. For a block read, the master ACK's each data byte except the last and continues to clock the interface. The I²C block will auto-increment the register address after each data byte.

When enabled, the CRC for a read transaction is calculated as follows:

- In a single-byte read transaction, the CRC is calculated after the second start and uses the slave address and data byte.
- In a block read transaction, the CRC for the first data byte is calculated after the second start and uses the slave address and data byte. The CRC for subsequent data bytes is calculated over the data byte only.

The CRC polynomial is x8 + x2 + x + 1, and the initial value is 0.

When the master detects a bad CRC, the I²C master will NACK the CRC, which causes the I²C slave to go to an idle state.

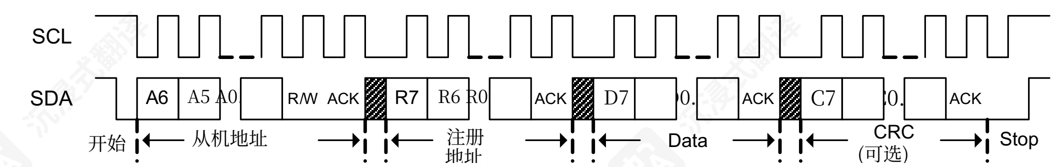
图 7-4 显示了一个写事务。通过在停止信号之前发送额外的数据字节，允许块写入。I²C 块会在每个数据字节后自动递增寄存器地址。

启用时，CRC 计算方法如下:

- 在单字节写事务中，CRC 计算范围覆盖从机地址、寄存器地址和数据。
- 在块写事务中，第一个数据字节的 CRC 计算范围覆盖从机地址、寄存器地址和数据。后续数据字节的 CRC 仅计算数据字节本身。

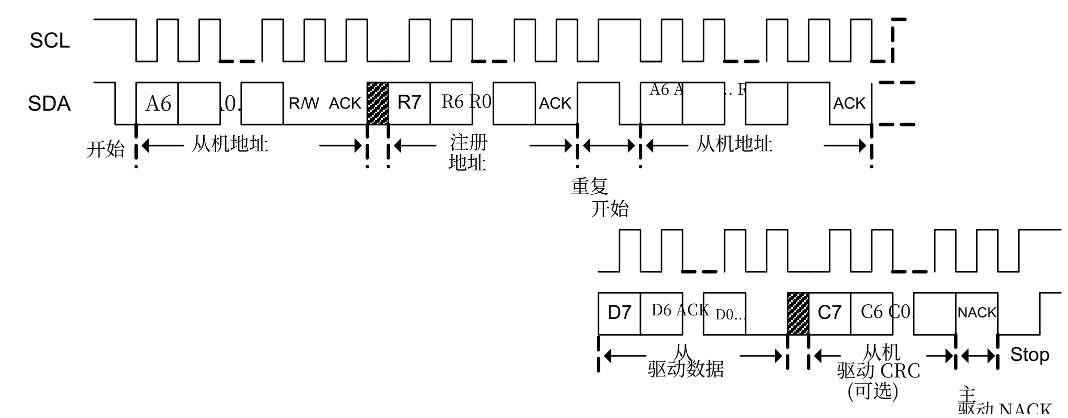
CRC 多项式为 x8 + x2 + x + 1，初始值为 0。

当从机检测到 CRC 错误时，I²C 从机会 NACK CRC，导致 I²C 从机进入空闲状态。



图图 7-4. I²C 写

图 7-5 展示了一个使用重复起始的读交易。



图图 7-5. I²C 带重复起始的读

图 7-6 展示了一个未使用重复起始的读交易，例如在硬件中不可用时。对于块读，主设备 ACK 每个数据字节（最后一个除外），并继续时钟接口。I²C 块会在每个数据字节后自动递增寄存器地址。

启用时，读交易的 CRC 计算如下:

- 在单字节读交易中，CRC 在第二个起始后计算，并使用从机地址和数据字节。
- 在块读取事务中，第一个数据字节的CRC是在第二个起始后计算的，使用从机地址和数据字节。后续数据字节的CRC仅对数据字节进行计算。

CRC多项式是x8 + x2 + x + 1，初始值为0。

当主机检测到CRC校验错误时，I²C主机会NACK CRC，这会导致I²C从机进入空闲状态。

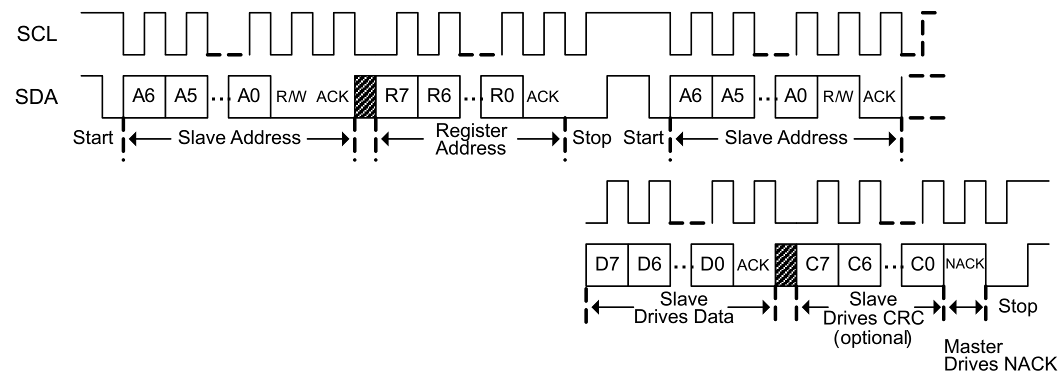


图 7-6. I²C Read Without Repeated Start

7.4 Device Functional Modes

Each bq769x0 device supports the following modes of operation.

表 7-2. Supported Power Modes

Mode	Description
NORMAL	Fully operational state. Both ADC and CC may be on, or disabled by host microcontroller. OV and UV protection enabled if ADC is on. OCD and SCD enabled. ADC and CC may be disabled to reduce power consumption, and CC may be operated in a “1-SHOT” mode for flexible power savings.
SHIP	Lowest possible power state, intended for pack assembly and/or long term pack storage. Must see a BOOT signal (> 1 VBOOT) on TS1 pin to boot from SHIP → NORMAL. Note that the device always enters SHIP mode upon POR.

7.4.1 NORMAL Mode

NORMAL mode represents the fully operational mode where all blocks are enabled and the device sees its highest current consumption. In this mode, certain blocks/functions may be disabled to save power—these include the ADC and CC. OV and UV are running continuously as long as the ADC is enabled. The OCD and SCD comparators may not be disabled in this mode.

Transitioning from NORMAL to SHIP mode is also initiated by the host, and requires consecutive writes to two bits in the SYS_CTRL1 register.

7.4.2 SHIP Mode

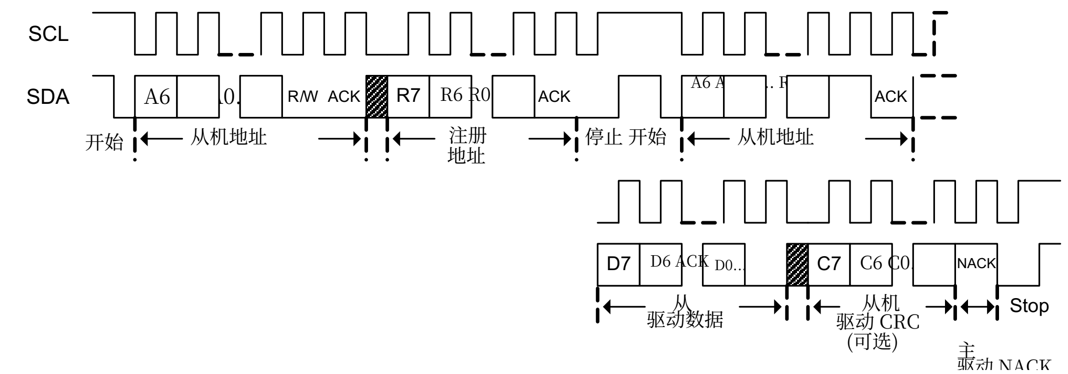
SHIP mode is the basic and lowest power mode that bq769x0 supports. SHIP mode is automatically entered during initial pack assembly and after every POR event. When the device is in NORMAL mode, it may enter SHIP by the host controller via a specific sequence of I²C commands.

In SHIP mode, only a minimum of blocks are turned on, including the VSTUP power supply and primal boot detector. Waking from SHIP mode to NORMAL mode requires pulling the TS1 pin greater than VBOOT, which triggers the device boot-up sequence.

To enter SHIP mode from NORMAL mode, the *[SHUT_A]* and *[SHUT_B]* bits in the SYS_CTRL1 register must be written with specific patterns across two consecutive writes:

- Write #1: *[SHUT_A]* = 0, *[SHUT_B]* = 1
- Write #2: *[SHUT_A]* = 1, *[SHUT_B]* = 0

Note that *[SHUT_A]* and *[SHUT_B]* should each be in a 0 state prior to executing the shutdown command above. If this specific sequence is entered into the device, the device transitions into SHIP mode. If any other sequence is written to the *[SHUT_A]* and *[SHUT_B]* bits or if either of the two patterns is not correctly entered, the device will not enter SHIP mode.



图图 7-6. I²C 无重复启动读取

7.4 设备功能模式

每个bq769x0设备支持以下工作模式。

表表 7-2. 支持的电源模式

Mode	描述
正常状态	完全运行状态。ADC 和 CC 可能都开启，或由主机微控制器禁用。如果 ADC 开启，则启用 OV 和 UV 保护。启用 OCD 和 SCD。ADC 和 CC 可能被禁用以降低功耗，并且 CC 可能以“1-SHOT”模式运行以实现灵活的节能。
SHIP	最低可能的功耗状态，用于电池组装和/或长期电池存储。必须看到 TS1 引脚上的 BOOT 信号（> 1 VBOOT）才能从 SHIP → NORMAL 启动。注意设备在上电复位（POR）后始终进入SHIP模式。

7.4.1 正常模式

正常模式代表设备完全运行的模式，所有模块均启用，此时设备功耗最高。在此模式下，某些模块/功能可能会被禁用以省电——包括ADC和CC。只要ADC启用，OV和UV就会持续运行。在此模式下，OCD和SCD比较器不能被禁用。

从正常模式切换到SHIP模式也由主机发起，需要向SYS_CTRL1寄存器中的两个位连续写入。

7.4.2 SHIP模式

SHIP模式是bq769x0支持的基本且功耗最低的模式。在初始电池组装期间和每次上电复位（POR）事件后，设备会自动进入SHIP模式。当设备处于正常模式时，可以通过主机控制器发送特定的I²C命令序列进入SHIP模式。

在SHIP模式下，仅开启最少数量的模块，包括VSTUP电源和初始启动检测器。从SHIP模式唤醒到NORMAL模式需要将TS1引脚拉高至VBOOT以上，从而触发设备启动序列。

要从NORMAL模式进入SHIP模式，SYS_CTRL1寄存器中的 *[SHUT_A]* 和 *[SHUT_B]* 位必须通过两次连续写入特定模式：

- 写入#1： *[SHUT_A]* = 0, *[SHUT_B]* = 1
- 写入#2： *[SHUT_A]* = 1, *[SHUT_B]* = 0

请注意， *[SHUT_A]* 和 *[SHUT_B]* 在执行上述关机命令之前应各自处于0状态。如果将此特定序列输入设备，设备将切换到SHIP模式。如果向 *[SHUT_A]* 和 *[SHUT_B]* 位写入任何其他序列，或者其中任一模式未正确输入，设备将不会进入SHIP模式。

CAUTION

DO NOT OPERATE THE DEVICE BELOW POR. When designing with the bq76940, the intermediate voltages (BAT–VC10x), (VC10x–VC5x), and (VC5x–VSS) must each never fall below V_{SHUT}. When this occurs, a full device reset must be initiated by powering down all three intermediate voltages (BAT–VC10x), (VC10x–VC5x), and (VC5x–VSS) below V_{SHUT} and rebooting by applying the appropriate VBOOT signal to the TS1 pin. When designing with the bq76930, the intermediate voltages (BAT–VC5x) and (VC5x–VSS) must each never fall below V_{SHUT}. If this occurs, a full device reset must be initiated by powering down both intermediate voltages (BAT–VC5x) and (VC5x–VSS) below V_{SHUT} and rebooting by applying the appropriate VBOOT signal to the TS1 pin.

The device will also enter SHIP mode during a POR event; however, this is not a recommended method of SHIP mode entry. If any of the supply-side voltages below fall below V_{SHUT} and then back up above VPORA, the device defaults into the SHIP mode state. This is similar to an initial pack assembly condition. In order to exit SHIP mode into NORMAL mode, the device must follow the standard boot sequence by applying a voltage greater than the VBOOT threshold on the TS1 pin.

TEXAS INSTRUMENTS
bq76920, bq76930, bq76940
www.ti.com.cn
ZHCSCE2F –2013年10月–修订于2015年11月

注意

禁止在低于 POR 状态下操作设备。在使用 bq76940 进行设计时，中间电压 (BAT–VC10x)、(VC10x–VC5x) 和 (VC5x–VSS) 均不得低于 V_{SHUT}。当出现这种情况时，必须通过将所有三个中间电压 (BAT–VC10x)、(VC10x–VC5x) 和 (VC5x–VSS) 降至 V_{SHUT} 以下来启动完整设备复位，并通过向 TS1 引脚施加适当的 VBOOT 信号来重启。在使用 bq76930 进行设计时，中间电压 (BAT–VC5x) 和 (VC5x–VSS) 均不得低于 V_{SHUT}。如果出现这种情况，必须通过将两个中间电压 (BAT–VC5x) 和 (VC5x–VSS) 降至以下来启动完整设备复位

下方 V 通过向其施加适当的 VBOOT 信号来关闭并重启
TS1 引脚。

设备在POR事件期间也会进入SHIP模式；但是，这不是推荐进入SHIP模式的方法。如果以下任一电源侧电压低于V_{SHUT} 然后回升到VPORA以上，设备将默认进入SHIP模式状态。这类似于初始电池组装条件。为了从SHIP模式退出到NORMAL模式，设备必须通过在TS1引脚上施加高于VBOOT阈值的电压来遵循标准启动序列。

7.5 Register Maps

Name	Addr	D7	D6	D5	D4	D3	D2	D1	D0
SYS_STAT	0x00	CC_READY	RSVD	DEVICE_ XREADY	OVRD_ ALERT	UV	OV	SCD	OCD
CELLBAL1	0x01	RSVD	RSVD	RSVD	CB<5:1>				
CELLBAL2(1)	0x02	RSVD	RSVD	RSVD	CB<10:6>				
CELLBAL3(2)	0x03	RSVD	RSVD	RSVD	CB<15:11>				
SYS_CTRL1	0x04	LOAD_ PRESENT	RSVD	RSVD	ADC_EN	TEMP_SEL	RSVD	SHUT_A	SHUT_B
SYS_CTRL2	0x05	DELAY_DIS	CC_EN	CC_ ONESHOT	RSVD			DSG_ON	CHG_ON
PROTECT1	0x06	RSNS	RSVD	RSVD	SCD_DELAY		SCD_THRESH		
PROTECT2	0x07	RSVD	OCD_DELAY			OCD_THRESH			
PROTECT3	0x08	UV_DELAY		OV_DELAY		RSVD			
OV_TRIP	0x09	OV_THRESH							
UV_TRIP	0x0A	UV_THRESH							
CC_CFG	0x0B	RSVD	RSVD	Must be programmed to 0x19					
VC1_HI	0x0C	RSVD	RSVD	<13:8>					
VC1_LO	0x0D	<7:0>							
VC2_HI	0x0E	RSVD	RSVD	<13:8>					
VC2_LO	0x0F	<7:0>							
VC3_HI	0x10	RSVD	RSVD	<13:8>					
VC3_LO	0x11	<7:0>							
VC4_HI	0x12	RSVD	RSVD	<13:8>					
VC4_LO	0x13	<7:0>							
VC5_HI	0x14	RSVD	RSVD	<13:8>					
VC5_LO	0x15	<7:0>							
VC6_HI(1)	0x16	RSVD	RSVD	<13:8>					
VC6_LO(1)	0x17	<7:0>							
VC7_HI(1)	0x18	RSVD	RSVD	<13:8>					
VC7_LO(1)	0x19	<7:0>							
VC8_HI(1)	0x1A	RSVD	RSVD	<13:8>					
VC8_LO(1)	0x1B	<7:0>							
VC9_HI(1)	0x1C	RSVD	RSVD	<13:8>					
VC9_LO(1)	0x1D	<7:0>							
VC10_HI(1)	0x1E	RSVD	RSVD	<13:8>					
VC10_LO(1)	0x1F	<7:0>							
VC11_HI(2)	0x20	RSVD	RSVD	<13:8>					
VC11_LO(2)	0x21	<7:0>							
VC12_HI(2)	0x22	RSVD	RSVD	<13:8>					
VC12_LO(2)	0x23	<7:0>							
VC13_HI(2)	0x24	RSVD	RSVD	<13:8>					
VC13_LO(2)	0x25	<7:0>							
VC14_HI(2)	0x26	RSVD	RSVD	<13:8>					
VC14_LO(2)	0x27	<7:0>							
VC15_HI(2)	0x28	RSVD	RSVD	<13:8>					
VC15_LO(2)	0x29	<7:0>							
BAT_HI	0x2A	<15:8>							
BAT_LO	0x2B	<7:0>							
TS1_HI	0x2C	RSVD	RSVD	<13:8>					
TS1_LO	0x2D	<7:0>							
TS2_HI(1)	0x2E	RSVD	RSVD	<13:8>					
TS2_LO(1)	0x2F	<7:0>							

(1) These registers are only valid for bq76930 and bq76940.
(2) These registers are only valid for bq76940.

7.5 寄存器映射

Name	Addr	D7	D6	D5	D4	D3	D2	D1	D0
SYS_STAT	0x00	CC_READY	RSVD	DEVICE_ XREADY	OVRD_ 警告	UV	OV	SCD	OCD
单元平衡1	0x01	RSVD	RSVD	RSVD	CB<5:1>				
单元平衡2(1)	0x02	RSVD	RSVD	RSVD	CB<10:6>				
单元平衡3(2)	0x03	RSVD	RSVD	RSVD	CB<15:11>				
SYS_CTRL1	0x04	加载当前	RSVD	RSVD	ADC_EN	TEMP_SEL	RSVD	SHUT_A	SHUT_B
SYS_CTRL2	0x05	DELAY_DIS	CC_EN	CC_ 单次触发	RSVD			DSG_ON	CHG_ON
PROTECT1	0x06	RSNS	RSVD	RSVD	SCD_DELAY		SCD_THRESH		
PROTECT2	0x07	RSVD	OCD_DELAY			OCD_THRESH			
PROTECT3	0x08	UV_DELAY		OV_DELAY		RSVD			
OV_TRIP	0x09	OV_THRESH							
UV_TRIP	0x0A	UV_THRESH							
CC_CFG	0x0B	RSVD	RSVD	必须编程为0x19					
VC1_HI	0x0C	RSVD	RSVD	<13:8>					
VC1_LO	0x0D	<7:0>							
VC2_HI	0x0E	RSVD	RSVD	<13:8>					
VC2_LO	0x0F	<7:0>							
VC3_HI	0x10	RSVD	RSVD	<13:8>					
VC3_LO	0x11	<7:0>							
VC4_HI	0x12	RSVD	RSVD	<13:8>					
VC4_LO	0x13	<7:0>							
VC5_HI	0x14	RSVD	RSVD	<13:8>					
VC5_LO	0x15	<7:0>							
VC6_HI(1)	0x16	RSVD	RSVD	<13:8>					
VC6_LO(1)	0x17	<7:0>							
VC7_HI(1)	0x18	RSVD	RSVD	<13:8>					
VC7_LO(1)	0x19	<7:0>							
VC8_HI(1)	0x1A	RSVD	RSVD	<13:8>					
VC8_LO(1)	0x1B	<7:0>							
VC9_HI(1)	0x1C	RSVD	RSVD	<13:8>					
VC9_LO(1)	0x1D	<7:0>							
VC10_HI(1)	0x1E	RSVD	RSVD	<13:8>					
VC10_LO(1)	0x1F	<7:0>							
VC11_HI(2)	0x20	RSVD	RSVD	<13:8>					
VC11_LO(2)	0x21	<7:0>							
VC12_HI(2)	0x22	RSVD	RSVD	<13:8>					
VC12_LO(2)	0x23	<7:0>							
VC13_HI(2)	0x24	RSVD	RSVD	<13:8>					
VC13_LO(2)	0x25	<7:0>							
VC14_HI(2)	0x26	RSVD	RSVD	<13:8>					
VC14_LO(2)	0x27	<7:0>							
VC15_HI(2)	0x28	RSVD	RSVD	<13:8>					
VC15_LO(2)	0x29	<7:0>							
BAT_HI	0x2A	<15:8>							
BAT_LO	0x2B	<7:0>							
TS1_HI	0x2C	RSVD	RSVD	<13:8>					
TS1_LO	0x2D	<7:0>							
TS2_HI(1)	0x2E	RSVD	RSVD	<13:8>					
TS2_LO(1)	0x2F	<7:0>							

(1) 这些寄存器仅适用于 bq76930 和 bq76940。 (2) 这些寄存器仅适用于 bq76940。

Name	Addr	D7	D6	D5	D4	D3	D2	D1	D0
TS3_HI [(2)]	0x30	RSVD	RSVD	<13:8>					
TS3_LO [(2)]	0x31	<7:0>							
CC_HI	0x32	<15:8>							
CC_LO	0x33	<7:0>							
ADCGAIN1	0x50	RSVD				ADCGAIN<4:3>		RSVD	
ADCOFFSET	0x51	ADCOFFSET<7:0>							
ADCGAIN2	0x59	ADCGAIN<2:0>			RSVD				

7.5.1 Register Details

表 7-3. SYS_STAT (0x00)

BIT	7	6	5	4	3	2	1	0
NAME	CC_READY	RSVD	DEVICE_ XREADY	OVRD_ ALERT	UV	OV	SCD	OCD
RESET	0	0	0	0	0	0	0	0
ACCESS	RW	RW	RW	RW	RW	RW	RW	RW

注

Bits in SYS_STAT may be cleared by writing a "1" to the corresponding bit.

Writing a "0" does not change the state of the corresponding bit.

CC_READY (Bit 7): Indicates that a fresh coulomb counter reading is available. Note that if this bit is not cleared between two adjacent CC readings becoming available, the bit remains latched to 1. This bit may only be cleared (and not set) by the host.

0 = Fresh CC reading not yet available or bit is cleared by host microcontroller.

1 = Fresh CC reading is available. Remains latched high until cleared by host.

RSVD (Bit 6): Reserved. Do not use.

DEVICE_XREADY (Bit 5): Internal chip fault indicator. When this bit is set to 1, it should be cleared by the host. May be set due to excessive system transients. This bit may only be cleared (and not set) by the host.

0 = Device is OK.

1 = Internal chip fault detected, recommend that host microcontroller clear this bit after waiting a few seconds. Remains latched high until cleared by the host.

OVRD_ALERT (Bit 4): External pull-up on the ALERT pin indicator. Only active when ALERT pin is not already being driven high by the AFE itself.

0 = No external override detected

1 = External override detected. Remains latched high until cleared by the host.

UV (Bit 3): Undervoltage fault event indicator.

0 = No UV fault is detected.

1 = UV fault is detected. Remains latched high until cleared by the host.

OV (Bit 2): Overvoltage fault event indicator.

0 = No OV fault is detected.

1 = OV fault is detected. Remains latched high until cleared by the host.

Name	Addr	D7	D6	D5	D4	D3	D2	D1	D0
TS3_HI [(2)]	0x30	RSVD	RSVD	<13:8>					
TS3_LO [(2)]	0x31	<7:0>							
CC_HI	0x32	<15:8>							
CC_LO	0x33	<7:0>							
ADCGAIN1	0x50	RSVD				ADCGAIN<4:3>		RSVD	
ADCOFFSET	0x51	ADCOFFSET<7:0>							
ADCGAIN2	0x59	ADCGAIN<2:0>			RSVD				

7.5.1 寄存器详情

表表 7-3. SYS_STAT (0x00)

BIT	7	6	5	4	3	2	1	0
NAME	CC_READY	RSVD	DEVICE_ XREADY	OVRD_ ALERT	UV	OV	SCD	OCD
RESET	0	0	0	0	0	0	0	0
ACCESS	RW	RW	RW	RW	RW	RW	RW	RW

注

SYS_STAT 中的位可以通过向对应位写入 "1" 来清除。

写入"0"不会改变相应位的状态。

CC_READY (位7)：指示有新的库仑计数器读数可用。注意，如果在这两个相邻的CC读数可用之间未清除该位，该位将保持锁定为1。该位只能由主机清除（不能设置）。

0 = 新鲜CC读取尚未可用或由主机微控制器清除该位。

1 = 新鲜CC读取可用。保持高电平锁定，直到由主机清除。

RSVD (位6)：保留。请勿使用。

DEVICE_XREADY (位5)：内部芯片故障指示器。当此位设置为1时，应由主机清除。可能由于系统瞬态过大而设置。此位只能由主机清除（不能设置）。

0 = 设备正常。

1 = 检测到内部芯片故障，建议主机微控制器在等待一段时间后清除此位。几秒钟。保持高电平状态，直到由主机清除。

OVRD_ALERT (位4)：ALERT引脚的外部上拉指示。仅在ALERT引脚未被AFE本身驱动为高电平时才激活。

0 = 未检测到外部覆盖

1 = 检测到外部覆盖。保持高电平状态，直到由主机清除。

UV (位3)：欠压故障事件指示器。

0 = 未检测到UV故障。

1 = 检测到UV故障。将保持锁定高电平，直到由主机清除。

OV (位2)：过压故障事件指示器。

0 = 未检测到过压故障。

1 = 检测到过压故障。保持高电平状态，直到由主机清除。

SCD (Bit 1): Short circuit in discharge fault event indicator.

0 = No SCD fault is detected.

1 = SCD fault is detected. Remains latched high until cleared by the host.

OCD (Bit 0): Over current in discharge fault event indicator.

0 = No OCD fault is detected.

1 = OCD fault is detected. Remains latched high until cleared by the host.

表 7-4. CELLBAL1 (0x01) for bq76920, bq76930, and bq76940

BIT	7	6	5	4	3	2	1	0
NAME	—	—	—	CB5	CB4	CB3	CB2	CB1
RESET	0	0	0	0	0	0	0	0
ACCESS	R	R	R	RW	RW	RW	RW	RW

CBx (Bits 4–0):

0 = Cell balancing on Cell “x” is disabled.

1 = Cell balancing on Cell “x” is enabled.

表 7-5. CELLBAL2 (0x02) for bq76930 and bq76940

BIT	7	6	5	4	3	2	1	0
NAME	—	—	—	CB10	CB9	CB8	CB7	CB6
RESET	0	0	0	0	0	0	0	0
ACCESS	R	R	R	RW	RW	RW	RW	RW

CBx (Bits 4–0):

0 = Cell balancing on Cell “x” is disabled.

1 = Cell balancing on Cell “x” is enabled.

表 7-6. CELLBAL3 (0x03) for bq76940

BIT	7	6	5	4	3	2	1	0
NAME	—	—	—	CB15	CB14	CB13	CB12	CB11
RESET	0	0	0	0	0	0	0	0
ACCESS	R	R	R	RW	RW	RW	RW	RW

CBx (Bits 4–0):

0 = Cell balancing on Cell “x” is disabled.

1 = Cell balancing on Cell “x” is enabled.

表 7-7. SYS_CTRL1 (0x04)

BIT	7	6	5	4	3	2	1	0
NAME	LOAD_ PRESENT	—	—	ADC_EN	TEMP_SEL	RSVD	SHUT_A	SHUT_B
RESET	0	0	0	0	0	0	0	0
ACCESS	R	R	R	RW	RW	RW	RW	RW

SCD (位1)：放电故障事件短路指示器

0 = 未检测到SCD故障。

1 = 检测到SCD故障。主机清除前将保持高电平锁定状态。

OCD（位0）：放电故障事件指示器，指示过流。

0 = 未检测到OCD故障。

1 = 检测到OCD故障。主机清除前将保持高电平锁定状态。

表表 7-4. CELLBAL1 (0x01) for bq76920, bq76930, and bq76940

BIT	7	6	5	4	3	2	1	0
NAME	—	—	—	CB5	CB4	CB3	CB2	CB1
复位	0	0	0	0	0	0	0	0
访问	R	R	R	RW	RW	RW	RW	RW

CBx (位 4–0)：

0 = 电池“x”的均衡功能已禁用。

1 = 电池“x”的均衡功能已启用。

表表 7-5. CELLBAL2 (0x02) for bq76930 and bq76940

BIT	7	6	5	4	3	2	1	0
NAME	—	—	—	CB10	CB9	CB8	CB7	CB6
复位	0	0	0	0	0	0	0	0
访问	R	R	R	RW	RW	RW	RW	RW

CBx (位 4–0)：

0 = 电池“x”的均衡功能已禁用。

1 = 电池“x”的均衡功能已启用。

表表 7-6. CELLBAL3 (0x03) for bq76940

BIT	7	6	5	4	3	2	1	0
NAME	—	—	—	CB15	CB14	CB13	CB12	CB11
复位	0	0	0	0	0	0	0	0
访问	R	R	R	RW	RW	RW	RW	RW

CBx (位4–0)：

0 = “x”单元的均衡功能已禁用。

1 = “x”单元的均衡功能已启用。

表表 7-7. SYS_CTRL1 (0x04)

BIT	7	6	5	4	3	2	1	0
NAME	加载_当前	—	—	ADC启用	温度选择	RSVD	关闭A	关闭B
重置	0	0	0	0	0	0	0	0
访问	R	R	R	RW	RW	RW	RW	RW

LOAD_PRESENT (Bit 7): Valid only when [CHG_ON] = 0. Is high if CHG pin is detected to exceed VLOAD_DETECT while CHG_ON = 0, which suggests that external load is present. Note this bit is read-only and automatically clears when load is removed.

0 = CHG pin < VLOAD_DETECT or [CHG_ON] = 1.

1 = CHG pin >VLOAD_DETECT, while [CHG_ON] = 0.

ADC_EN (Bit 4): ADC enable command

0 = Disable voltage and temperature ADC readings (also disables OV protection)

1 = Enable voltage and temperature ADC readings (also enables OV protection)

TEMP_SEL (Bit 3): TSx_HI and TSx_LO temperature source

0 = Store internal die temperature voltage reading in TSx_HI and TSx_LO

1 = Store thermistor reading in TSx_HI and TSx_LO (all thermistor ports)

RSVD (Bit 2): Reserved, do not set to 1.

SHUT_A, SHUT_B (Bits 1–0): Shutdown command from host microcontroller. Must be written in a specific sequence, shown below:

Starting from: *[SHUT_A]* = 0, *[SHUT_B]* = 0

Write #1: *[SHUT_A]* = 0, *[SHUT_B]* = 1

Write #2: *[SHUT_A]* = 1, *[SHUT_B]* = 0

Other writes cause the command to be ignored.

表 7-8. SYS_CTRL2 (0x05)

BIT	7	6	5	4	3	2	1	0
NAME	DELAY_DIS	CC_EN	CC_ ONESHOT	RSVD	RSVD	RSVD	DSG_ON	CHG_ON
RESET	0	0	0	0	0	0	0	0
ACCESS	RW	RW	RW	RW	RW	RW	RW	RW

DELAY_DIS (Bit 7): Disable OV, UV, OCD, and SCD delays for faster customer production testing.

0 = Normal delay settings

1 = OV, UV, OCD, and SCD delay circuit is bypassed, creating zero delay (approximately 250 ms).

CC_EN (Bit 6): Coulomb counter continuous operation enable command. If set high, *[CC_ONESHOT]* bit is ignored.

0 = Disable CC continuous readings

1 = Enable CC continuous readings and ignore *[CC_ONESHOT]* state

CC_ONESHOT (Bit 5): Coulomb counter single 250-ms reading trigger command. If set to 1, the coulomb counter will be activated for a single 250-ms reading, and then turned back off. *[CC_ONESHOT]* will also be cleared at the conclusion of this reading, while *[CC_READY]* bit will be set to 1.

0 = No action

1 = Enable single CC reading (only valid if *[CC_EN]* = 0), and *[CC_READY]* = 0)

RSVD (Bit 4–2): Reserved. Do not use.

LOAD_PRESENT (位7)：仅在 [CHG_ON] = 0 时有效。若 CHG 引脚在 CHG_ON = 0 时检测到电压超过 VLOAD_DETECT，该位将置高，表明外部负载存在。注意此位为只读位，当负载移除时会自动清除。

0 = CHG 引脚 < VLOAD_DETECT 或 [CHG_ON] = 1.

1 = CHG 引脚 >VLOAD_DETECT，同时 [CHG_ON] = 0.

ADC_EN (位 4): ADC 使能命令

0 = 禁用电压和温度 ADC 读取（同时禁用 OV 保护）

1 = 启用电压和温度 ADC 读取（同时启用 OV 保护）

TEMP_SEL (位 3)：TSx_HI 和 TSx_LO 温度源

0 = 将内部芯片温度电压读数存储在 TSx_HI 和 TSx_LO

1 = 将热敏电阻读数存储在 TSx_HI 和 TSx_LO（所有热敏电阻端口）

RSVD (位 2)：保留，不要设置为 1。

SHUT_A、SHUT_B (位 1–0)：来自主机微控制器的关断命令。必须按特定顺序写入，如下所示：

从：*[SHUT_A]* = 0、*[SHUT_B]* = 0

Write #1: *[SHUT_A]* = 0, *[SHUT_B]* = 1

Write #2: *[SHUT_A]* = 1, *[SHUT_B]* = 0

其他写入会导致命令被忽略。

表表 7-8. SYS_CTRL2 (0x05)

BIT	7	6	5	4	3	2	1	0
NAME	DELAY_DIS	CC_EN	CC_ ONESHOT	RSVD	RSVD	RSVD	DSG_ON	CHG_ON
RESET	0	0	0	0	0	0	0	0
ACCESS	RW	RW	RW	RW	RW	RW	RW	RW

DELAY_DIS (位7)：禁用OV、UV、OCD和SCD延迟，以便更快地进行客户生产测试。

0 = 正常延迟设置

1 = OV、UV、OCD和SCD延迟电路被旁路，产生零延迟（约250 ms）。

CC_EN (位6): 库仑计连续运行使能命令。如果设为高电平，*[CC_ONESHOT]* 位将被忽略。

0 = 禁用库仑计连续读取

1 = 启用库仑计连续读取并忽略 *[CC_ONESHOT]* 状态

CC_ONESHOT (位5): 库仑计单次250毫秒读取触发命令。如果设为1，库仑计将被激活进行单次250毫秒读取，然后关闭。
[CC_ONESHOT] 将在此次读取结束时也被清除，同时 *[CC_READY]* 位将被设为1。

0 = 无操作

1 = 启用单次CC读取（仅当 *[CC_EN]* = 0且 *[CC_READY]* = 0时有效）

RSVD (位4–2)：保留。请勿使用。

DSG_ON (Bit 1): Discharge FET driver (low side NCH) or discharge signal control

0 = DSG is off.

1 = DSG is on.

CHG_ON (Bit 0): Discharge FET driver (low side NCH) or discharge signal control

0 = CHG is off.

1 = CHG is on.

表 7-9. PROTECT1 (0x06)

BIT	7	6	5	4	3	2	1	0
NAME	RSNS	—	RSVD	SCD_D1	SCD_D0	SCD_T2	SCD_T1	SCD_T0
RESET	0	0	0	0	0	0	0	0
ACCESS	RW	R	RW	RW	RW	RW	RW	RW

RSNS (Bit 7): Allows for doubling the OCD and SCD thresholds simultaneously

0 = OCD and SCD thresholds at lower input range

1 = OCD and SCD thresholds at upper input range

RSVD (Bit 5): Reserved, do not set to 1.

SCD_D1:0 (Bits 4–3):
Short circuit in discharge delay setting. A 400-µs setting is recommended only in systems using maximum cell measurement input resistance, Rc, of 1 kΩ (which corresponds to minimum internal cell balancing current or external cell balancing configuration).

Code	(in µs)
0x0	70
0x1	100
0x2	200
0x3	400

SCD_T2:0 (Bits 2–0): Short circuit in discharge threshold setting

Code	RSNS = 1 (in mV)	RSNS = 0 (in mV)
0x0	44	22
0x1	67	33
0x2	89	44
0x3	111	56
0x4	133	67
0x5	155	78
0x6	178	89
0x7	200	100

表 7-10. PROTECT2 (0x07)

BIT	7	6	5	4	3	2	1	0
NAME	—	OCD_D2	OCD_D1	OCD_D0	OCD_T3	OCD_T2	OCD_T1	OCD_T0
RESET	0	0	0	0	0	0	0	0
ACCESS	R	RW	RW	RW	RW	RW	RW	RW

DSG_ON (位1)：放电FET驱动器（低端NCH）或放电信号控制

0 = DSG已关闭。

1 = DSG已开启。

CHG_ON (位0)：放电FET驱动器（低端NCH）或放电信号控制

0 = CHG已关闭。

1 = CHG已开启。

表表 7-9. PROTECT1 (0x06)

BIT	7	6	5	4	3	2	1	0
NAME	RSNS	—	RSVD	SCD_D1	SCD_D0	SCD_T2	SCD_T1	SCD_T0
重置	0	0	0	0	0	0	0	0
访问	RW	R	RW	RW	RW	RW	RW	RW

RSNS (位7)：允许同时将OCD和SCD阈值提高一倍

0 = 在输入范围较低时OCD和SCD阈值

1 = 在输入范围较高时OCD和SCD阈值

RSVD (位5)：保留，不要设置为1

SCD_D1:0 (位4–3):
放电延迟设置中的短路。建议仅在系统使用最大电池测量输入电阻Rc为1 kΩ （这对应于最小内部电池均衡电流或外部电池均衡配置）时设置400-µs。

Code	(以µs为单位)
0x0	70
0x1	100
0x2	200
0x3	400

SCD_T2:0 (位2–0): 放电阈值设置中的短路

Code	RSNS = 1 (以mV为单位)	RSNS = 0 (以mV为单位)
0x0	44	22
0x1	67	33
0x2	89	44
0x3	111	56
0x4	133	67
0x5	155	78
0x6	178	89
0x7	200	100

表表 7-10. PROTECT2 (0x07)

BIT	7	6	5	4	3	2	1	0
NAME	—	OCD_D2	OCD_D1	OCD_D0	OCD_T3	OCD_T2	OCD_T1	OCD_T0
RESET	0	0	0	0	0	0	0	0
ACCESS	R	RW	RW	RW	RW	RW	RW	RW

OCD_D2:0 (Bits 6–4): Overcurrent in discharge delay setting

Code	(in ms)
0x0	8
0x1	20
0x2	40
0x3	80
0x4	160
0x5	320
0x6	640
0x7	1280

OCD_T3:0 (Bits 3–0): Overcurrent in discharge threshold setting.

Code	RSNS = 1 (in mV)	(RSNS = 0 (in mV)
0x0	17	8
0x1	22	11
0x2	28	14
0x3	33	17
0x4	39	19
0x5	44	22
0x6	50	25
0x7	56	28
0x8	61	31
0x9	67	33
0xA	72	36
0xB	78	39
0xC	83	42
0xD	89	44
0xE	94	47
0xF	100	50

表 7-11. PROTECT3 (0x08)

BIT	7	6	5	4	3	2	1	0
NAME	UV_D1	UV_D0	OV_D1	OV_D0	RSVD	RSVD	RSVD	RSVD
RESET	0	0	0	0	0	0	0	0
ACCESS	RW	RW	RW	RW	RW	RW	RW	RW

UV_D1:0 (Bits 7–6): Undervoltage delay setting

Code	(in s)
0x0	1
0x1	4
0x2	8
0x3	16

OCD_D2:0 (位 6–4): 放电延迟设置中的过流

Code	(毫秒)
0x0	8
0x1	20
0x2	40
0x3	80
0x4	160
0x5	320
0x6	640
0x7	1280

OCD_T3:0 (位 3–0): 放电阈值设置中的过流。

Code	RSNS = 1 (毫伏)	(RSNS = 0 (毫伏)
0x0	17	8
0x1	22	11
0x2	28	14
0x3	33	17
0x4	39	19
0x5	44	22
0x6	50	25
0x7	56	28
0x8	61	31
0x9	67	33
0xA	72	36
0xB	78	39
0xC	83	42
0xD	89	44
0xE	94	47
0xF	100	50

表表 7-11. PROTECT3 (0x08)

BIT	7	6	5	4	3	2	1	0
NAME	UV_D1	UV_D0	OV_D1	OV_D0	RSVD	RSVD	RSVD	RSVD
RESET	0	0	0	0	0	0	0	0
访问	RW	RW	RW	RW	RW	RW	RW	RW

UV_D1:0 (位 7–6): 低压延迟设置

Code	(秒)
0x0	1
0x1	4
0x2	8
0x3	16

OV_D1:0 (Bits 5–4): Overvoltage delay setting

Code	(in s)
0x0	1
0x1	2
0x2	4
0x3	8

RSVD (Bits 3–0): These bits are for TI internal debug use only and must be configured to the default settings indicated.

表 7-12. OV_TRIP (0x09)

BIT	7	6	5	4	3	2	1	0
NAME	OV_T7	OV_T6	OV_T5	OV_T4	OV_T3	OV_T2	OV_T1	OV_T0
RESET	1	0	1	0	1	1	0	0
ACCESS	RW	RW	RW	RW	RW	RW	RW	RW

OV_T7:0 (Bits 7–0): Middle 8 bits of the direct ADC mapping of the desired OV protection threshold, with upper 2 MSB set to 10 and lower 2 LSB set to 1000. The equivalent OV threshold is mapped to: 10-OV_T<7:0>1000.

By default, OV_TRIP is configured to a 0xAC setting.

Note that OV_TRIP is based on the ADC voltage, which requires back-calculation using the GAIN and OFFSET values stored in ADCGAIN<4:0>and ADCOFFSET<7:0>.

表 7-13. UV_TRIP (0x0A)

BIT	7	6	5	4	3	2	1	0
NAME	UV_T7	UV_T6	UV_T5	UV_T4	UV_T3	UV_T2	UV_T1	UV_T0
RESET	1	0	0	1	0	1	1	1
ACCESS	RW	RW	RW	RW	RW	RW	RW	RW

UV_T7:0 (Bits 7–0): Middle 8 bits of the direct ADC mapping of the desired UV protection threshold, with upper 2 MSB set to 01 and lower 4 LSB set to 0000. In other words, the equivalent OV threshold is mapped to: 01-UV_T<7:0>–0000.

By default, UV_TRIP is configured to a 0x97 setting. .

Note that UV_TRIP is based on the ADC voltage, which requires back-calculation using the GAIN and OFFSET values stored in ADCGAIN<4:0>and ADCOFFSET<7:0>.

表 7-14. CC_CFG REGISTER (0x0B)

BIT	7	6	5	4	3	2	1	0
NAME	—	—	CC_CFG5	CC_CFG4	CC_CFG3	CC_CFG2	CC_CFG1	CC_CFG0
RESET	0	0	0	0	0	0	0	0
ACCESS	R	R	RW	RW	RW	RW	RW	RW

OV_D1:0 (位 5–4): 过压延迟设置

Code	(秒)
0x0	1
0x1	2
0x2	4
0x3	8

RSVD (位 3–0): 这些位仅用于德州仪器内部调试，必须配置为指示的默认设置。

表表 7-12. OV_TRIP (0x09)

BIT	7	6	5	4	3	2	1	0
NAME	OV_T7	OV_T6	OV_T5	OV_T4	OV_T3	OV_T2	OV_T1	OV_T0
重置	1	0	1	0	1	1	0	0
访问	RW	RW	RW	RW	RW	RW	RW	RW

OV_T7:0 (位 7–0): 目标 OV 保护阈值直接 ADC 映射的中 8 位，其中高 2 位 MSB 设置为 10，低 2 位 LSB 设置为 1000。等效的 OV 阈值映射为：10-OV_T<7:0>1000。

默认情况下，OV_TRIP 配置为 0xAC 设置。

请注意，OV_TRIP 基于 ADC 电压，需要使用存储在 ADCGAIN<4:0> 和 ADCOFFSET<7:0> 中的 GAIN 和 OFFSET 值进行反向计算。

表7-13. UV_TRIP (0x0A)

BIT	7	6	5	4	3	2	1	0
NAME	UV_T7	UV_T6	UV_T5	UV_T4	UV_T3	UV_T2	UV_T1	UV_T0
RESET	1	0	0	1	0	1	1	1
ACCESS	RW	RW	RW	RW	RW	RW	RW	RW

UV_T7:0 (位7–0): 中间8位是所需UV保护阈值的直接ADC映射，其中高2位MSB设为01，低4位LSB设为0000。换句话说，等效的OV阈值映射为：01-UV_T<7:0>-0000。

默认情况下，UV_TRIP 配置为 0x97 设置。

请注意，UV_TRIP 基于ADC电压，需要使用存储在ADCGAIN<4:0>和ADCOFFSET<7:0>中的GAIN和OFFSET值进行反向计算。

表表 7-14. CC_CFG 寄存器 (0x0B)

BIT	7	6	5	4	3	2	1	0
NAME	—	—	CC_CFG5	CC_CFG4	CC_CFG3	CC_CFG2	CC_CFG1	CC_CFG0
重置	0	0	0	0	0	0	0	0
访问	R	R	RW	RW	RW	RW	RW	RW

CC_CFG5:0 (Bits 5–0): For optimal performance, these bits should be programmed to 0x19 upon device startup.

7.5.2 *Read-Only Registers*

表 7-15. CELL VOLTAGE REGISTERS

VC1_HI, _LO (0x0C–0x0D), VC2_HI, _LO (0x0E–0x0F), VC3_HI, _LO (0x10–0x11), VC4_HI, _LO (0x12–0x13), VC5_HI, _LO (0x14–0x15) / bq76930, bq76940: VC6_HI, _LO (0x16–0x17), VC7_HI, _LO (0x18–0x19), VC8_HI, _LO (0x1A–0x1B), VC9_HI, _LO (0x1C–0x1D), VC10_HI, _LO (0x1E–0x1F) / bq76940: VC11_HI, _LO (0x20–0x21), VC12_HI, _LO (0x22–0x23), VC13_HI, _LO (0x24–0x25), VC14_HI, _LO (0x26–0x27), VC15_HI, _LO (0x28–0x29)								
BIT	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
NAME	—	—	D13	D12	D11	D10	D9	D8
RESET	0	0	0	0	0	0	0	0
NAME	D7	D6	D5	D4	D3	D2	D1	D0
RESET	0	0	0	0	0	0	0	0

D11:8 (Bits 3–0): Cell "x" ADC reading, upper 6 MSB. Always returned as an atomic value if both high and low registers are read in the same transaction (using address auto-increment).

D7:0 (Bits 7–0): Cell "x" ADC reading, lower 8 LSB.

表 7-16. BAT_HI (0x2A) and BAT_LO (0x2B)

BIT	7	6	5	4	3	2	1	0
NAME	D15	D14	D13	D12	D11	D10	D9	D8
RESET	0	0	0	0	0	0	0	0
NAME	D7	D6	D5	D4	D3	D2	D1	D0
RESET	0	0	0	0	0	0	0	0

D15:8 (Bits 7–0): BAT calculation based on adding up Cells 1–15, upper 8 MSB. Always returned as an atomic value if both high and low registers are read in the same transaction (using address auto-increment).

D7:0 (Bits 7–0): BAT calculation based on adding up Cells 1–15, lower 8 LSB

表 7-17. TS1_HI (0x2C) and TS1_LO (0x2D)

BIT	7	6	5	4	3	2	1	0
NAME	—	—	D13	D12	D11	D10	D9	D8
RESET	0	0	0	0	0	0	0	0
NAME	D7	D6	D5	D4	D3	D2	D1	D0
RESET	0	0	0	0	0	0	0	0

D11:8 (Bits 3–0): TS1 or DIETEMP ADC reading, upper 6 MSB. Always returned as an atomic value if both high and low registers are read in the same transaction (using address auto-increment).

D7:0 (Bits 7–0): TS1 or DIETEMP ADC reading, lower 8 LSB

CC_CFG5:0 (位5–0)：为了获得最佳性能，这些位应在设备启动时编程为0x19。

7.5.2 只读寄存器

表 7-15. 电池电压寄存器

VC1_HI, _LO (0x0C–0x0D), VC2_HI, _LO (0x0E–0x0F), VC3_HI, _LO (0x10–0x11), VC4_HI, _LO (0x12–0x13), VC5_HI, _LO (0x14–0x15) / bq76930, bq76940: VC6_HI, _LO (0x16–0x17), VC7_HI, _LO (0x18–0x19), VC8_HI, _LO (0x1A–0x1B), VC9_HI, _LO (0x1C–0x1D), VC10_HI, _LO (0x1E–0x1F) / bq76940: VC11_HI, _LO 0x20–0x21), VC12_HI, _LO (0x22–0x23), VC13_HI, _LO (0x24–0x25), VC14_HI, _LO (0x26–0x27), VC15_HI, _LO (0x28–0x29)								
BIT	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
NAME	—	—	D13	D12	D11	D10	D9	D8
复位	0	0	0	0	0	0	0	0
NAME	D7	D6	D5	D4	D3	D2	D1	D0
复位	0	0	0	0	0	0	0	0

D11:8 (位3–0): 单元“x”的ADC读数，高6位MSB。如果高低寄存器在同一事务中读取（使用地址自动递增），则始终以原子值返回。

D7:0 (位7–0): 单元“x”的ADC读数，低8位LSB。

表表 7-16. BAT_HI (0x2A) 和 BAT_LO (0x2B)

BIT	7	6	5	4	3	2	1	0
NAME	D15	D14	D13	D12	D11	D10	D9	D8
RESET	0	0	0	0	0	0	0	0
NAME	D7	D6	D5	D4	D3	D2	D1	D0
RESET	0	0	0	0	0	0	0	0

D15:8 (位 7–0): BAT 计算基于 1–15 单元累加，高 8 位 MSB。如果高低寄存器在同一事务中读取（使用地址自动增量），则始终作为原子值返回。

D7:0 (位 7–0): BAT 计算基于 1–15 单元累加，低 8 位 LSB

表表 7-17. TS1_HI (0x2C) 和 TS1_LO (0x2D)

BIT	7	6	5	4	3	2	1	0
NAME	—	—	D13	D12	D11	D10	D9	D8
重置	0	0	0	0	0	0	0	0
NAME	D7	D6	D5	D4	D3	D2	D1	D0
重置	0	0	0	0	0	0	0	0

D11:8 (位3–0): TS1或DIETEMP ADC读数，高6位MSB。如果在同一事务中同时读取高低寄存器（使用地址自动递增），则始终以原子值返回。

D7:0 (位7–0): TS1或DIETEMP ADC读数，低8位LSB

表 7-18. TS2_HI (0x2E) and TS2_LO (0x2F)

BIT	7	6	5	4	3	2	1	0
NAME	—	—	D13	D12	D11	D10	D9	D8
RESET	0	0	0	0	0	0	0	0
NAME	D7	D6	D5	D4	D3	D2	D1	D0
RESET	0	0	0	0	0	0	0	0

D11:8 (Bits 3–0): TS2 ADC reading, upper 6 MSB. Always returned as an atomic value if both high and low registers are read in the same transaction (using address auto-increment).

D7:0 (Bits 7–0): TS2 ADC reading, lower 8 LSB

表 7-19. TS3_HI (0x30) and TS3_LO (0x31)

BIT	7	6	5	4	3	2	1	0
NAME	—	—	D13	D12	D11	D10	D9	D8
RESET	0	0	0	0	0	0	0	0
NAME	D7	D6	D5	D4	D3	D2	D1	D0
RESET	0	0	0	0	0	0	0	0

D11:8 (Bits 3–0): TS3 ADC reading, upper 6 MSB. Always returned as an atomic value if both high and low registers are read in the same transaction (using address auto-increment).

D7:0 (Bits 7–0): TS3 ADC reading, lower 8 LSB

表 7-20. CC_HI (0x32) and CC_LO (0x33)

BIT	7	6	5	4	3	2	1	0
NAME	CC15	CC14	CC13	CC12	CC11	CC10	CC9	CC8
RESET	0	0	0	0	0	0	0	0
NAME	CC7	CC6	CC5	CC4	CC3	CC2	CC1	CC0
RESET	0	0	0	0	0	0	0	0

CC15:8 (Bits 7–0): Coulomb counter upper 8 MSB. Always returned as an atomic value if both high and low registers are read in the same transaction (using address auto-increment).

CC7:0 (Bits 7–0): Coulomb counter lower 8 LSB

表 7-21. ADCGAIN1 (0x50)

BIT	7	6	5	4	3	2	1	0
NAME	—	—	—	—	ADCGAIN4	ADCGAIN3	—	—
RESET	—	—	—	—	—	—	—	—
ACCESS	R	R	R	R	R	R	R	R

表 7-22. ADCGAIN2 (0x59)

BIT	7	6	5	4	3	2	1	0
NAME	ADCGAIN2	ADCGAIN1	ADCGAIN0	—	—	—	—	—
RESET	—	—	—	—	—	—	—	—
ACCESS	R	R	R	R	R	R	R	R

ADCGAIN4:3 (Bits 3–2, address 0x50):

ADC GAIN
offset upper 2 MSB

表表 7-18. TS2_HI (0x2E) 和 TS2_LO (0x2F)

BIT	7	6	5	4	3	2	1	0
NAME	—	—	D13	D12	D11	D10	D9	D8
复位	0	0	0	0	0	0	0	0
NAME	D7	D6	D5	D4	D3	D2	D1	D0
复位	0	0	0	0	0	0	0	0

D11:8 (位3–0): TS2 ADC读数，高6位MSB。如果高低寄存器在同一事务中读取（使用地址自动递增），则始终以原子值形式返回。

D7:0 (位 7–0): TS2 ADC 读取值，低 8 位 LSB

表表 7-19. TS3_HI (0x30) 和 TS3_LO (0x31)

BIT	7	6	5	4	3	2	1	0
NAME	—	—	D13	D12	D11	D10	D9	D8
重置	0	0	0	0	0	0	0	0
NAME	D7	D6	D5	D4	D3	D2	D1	D0
重置	0	0	0	0	0	0	0	0

D11:8 (位 3–0): TS3 ADC 读取值，高 6 位 MSB。如果高低寄存器在同一事务中读取（使用地址自动递增），则始终以原子值返回。

D7:0 (位 7–0): TS3 ADC 读取值，低 8 位 LSB

表表 7-20. CC_HI (0x32) 和 CC_LO (0x33)

BIT	7	6	5	4	3	2	1	0
NAME	CC15	CC14	CC13	CC12	CC11	CC10	CC9	CC8
重置	0	0	0	0	0	0	0	0
NAME	CC7	CC6	CC5	CC4	CC3	CC2	CC1	CC0
重置	0	0	0	0	0	0	0	0

CC15:8 (位 7–0): 电池计数器高 8 位 MSB。如果高低寄存器在同一事务中读取（使用地址自动递增），则始终以原子值返回。

CC7:0 (位 7–0): 电池计数器低 8 位 LSB

表 7-21. ADCGAIN1 (0x50)

BIT	7	6	5	4	3	2	1	0
NAME	—	—	—	—	ADCGAIN4	ADCGAIN3	—	—
重置	—	—	—	—	—	—	—	—
访问	R	R	R	R	R	R	R	R

表表 7-22. ADCGAIN2 (0x59)

BIT	7	6	5	4	3	2	1	0
NAME	ADCGAIN2	ADCGAIN1	ADCGAIN0	—	—	—	—	—
复位	—	—	—	—	—	—	—	—
访问	R	R	R	R	R	R	R	R

ADCGAIN4:3 (位3-2，地址0x50)：

ADC增益
偏移量高位2位

ADCGAIN2:0 (Bits 7–5, address 0x59):

ADC GAIN offset lower 3 LSB

ADCGAIN<4:0> is a production-trimmed value for the ADC transfer function, in units of µV/LSB. The range is 365 µV/LSB to 396 µV/LSB, in steps of 1 µV/LSB, and may be calculated as follows:

GAIN = 365 µV/LSB + (ADCGAIN<4:0>in decimal) × (1 µV/LSB)

Alternately, a conversion table is provided below:

ADC GAIN	Gain (µV/LSB)	ADC GAIN	Gain (µV/LSB)
0x00	365	0x10	381
0x01	366	0x11	382
0x02	367	0x12	383
0x03	368	0x13	384
0x04	369	0x14	385
0x05	370	0x15	386
0x06	371	0x16	387
0x07	372	0x17	388
0x08	373	0x18	389
0x09	374	0x19	390
0x0A	375	0x1A	391
0x0B	376	0x1B	392
0x0C	377	0x1C	393
0x0D	378	0x1D	394
0x0E	379	0x1E	395
0x0F	380	0x1F	396

表 7-23. ADCOFFSET (0x51)

BIT	7	6	5	4	3	2	1	0
NAME	ADC OFFSET7	ADC OFFSET6	ADC OFFSET5	ADC OFFSET4	ADC OFFSET3	ADC OFFSET2	ADC OFFSET1	ADC OFFSET0
RESET	—	—	—	—	—	—	—	—
ACCESS	R	R	R	R	R	R	R	R

ADCOFFSET7:0 (Bits 7–0):

ADC offset, stored in 2's complement format in mV units. Positive full-scale range corresponds to 0x7F and negative full-scale corresponds to 0x80. The full-scale input range is –128 mV to 127 mV, with an LSB of 1 mV.

The table below shows example offsets.

ADCOFFSET	Offset (mV)
0x00	0
0x01	1
0x7F	127
0x80	–128
0x81	–127
0xFF	–1

ADCGAIN2:0 (位 7–5, 地址 0x59):

ADC增益偏移量低位3位

ADCGAIN<4:0>是 ADC 转换函数的生产调整值，单位为 µV/LSB。范围是 365 µV/LSB 至 396 µV/LSB，步长为 1 µV/LSB，计算方法如下：

GAIN = 365 µV/LSB + (ADCGAIN<4:0>in decimal) × (1 µV/LSB)

或者，下方提供了转换表：

ADC 增益	增益 (µV/LSB)	ADC 增益	增益 (µV/LSB)
0x00	365	0x10	381
0x01	366	0x11	382
0x02	367	0x12	383
0x03	368	0x13	384
0x04	369	0x14	385
0x05	370	0x15	386
0x06	371	0x16	387
0x07	372	0x17	388
0x08	373	0x18	389
0x09	374	0x19	390
0x0A	375	0x1A	391
0x0B	376	0x1B	392
0x0C	377	0x1C	393
0x0D	378	0x1D	394
0x0E	379	0x1E	395
0x0F	380	0x1F	396

表 7-23. ADCOFFSET (0x51)

BIT	7	6	5	4	3	2	1	0
NAME	ADC OFFSET7	ADC OFFSET6	ADC OFFSET5	ADC 偏移4	ADC 偏移3	ADC 偏移2	ADC 偏移1	ADC 偏移0
复位	—	—	—	—	—	—	—	—
访问	R	R	R	R	R	R	R	R

ADCOFFSET7:0 (位 7–0):

ADC偏移量，以mV单位以二进制补码格式存储。正满量程对应0x7F，负满量程对应0x80。满量程输入范围为–128 mV至127 mV，LSB为1 mV。

下表展示了示例偏移量。

ADCOFFSET	偏移量 (mV)
0x00	0
0x01	1
0x7F	127
0x80	–128
0x81	–127
0xFF	–1

8 Application and Implementation

注

Information in the following application section is not part of the TI component specification, and TI does not warrant its accuracy or completeness. TI's customers are responsible for determining suitability of components for their purposes. Customers should validate and test their design implementation to confirm system functionality.

8.1 Application Information

The bq769x0 family of battery monitoring AFEs enabling cell parametric measurement and protection is a variety of 3-series to 15-series Li-Ion/Li Polymer battery packs.

To evaluate the performance and configurations of the device users need the bq76940/bq76930/bq76920 Evaluation Software, (SLUCC539) tool to configure the internal registers for a specific battery pack and application. The Evaluation Software tool is a graphical user-interface tool installed on a PC during development. This can be used in conjunction with the bq76920EVM, bq76930EVM or bq76940EVM.

The bq769x0 devices are expected to be implemented in a system with a microcontroller that can perform additional functions based on the data made collected. The bq78350 is one example of a companion to the bq769x0 family.

8.1.1 Configuring Alternative Cell Counts

Each bq769x0 family of IC's support a variety of cell counts. The following tables provide guidance on which device and which input pins to use, depending on the number of cells in the pack.

表 8-1. Cell Connections for bq76920

Cell Input	3 Cells	4 Cells	5 Cells
VC5–VC4	CELL 3	CELL 4	CELL 5
VC4–VC3	short	short	CELL 4
VC3–VC2	short	CELL 3	CELL 3
VC2–VC1	CELL 2	CELL 2	CELL 2
VC1–VC0	CELL 1	CELL 1	CELL 1

表 8-2. Cell Connections for bq76930

Cell Input	6 Cells	7 Cells	8 Cells	9 Cells	10 Cells
VC10–VC9	CELL 6	CELL 7	CELL 8	CELL 9	CELL 10
VC9–VC8	short	short	short	short	CELL 9
VC8–VC7	short	short	CELL 7	CELL 8	CELL 8
VC7–VC6	CELL 5	CELL 6	CELL 6	CELL 7	CELL 7
VC6–VC5b	CELL 4	CELL 5	CELL 5	CELL 6	CELL 6
VC5–VC4	CELL 3	CELL 4	CELL 4	CELL 5	CELL 5
VC4–VC3	short	short	short	CELL 4	CELL 4
VC3–VC2	short	CELL 3	CELL 3	CELL 3	CELL 3
VC2–VC1	CELL 2	CELL 2	CELL 2	CELL 2	CELL 2
VC1–VC0	CELL 1	CELL 1	CELL 1	CELL 1	CELL 1

8 应用与实现

注

以下应用部分中的信息不属于TI组件规范的一部分，TI不保证其准确性或完整性。TI的客户负责确定组件的适用性。客户应验证和测试其设计实现，以确认系统功能。

8.1 应用信息

bq769x0 系列电池监控 AFE 能够实现电芯参数测量和保护，适用于 3 系列至 15 系列锂离子/锂聚合物电池组。

为了评估设备的性能和配置，用户需要使用 bq76940/bq76930/bq76920 评估软件（SLUCC539）工具来配置特定电池组和应用的内部寄存器。评估软件工具是一款图形用户界面工具，在开发过程中安装在 PC 上。这可以与 bq76920EVM、bq76930EVM 或 bq76940EVM 一起使用。

bq769x0 设备预计将在带有微控制器的系统中实现，该微控制器可以根据收集的数据执行附加功能。bq78350 是与 bq769x0 系列配合使用的其中一个示例。

8.1.1 配置备用 CellCounts

每个 bq769x0 系列的 IC 都支持多种电芯数量。以下表格提供了关于根据电池组中电芯数量应使用哪些设备和哪些输入引脚的指导。

表 8-1. bq76920 的 Cell 连接

Cell 输入	3 个 Cell	4 个 Cell	5 个 Cell
VC5–VC4	单元格 3	单元格 4	单元格 5
VC4–VC3	短	短	单元格 4
VC3–VC2	短	单元格 3	单元格 3
VC2–VC1	单元格 2	单元格 2	单元格 2
VC1–VC0	单元格 1	单元格 1	单元格 1

表表 8-2. bq76930 的单元格连接

单元格输入	6 个单元格	7 个单元格	8 单元格	9 单元格	10 单元格
VC10–VC9	单元格 6	单元格 7	单元格 8	单元格 9	单元格 10
VC9–VC8	短	短	短	短	单元格 9
VC8–VC7	短	短	单元格 7	单元格 8	单元格 8
VC7–VC6	单元格 5	单元格 6	单元格 6	单元格 7	单元格 7
VC6–VC5b	单元格 4	单元格 5	单元格 5	单元格 6	单元格 6
VC5–VC4	单元格 3	单元格 4	单元格 4	单元格 5	单元格 5
VC4–VC3	短	短	短	单元格 4	单元格 4
VC3–VC2	短	单元格 3	单元格 3	单元格 3	单元格 3
VC2–VC1	单元格 2	单元格 2	单元格 2	单元格 2	单元格 2
VC1–VC0	单元格 1	单元格 1	单元格 1	单元格 1	单元格 1

表 8-3. Cell Connections for bq76940

Cell Input	9 Cells	10 Cells	11 Cells	12 Cells	13 Cells	14 Cells	15 Cells
VC15–VC14	CELL 9	CELL 10	CELL 11	CELL 12	CELL 13	CELL 14	CELL 15
VC14–VC13	short	short	short	short	short	short	CELL 14
VC13–VC12	short	short	short	CELL 11	CELL 12	CELL 13	CELL 13
VC12–VC11	CELL 8	CELL 9	CELL 10	CELL 10	CELL 11	CELL 12	CELL 12
VC11–VC10b	CELL 7	CELL 8	CELL 9	CELL 9	CELL 10	CELL 11	CELL 11
VC10–VC9	CELL 6	CELL 7	CELL 8	CELL 8	CELL 9	CELL 10	CELL 10
VC9–VC8	short	short	short	short	short	CELL 9	CELL 9
VC8–VC7	short	short	CELL 7	CELL 7	CELL 8	CELL 8	CELL 8
VC7–VC6	CELL 5	CELL 6	CELL 6	CELL 6	CELL 7	CELL 7	CELL 7
VC6–VC5b	CELL 4	CELL 5	CELL 5	CELL 5	CELL 6	CELL 6	CELL 6
VC5–VC4	CELL 3	CELL 4	CELL 4	CELL 4	CELL 5	CELL 5	CELL 5
VC4–VC3	short	short	short	short	CELL 4	CELL 4	CELL 4
VC3–VC2	short	CELL 3	CELL 3	CELL 3	CELL 3	CELL 3	CELL 3
VC2–VC1	CELL 2	CELL 2	CELL 2	CELL 2	CELL 2	CELL 2	CELL 2
VC1–VC0	CELL 1	CELL 1	CELL 1	CELL 1	CELL 1	CELL 1	CELL 1

8.2 Typical Applications

CAUTION

The external circuitries in the following schematics show minimum requirements to ensure device robustness during cell connection to the PCB and normal operation.

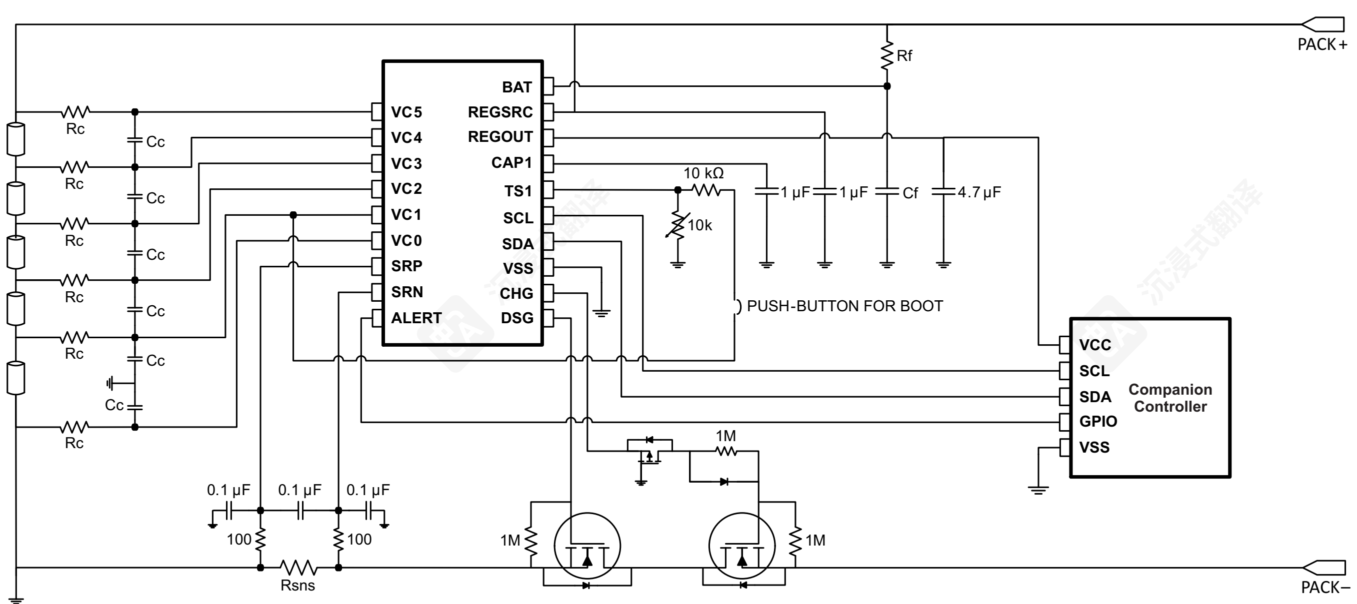


图 8-1. bq76920 with bq78350 Companion Controller IC

表8-3. bq76940的单元连接

单元输入	9个单元	10个单元	11 个单元格	12 个单元格	13 个单元格	14 个单元格	15 个单元格
VC15–VC14	单元格 9	单元格 10	单元格 11	单元格 12	单元格 13	单元格 14	单元格 15
VC14–VC13	短	短	短	短	短	短	单元格 14
VC13–VC12	短	短	短	单元格 11	单元格 12	单元格 13	单元格 13
VC12–VC11	单元格 8	单元格 9	单元格 10	单元格 10	单元格 11	单元格 12	单元格 12
VC11–VC10b	单元格 7	单元格 8	单元格 9	单元格 9	单元格 10	单元格 11	单元格 11
VC10–VC9	单元格 6	单元格 7	单元格 8	单元格 8	单元格 9	单元格 10	单元格 10
VC9–VC8	短	短	短	短	短	单元格 9	单元格 9
VC8–VC7	短	短	单元格 7	单元格 7	单元格 8	单元格 8	单元格 8
VC7–VC6	单元格 5	单元格 6	单元格 6	单元格 6	单元格 7	单元格 7	单元格 7
VC6–VC5b	单元格 4	单元格 5	单元格 5	单元格 5	单元格 6	单元格 6	单元格 6
VC5–VC4	单元格 3	单元格 4	单元格 4	单元格 4	单元格 5	单元格 5	单元格 5
VC4–VC3	短	短	短	短	单元格 4	单元格 4	单元格 4
VC3–VC2	短	单元格 3	单元格 3	单元格 3	单元格 3	单元格 3	单元格 3
VC2–VC1	单元格 2	单元格 2	单元格 2	单元格 2	单元格 2	单元格 2	单元格 2
VC1–VC0	单元格 1	单元格 1	单元格 1	单元格 1	单元格 1	单元格 1	单元格 1

8.2 典型应用

注意

以下原理图中所示的外部电路显示了确保电芯连接到 PCB 和正常操作期间设备稳健运行所需的最小要求。

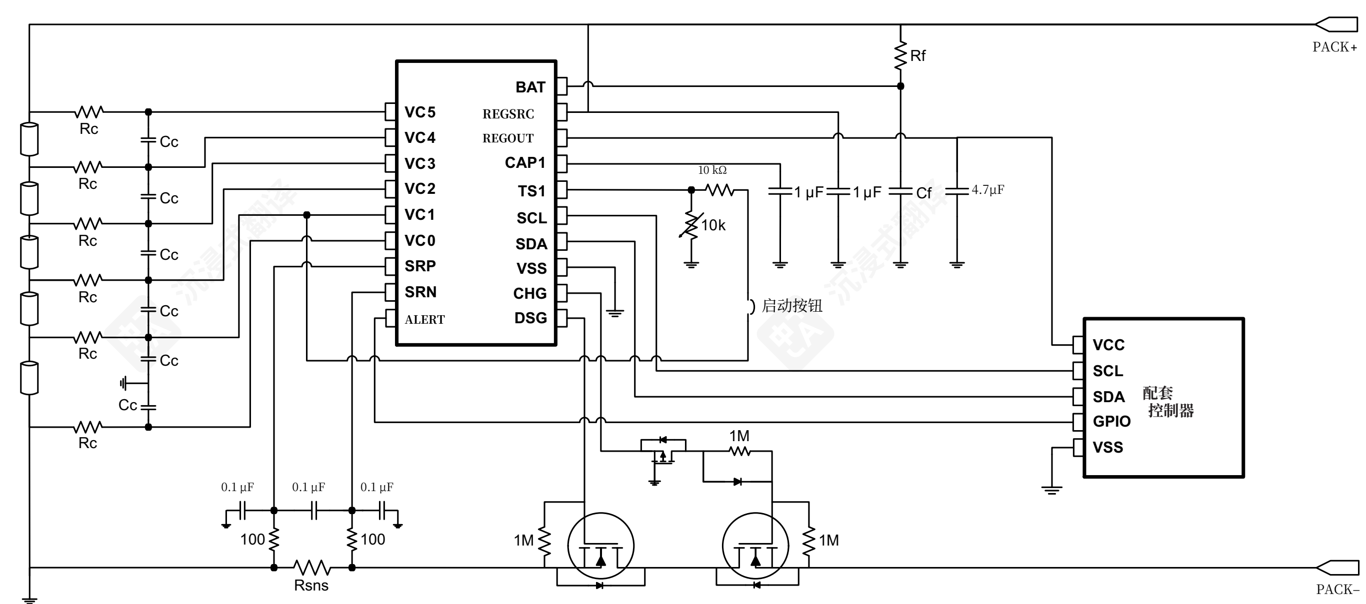


图8-1. bq76920与bq78350伴侣控制器IC

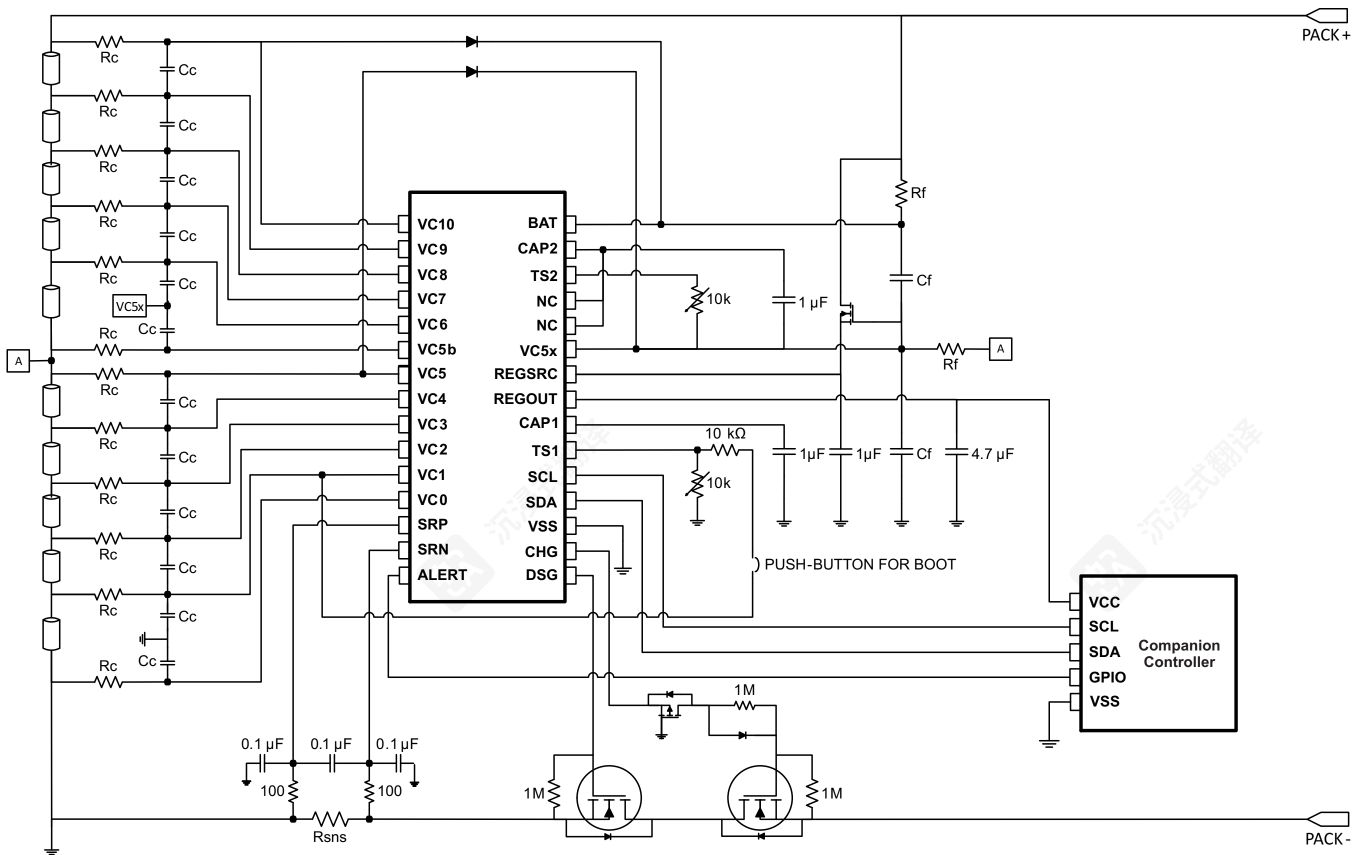
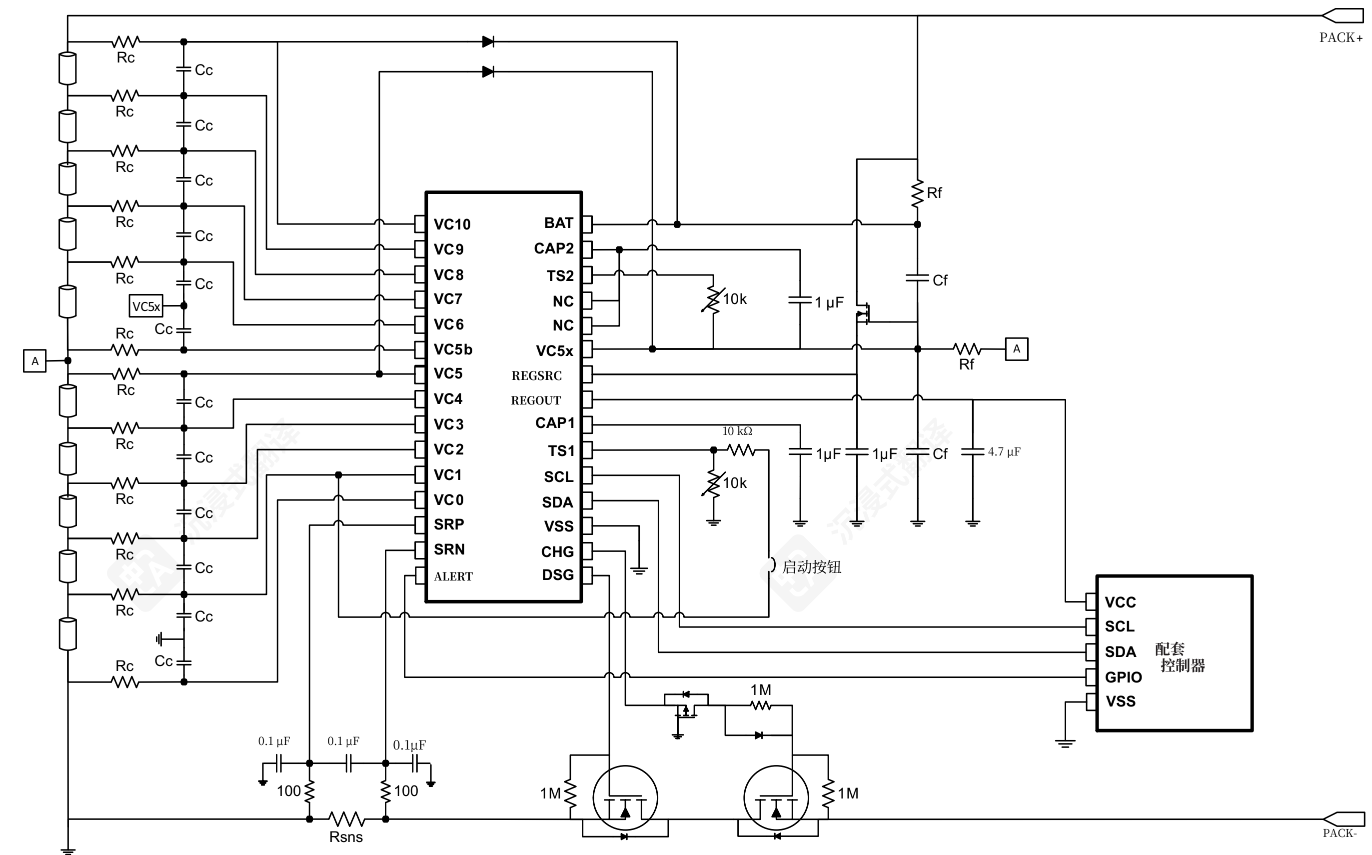


图 8-2. bq76930 with bq78350 Companion Controller IC



图图 8-2. bq76930 与 bq78350 伴侣控制器 IC

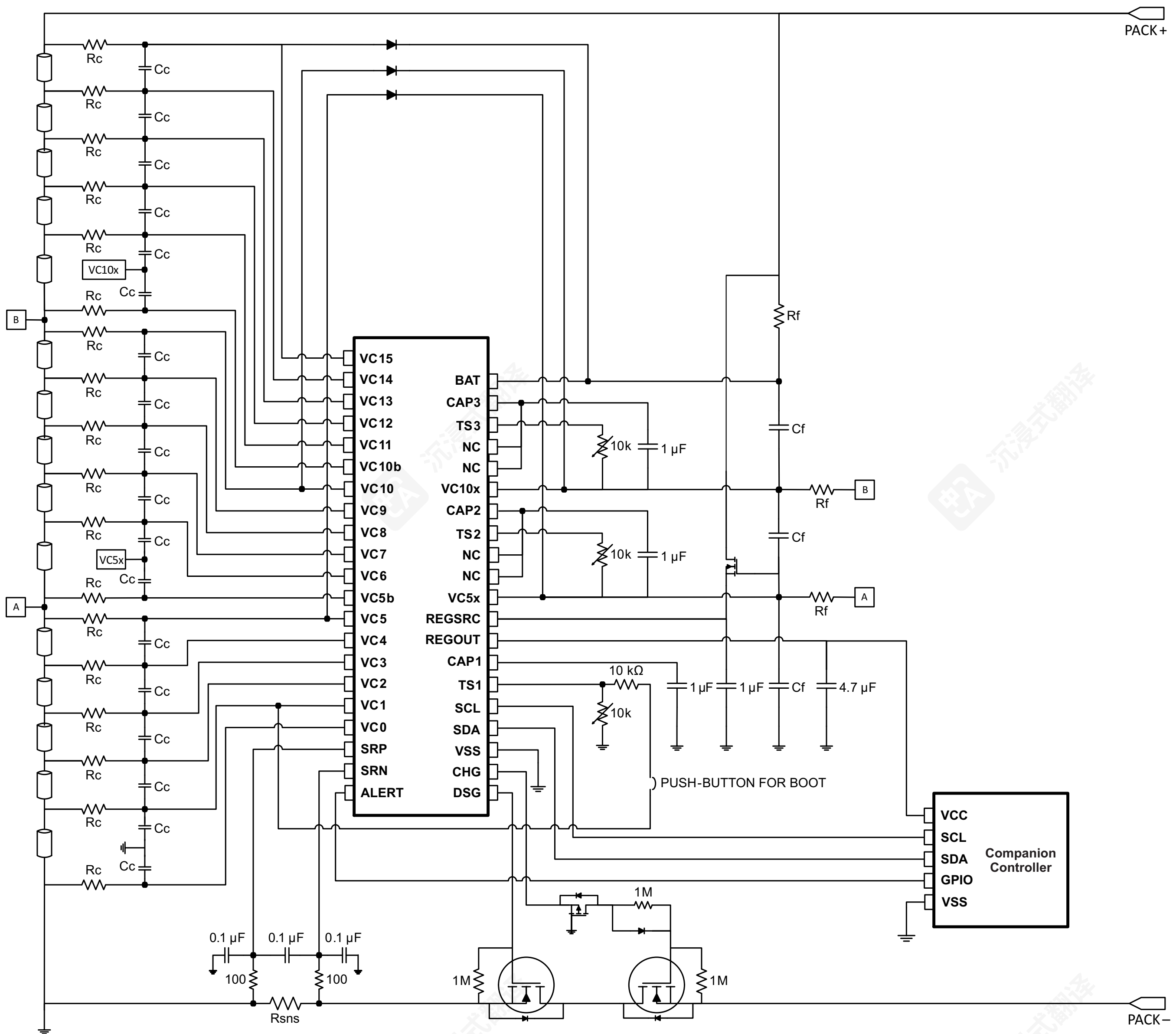


图 8-3. bq76940 with bq78350 Companion Controller IC

8.2.1 Design Requirements

表 8-4. bq769x0 Design Requirements

DESIGN PARAMETER	EXAMPLE VALUE at TA = 25°C
Minimum system operating voltage	24 V
Cell minimum operating voltage	3.0 V
Series Cell Count	8
Charge Voltage	33.6 V
Maximum Charge Current	3.0 A
Peak Discharge Current	10.0 A
OV Protection Threshold	4.30 V
OV Protection Delay	2s
UV Protection Threshold	2.5 V

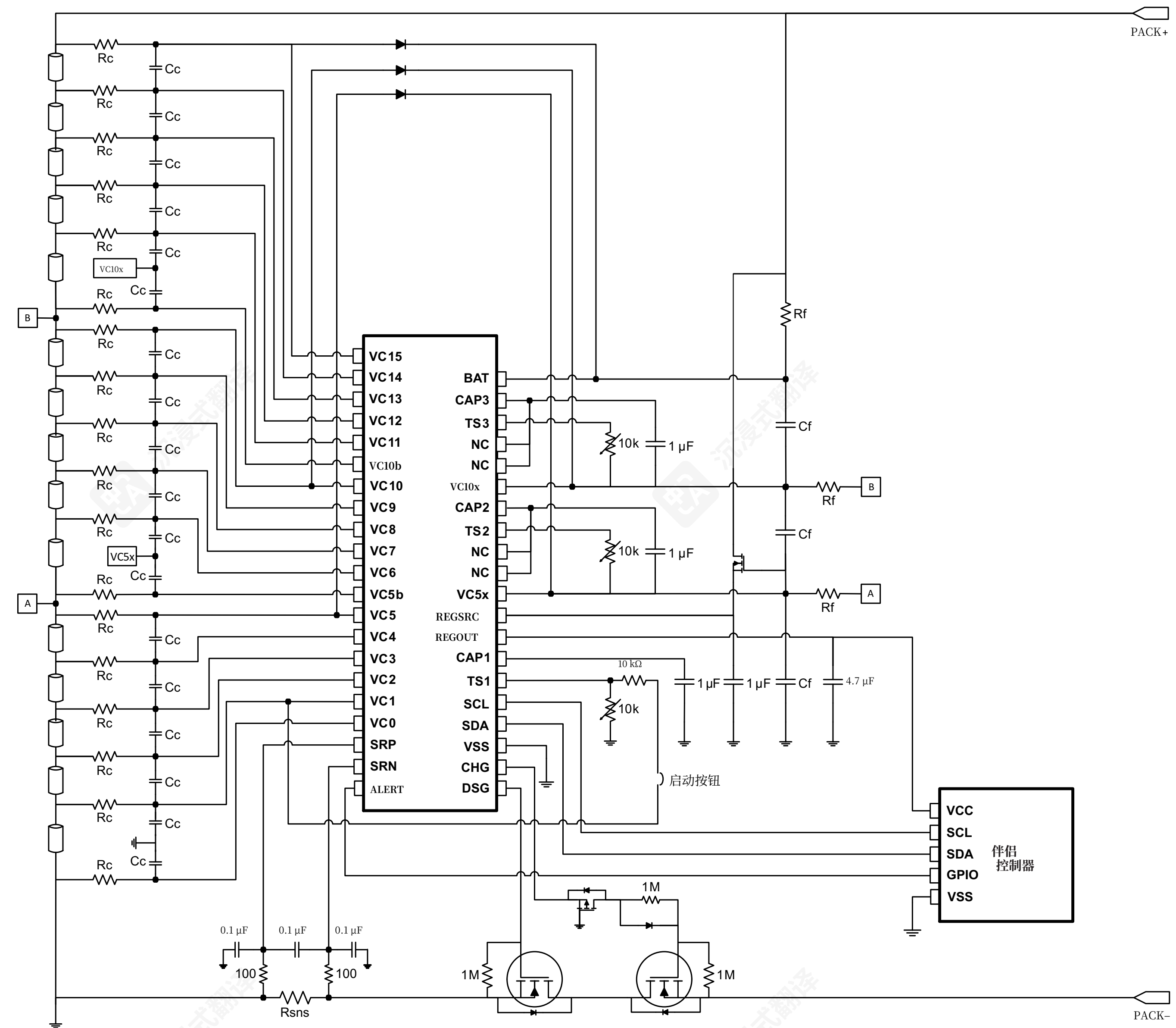


图8-3. bq76940与bq78350伴侣控制器IC

8.2.1 设计要求

表8-4. bq769x0设计要求

设计参数	TA = 25°C时的示例值
最小系统工作电压	24 V
电池最小工作电压	3.0 V
系列电芯数量	8
充电电压	33.6 V
最大充电电流	3.0 A
峰值放电电流	10.0 A
过压保护阈值	4.30 V
过压保护延时	2s
UV保护阈值	2.5 V

表 8-4. bq769x0 Design Requirements (continued)

DESIGN PARAMETER	EXAMPLE VALUE at TA = 25°C
UV Protection Delay	4s
OCD Protection Threshold Max	15 A
OCD Protection Delay Time	320 ms
SCD Protection Threshold Max	25 A
SCD Protection Delay Time	100 µs

8.2.2 Detailed Design Procedure

To begin the design process, there are some key steps required for component selection and protection configuration.

8.2.2.1 Step-by-Step Design Procedure

- Determine the number of series cells.
  - This value depends on the cell chemistry and the load requirements of the system. For example, to support a minimum battery voltage of 24 V using Li-CO2 type cells with a cell minimum voltage of 3.0 V, there needs to be at least 8-series cells.
- Select the correct bq769x0 device.
  - For 8-series cells, the bq76930 is needed.
  - For the correct cell connections, see 表 8-2.
- Select the correct protection FETs.
  - The bq76930 uses a low-side drive suitable for N-CH FETs.
  - These FETs should be rated for the maximum:
    - Voltage, which should be approximately 5 V (DC) 10 V (peak) per series cell: for example, 40 V.
    - Current, which should be calculated based on both the maximum DC current and the maximum transient current with some margin: for example, 30 A.
    - Power Dissipation, which can be a factor of the RDS(ON) rating of the FET, the FET package, and the PCB design: for example, 5 W, assuming 5 mΩ RDS(ON).
- Select the correct sense resistor.
  - The resistance value should be selected to maximize the input bandwidth use of the coulomb counter range, CC_{RANGE}, but not exceed the absolute maximum ratings.
    - Using the normal max discharge current, RSNS = 200 mV / 10.0 A = 20 mΩ.
    - However, considering I_{SCD} of 25 A and Abs Max SRP–SRN input of –300 mV, RSNS = 300 mV / 25 A = 7.5 mΩ
    - The maximum operating current of the system should also be considered as this should be below the maximum OCD Threshold, which with a RSNS of 7.5 mΩ gives a max OCD current setting of 13.3 A.
  - Further tolerance analysis (value tolerance, temperature variation, and so on) and PCB design margin should also be considered, so RSNS of 5 mΩ would be suitable with a 75-ppm temperature coefficient and power rating of 5 W.
- The bq76930 is chosen, and so the REGSRC pin needs to be powered through a source follower circuit where the FET is used to provide current for REGSRC from the battery positive terminal while reducing the voltage to a suitable value for the IC.
  - The FET also dissipates the power resulting from the load current and dropped voltage external to the IC and care should be taken to ensure the correct dissipation ratings are specified by the chosen FET.

表8-4. bq769x0设计要求（续）

设计参数	示例值（TA = 25℃）
紫外线防护延迟	4s
OCD防护阈值最大值	15 A
OCD保护延迟时间	320 ms
SCD保护阈值最大值	25 A
SCD保护延迟时间	100 µs

8.2.2 详细设计步骤

开始设计流程时，组件选择和保护配置需要一些关键步骤。

8.2.2.1 步骤化设计流程

- 确定串联电芯的数量。
  – 此值取决于电芯化学类型和系统的负载需求。例如，使用最小电压为3.0V的Li-CO2类型电芯支持最低电池电压24V，至少需要8串电芯。
- 选择正确的bq769x0设备。
  – 对于8串电芯，需要bq76930。
  – 对于正确的电芯连接，见表8-2。
- 选择正确的保护FET。
  – Thebq76930使用适合N-CHFETs的低边驱动。
  – 这些FET应额定为最大：
  • 电压，应为每个系列电池约5 V (DC) 10 V (峰值)：例如，40 V。
  • 电流，应根据最大直流电流和最大瞬态电流加上一定余量计算：例如，30 A。
  • 功耗，可以是FET的RDS(ON)额定值的因素、FET封装和PCB设计：例如，5 W，假设5 mΩ RDS(ON)。
- 选择正确的电流检测电阻。
  – 电阻值应选择以最大化库仑计数器范围 CC_{RANGE} 的输入带宽使用，但不能超过绝对最大额定值。
  • 使用正常最大放电电流，RSNS = 200 mV / 10.0 A = 20 mΩ。
  • 但是，考虑到 I SCD 为 25 A，以及绝对最大 SRP–SRN 输入为 –300 mV，RSNS = 300 mV / 25 A = 7.5 mΩ
  • 系统的最大工作电流也应考虑，因为这应低于最大OCD阈值，而具有7.5 mΩ 的RSNS时，最大OCD电流设置为13.3A。
  – 进一步的容差分析（值容差、温度变化等）和PCB设计余量也应考虑，因此5 mΩ 的RSNS是合适的，具有75-ppm温度系数和5 W的功率额定值。
- 选择bq76930，因此REGSRC引脚需要通过一个源跟随电路供电，其中FET用于从电池正极提供电流给REGSRC，同时将电压降低到适合IC的值。
  – FET还会耗散由负载电流和IC外部电压降产生的功率，因此应确保所选FET的耗散额定值正确。

- Configure the Current-based protection settings through PROTECT1 and PROTECT2:
  - Ideal SCD Threshold = 25 A × 5 mΩ = 125 mV.
    - However, the closest options are 111 mV (0x03) and 133 mV (0x04) providing 22.2 A and 26.6 A, respectively. Both options have the RSNS bit = 1.
    - 0x03 (22.2 A) will be used in this example.
  - The SCD delay threshold setting for a 100 µs delay is 0x01.
  - Therefore, PROTECT1 should be programmed with 0x8C.
  - Ideal OCD Threshold = 15 A × 5 mΩ = 75 mV.
    - However, the closest options are 72 mV (0x0A) and 78 mV (0x0B), providing 14.4 A and 15.6 A, respectively. Both options have the RSNS bit = 1.
    - 0x0A (14,4A) will be used in this example.
  - The OCD delay threshold setting for a 320-ms delay is 0x05.
  - Therefore, PROTECT2 should be programmed with 0x5B.

注

Care should be taken when determining the setting of OV_TRIP and UV_TRIP as these are ADC value outputs and correlation to cell voltage also requires consideration of the ADC GAIN and ADC OFFSET registers. More specific details can be found in 节 7.3.1.2.

- Configure the Voltage-based protection settings through OV_TRIP, UV_TRIP and PROTECT3:
  - The selected OV Threshold is 4.30 V.
    - Therefore, OV_TRIP should be programmed with 0xC9.
  - The selected UV Threshold is 2.5 V.
    - Therefore, UV_TRIP should be programmed with 0x1A.
  - The selected OV Delay is 2 s and the selected UV Delay is 4 s.
    - Therefore, PROTECT3 should be programmed with 0x50.

- 通过PROTECT1和PROTECT2配置基于电流的保护设置:
  - 理想SCD阈值 = 25 A × 5 mΩ = 125 mV。
    - 但是，最接近的选项是111 mV (0x03)和133 mV (0x04)，分别提供22.2 A和26.6 A。这两个选项都有RSNSbit = 1。
    - 0x03 (22.2 A) 将在本例中使用。
  - 100 µs延迟的SCD延迟阈值设置为0x01。
  - 因此，PROTECT1 应该使用 0x8C 编程。
  - 理想的 OCD 阈值 = 15 A × 5 mΩ = 75 mV。
    - 然而，最接近的选项是 72 mV (0x0A) 和 78 mV (0x0B)，分别提供 14.4 A 和 15.6 A。这两个选项都有 RSNSbit = 1。
    - 00x0A (14,4A) 将在本例中使用。
  - 对于 320-ms 延迟，OCDdelaythreshold 设置为 0x05。
  - 因此，PROTECT2 应该使用 0x5B 编程。

注

在确定OV_TRIP和UV_TRIP设置时需谨慎，因为它们是ADC值输出，与电池电压的相关性还需要考虑ADC GAIN和ADC OFFSET寄存器。更多详细信息可参考节7.3.1.2。

- 通过 OV_TRIP、UV_TRIP 和 PROTECT3 配置基于电压的保护设置:
  - 选择的 OV 阈值是 4.30 V。
    - 因此，OV_TRIP 应该使用 0xC9 进行编程。
  - 选择的 UV 阈值是 2.5 V。
    - 因此，UV_TRIP 应该使用 0x1A 进行编程。
  - 选择的 OV 延迟是 2 s，选择的 UVDelay 是 4 s。
    - 因此，PROTECT3 应该用 0x50 编程。

8.2.3 Application Curves

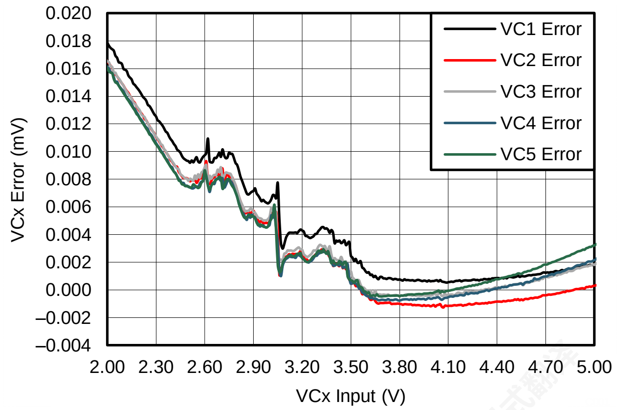


图 8-4. bq76930 VCx Error Across Input Range @ 25°C

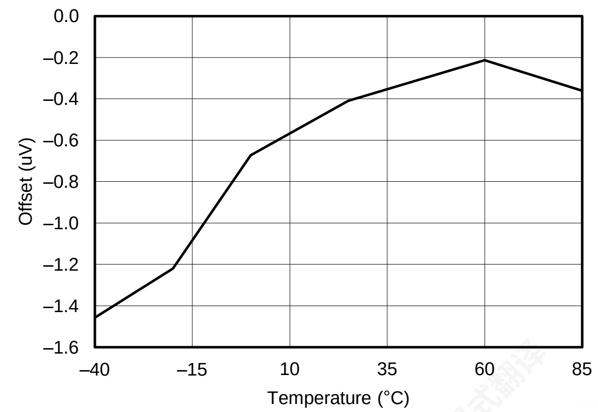
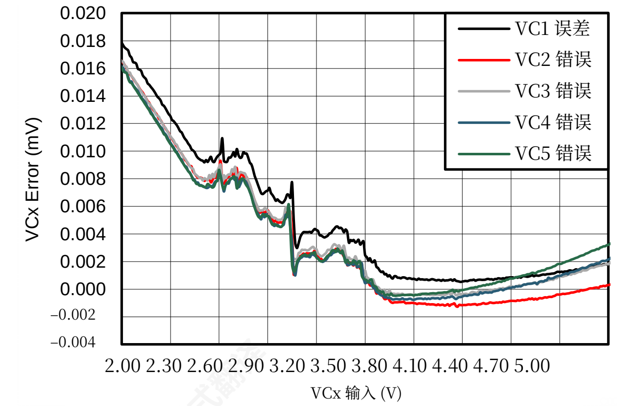
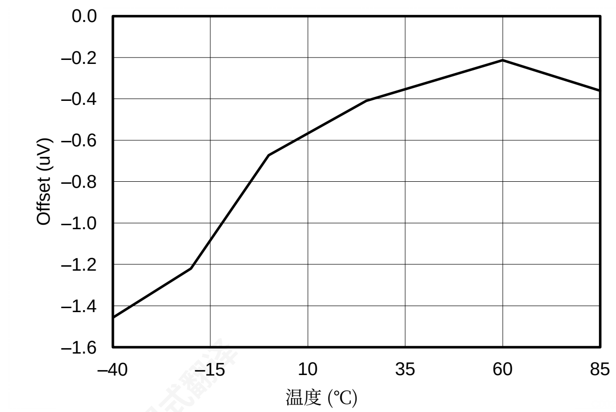


图 8-5. Coulomb Counter Offset

8.2.3 应用曲线



图图 8-4. bq76930 VCx 错误跨输入范围 @ 25°C



图图 8-5. 库仑计数器偏移

9 Power Supply Recommendations

The bq769x0 devices are powered through the BAT and REGSRC pins but the bq76930 and bq76940 have additional 'Power' pins to provide the power to the entire device in the higher cell configurations.

The use of Rf and Cf connected to the BAT pin, noted in the typical application diagrams, are required to filter system transients from disturbing the device power supply. These components should be placed as close as to the IC as possible.

Additionally, for the bq76930 and bq76940 there are additional requirements to ensure a stable power supply to the device. The REGSRC pin is powered through a source follower circuit where the FET is used to provide current for REGSRC from the battery positive terminal while reducing the voltage to a suitable value for the IC. The FET also dissipates the power resulting from the load current and dropped voltage external to the IC and care should be taken to ensure the correct dissipation ratings are specified by the chosen FET.

The bq76920 does not use a FET because the battery voltage is within the REGSRC range.

9 电源建议

bq769x0 器件通过 BAT 和 REGSRC 引脚供电，但 bq76930 和 bq76940 还具有额外的‘电源’引脚，用于在较高电池配置中为整个器件供电。

典型应用图中标注的连接到 BAT 引脚的 Rf 和 Cf 的使用是必要的，用于过滤系统瞬变，以免干扰器件电源。这些元件应尽可能靠近 IC 放置。

此外，对于 bq76930 和 bq76940，还有额外的要求以确保向器件提供稳定的电源。REGSRC 引脚通过源跟随电路供电，其中 FET 用于从电池正极端为 REGSRC 提供电流，同时将电压降低到适合 IC 的值。FET 还会耗散器件外部负载电流和电压降产生的功率，因此应确保所选 FET 的正确耗散额定值。

bq76920 不使用 FET，因为电池电压在 REGSRC 范围内。

10 Layout

10.1 Layout Guidelines

It is strongly recommended for best measurement performance to keep high current signals from interfering with the measurement system inputs and ground.

A second key recommendation is to ensure that the bq769x0 input filtering capacitors and power capacitors are connected to a common ground with as little parasitic resistance between the connections as possible.

10.2 Layout Example

图 10-1 shows a guideline of how to place key components compared to respective ground zones, based on the bq76920/30/40 EVMs.

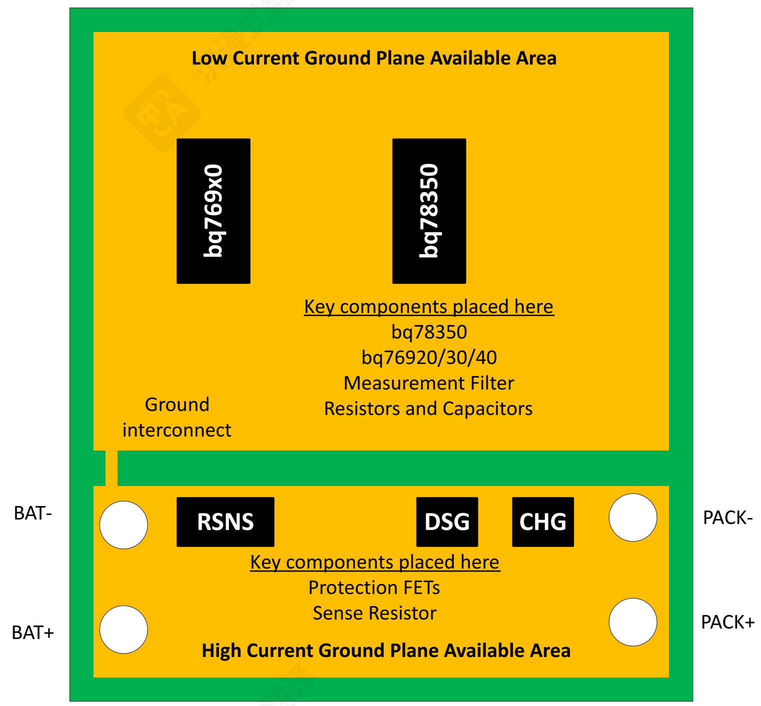


图 10-1. System Component Placement Layout vs. Ground Zone Guide

> **CAUTION**
>
> Care should be taken when placing key power pin capacitors to minimize PCB trace impedances. These impedances could result in device resets or other unexpected operations when the device is at peak power consumption.
>
> Although not shown in the diagrams, this caution also applies to the resistor and capacitor network surrounding the current sense resistor.

10 布局

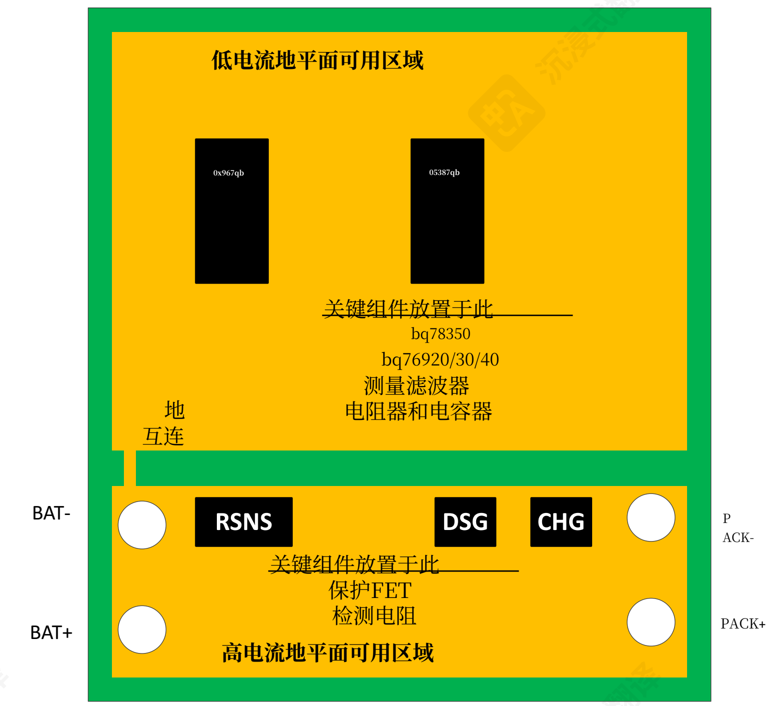
10.1 布局指南

为获得最佳测量性能，强烈建议将高电流信号与测量系统输入和接地隔离开。

第二个关键建议是确保 bq769x0 输入滤波电容和电源电容尽可能以最小的寄生电阻连接到公共地。

10.2 布局示例

图 10-1 展示了基于 bq76920/30/40EVM 的关键元件相对于各自接地区域的位置指南。



图图 10-1. 系统元件布局与接地区域指南

> 注意
>
> 放置关键电源引脚电容器时需谨慎，以尽量减小PCB走线阻抗。这些阻抗可能导致设备在峰值功耗时重置或出现其他意外操作。
>
> 虽然图中未显示，但此注意事项也适用于围绕电流检测电阻的电阻和电容网络。

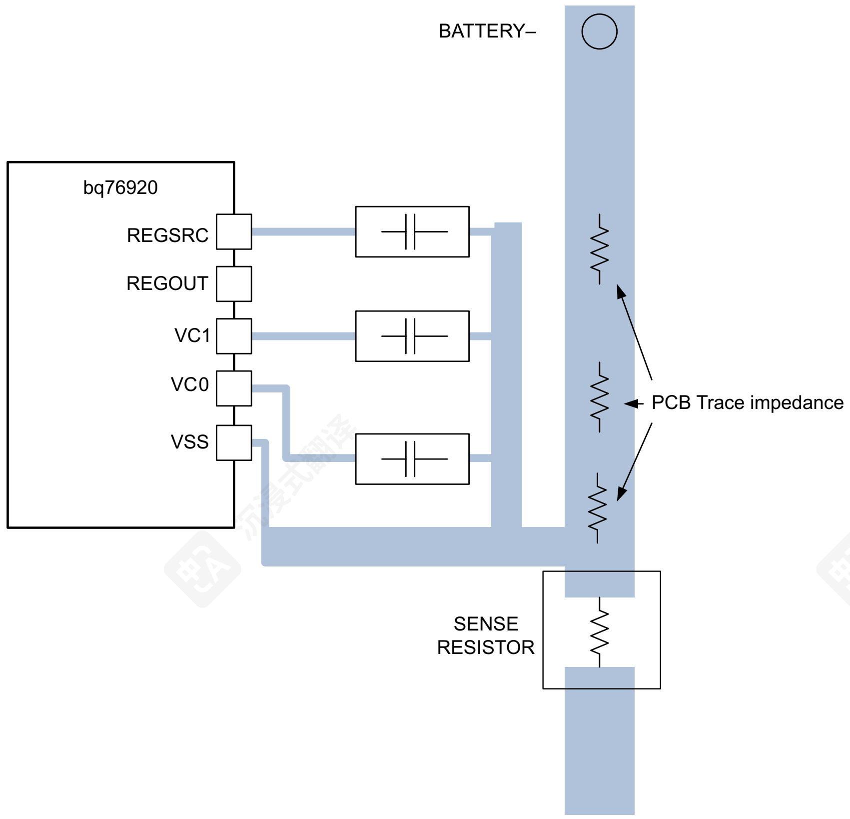
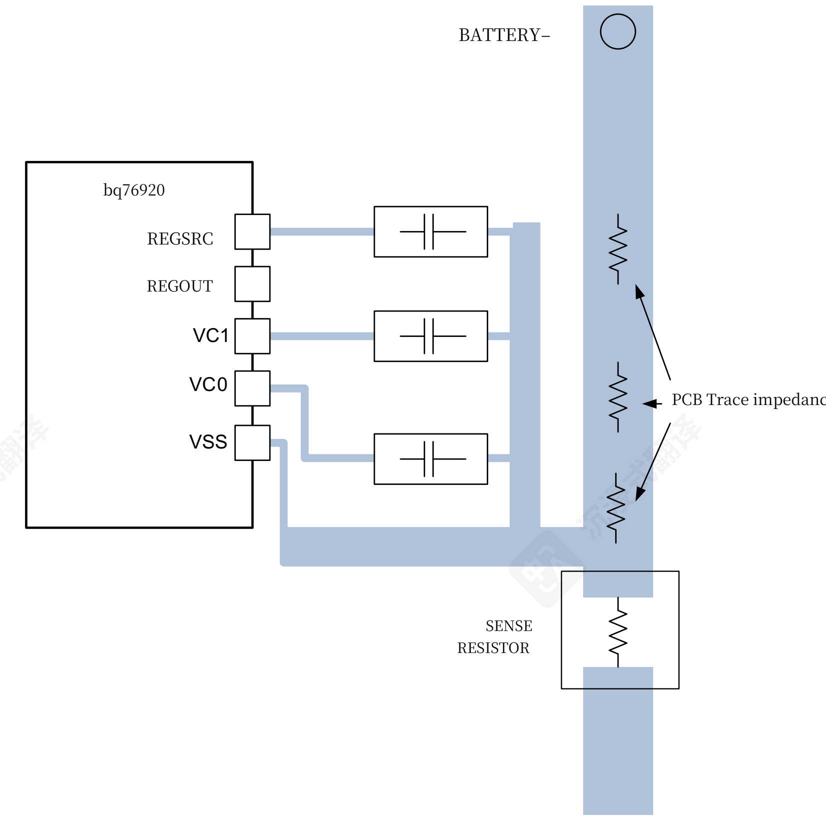


图 10-2.
Good Layout: Input Capacitor Grounding with Low Parasitic PCB Impedance



图图 10-2.
良好布局：输入电容接地与低寄生PCB阻抗

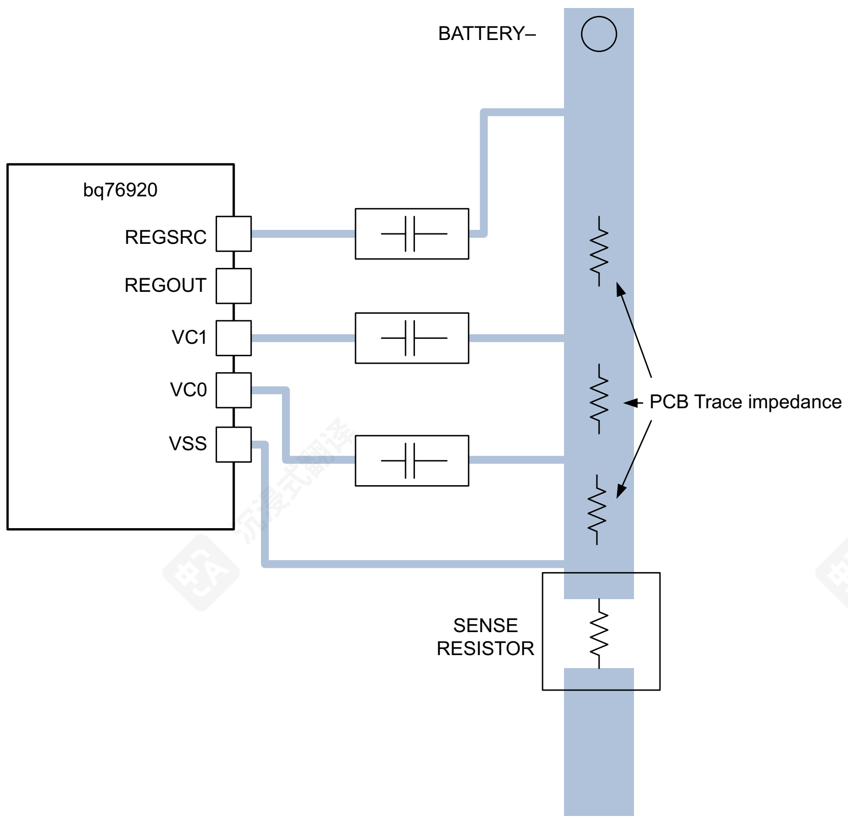


图 10-3.
Weak Layout: Input Capacitor Grounding with High Parasitic PCB Impedance

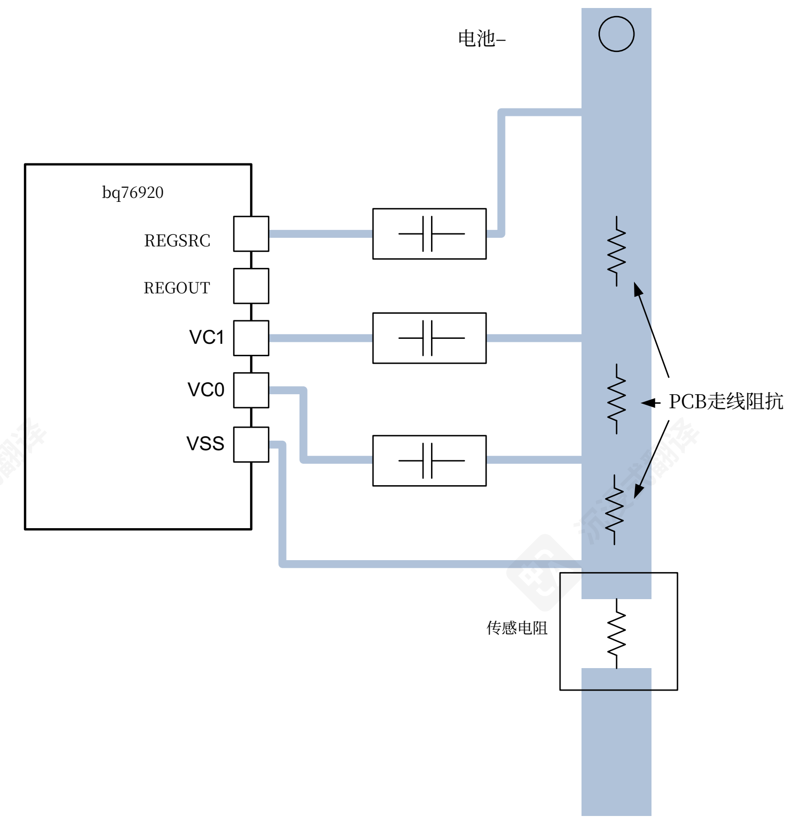


图 10-3.
弱布局：输入电容接地与高寄生 PCB 阻抗

11 器件和文档支持

11.1 文档支持

11.1.1 相关文档

相关文档如下：《*bq76920* 评估模块用户指南》（文献编号：SLVU924）和《bq76920、bq76930、bq76940 AFE FAQ》（文献编号：SLUUB41）。

11.2 相关链接

下面的表格列出了快速访问链接。范围包括技术文档、支持和社区资源、工具和软件，以及样片或购买的快速访问。

表 **11-1.** 相关链接

器件	产品文件夹	样片与购买	技术文档	工具与软件	支持与社区
bq76920	请单击此处	请单击此处	请单击此处	请单击此处	请单击此处
bq76930	请单击此处	请单击此处	请单击此处	请单击此处	请单击此处
bq76940	请单击此处	请单击此处	请单击此处	请单击此处	请单击此处

11.2.1 社区资源

下列链接提供到 TI 社区资源的连接。 链接的内容由各个分销商“按照原样”提供。 这些内容并不构成 TI 技术规范和标准且不一定反映 TI 的观点；请见 TI 的使用条款。

TI E2E™ 在线社区 ***TI* 工程师对工程师 *(E2E)* 社区。** 此社区的创建目的是为了促进工程师之间协作。 在 e2e.ti.com 中，您可以咨询问题、共享知识、探索思路，在同领域工程师的帮助下解决问题。

德州仪器 (TI) 嵌入式处理器维基网站 *德州仪器 (TI) 嵌入式处理器维基网站。* 此网站的建立是为了帮助开发人员从德州仪器 (TI) 的嵌入式处理器入门并且也为了促进与这些器件相关的硬件和软件的总体知识的创新和增长。

11.3 商标

E2E is a trademark of Texas Instruments.
I^2C is a trademark of NXP B.V. Corporation.
All other trademarks are the property of their respective owners.

11.4 静电放电警告

ESD 可能会损坏该集成电路。德州仪器 (TI) 建议通过适当的预防措施处理所有集成电路。如果不遵守正确的处理措施和安装程序，可能会损坏集成电路。

ESD 的损坏小至导致微小的性能降级，大至整个器件故障。 精密的集成电路可能更容易受到损坏，这是因为非常细微的参数更改都可能会导致器件与其发布的规格不相符。

11.5 Glossary

TI Glossary This glossary lists and explains terms, acronyms, and definitions.

12 机械、封装和可订购信息

以下页中包括机械、封装和可订购信息。这些信息是针对指定器件可提供的最新数据。这些数据会在无通知且不对本文档进行修订的情况下发生改变。欲获得该数据表的浏览器版本，请查阅左侧的导航栏。

TEXAS INSTRUMENTS
bq76920, bq76930, bq76940
www.ti.com.cn ZHCSCE2F –2013年10月–修订于2015年11月

11 器件和文档支持

11.1 文档支持

11.1.1 相关文档

相关文档如下:《bq76920 评估模块用户指南》(文献编号:SLVU924)和《bq76920、bq76930、 bq76940AFE FAQ》(文献编号:SLUUB41)。

11.2 相关链接

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bq76920	请单击此处	请单击此处	请单击此处	请单击此处	请单击此处
bq76930	请单击此处	请单击此处	请单击此处	请单击此处	请单击此处
bq76940	请单击此处	请单击此处	请单击此处	请单击此处	请单击此处

11.2.1 社区资源

下列链接提供到 TI 社区资源的连接。 链接的内容由各个分销商“按照原样”提供。 这些内容并不构成 TI 技术 规范和标准且不一定反映 TI的观点;请见 TI 的使用条款。

TI E2E 在线社区 TI 工程师对工程师 (E2E) 社区。 此社区的创建目的是为了促进工程师之间协作。 在 e2e.ti.com 中,您可以咨询问题、共享知识、探索思路,在同领域工程师的帮助下解决问题。

德州仪器 (TI) 嵌入式处理器维基网站 德州仪器 (TI) 嵌入式处理器维基网站。 此网站的建立是为了帮助开发人员从德州仪器 (TI) 的嵌入式处理器入门并且也为了促进与这些器件相关的硬件和软件的总体知识的创新和增长。

11.3 商商标标

E2E 是德州仪器的商标。I^2C 是 NXP B.V. 公司的
商标。所有其他商标均为其各自所有者的财产。
各自所有者。

11.4 静电放电警告

ESD 可能会损坏该集成电路。德州仪器 (TI) 建议通过适当的预防措施处理所有集成电路。如果不遵守正确的处理措施和安装程序，可能会损坏集成电路。

ESD 的损坏小至导致微小的性能降级，大至整个器件故障。精密的集成电路可能更容易受到损坏，这是因为非常细微的参数更改都可能导致器件与其发布的规格不相符。

11.5 术语表

TI 术语表 本术语表列出了术语、缩写和定义，并进行了解释。

12 机械、封装和可订购信息

以下页中包括机械、封装和可订购信息。这些信息是针对指定器件可提供的最新数据。这些数据会在无通知 且不对本文档进行修订的情况下发生改变。欲获得该数据表的浏览器版本,请查阅左侧的导航栏。

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提交文档反馈意见
产品主页链接: bq76920 bq76930 bq76940

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只有那些 TI 特别注明属于军用等级或“增强型塑料”的 TI 组件才是设计或专门用于军事/航空应用或环境的。购买者认可并同 意，对并非指定面向军事或航空航天用途的 TI 组件进行军事或航空航天方面的应用，其风险由客户单独承担，并且由客户独 力负责满足与此类使用相关的所有法律和法规要求。

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	产品		应用
数字音频	www.ti.com.cn/audio	通信与电信	www.ti.com.cn/telecom
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数据转换器	www.ti.com.cn/dataconverters	消费电子	www.ti.com/consumer-apps
DLP® 产品	www.dlp.com	能源	www.ti.com/energy
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微控制器 (MCU)	www.ti.com.cn/microcontrollers		
RFID 系统	www.ti.com.cn/rfidsys		
OMAP应用处理器	www.ti.com/omap		
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数据转换器	www.ti.com.cn/dataconverters	消费电子	www.ti.com/consumer-apps
DLP® 产品	www.dlp.com	能源	www.ti.com/energy
DSP - 数字信号处理器	www.ti.com.cn/dsp	工业应用	www.ti.com.cn/industrial
时钟和计时器	www.ti.com.cn/clockandtimers	医疗电子	www.ti.com.cn/medical
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PACKAGING INFORMATION

Orderable Device	Status (1)	Package Type	Package Drawing	Pins	Package Qty	Eco Plan (2)	Lead/Ball Finish (6)	MSL Peak Temp (3)	Op Temp (°C)	Device Marking (4/5)	Samples
BQ7692000PW	ACTIVE	TSSOP	PW	20	70	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-2-260C-1 YEAR	-40 to 85	BQ7692000	Samples
BQ7692000PWR	ACTIVE	TSSOP	PW	20	2000	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-2-260C-1 YEAR	-40 to 85	BQ7692000	Samples
BQ7692003PW	ACTIVE	TSSOP	PW	20	70	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-2-260C-1 YEAR	-40 to 85	BQ7692003	Samples
BQ7692003PWR	ACTIVE	TSSOP	PW	20	2000	Green (RoHS & no Sb/Br)	CU NIPDAU | Call TI	Level-2-260C-1 YEAR	-40 to 85	BQ7692003	Samples
BQ7692006PW	ACTIVE	TSSOP	PW	20	70	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-2-260C-1 YEAR	-40 to 85	BQ7692006	Samples
BQ7692006PWR	ACTIVE	TSSOP	PW	20	2000	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-2-260C-1 YEAR	-40 to 85	BQ7692006	Samples
BQ7693000DBT	ACTIVE	TSSOP	DBT	30	60	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-2-260C-1 YEAR	-40 to 85	BQ7693000	Samples
BQ7693000DBTR	ACTIVE	TSSOP	DBT	30	2000	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-2-260C-1 YEAR	-40 to 85	BQ7693000	Samples
BQ7693001DBT	ACTIVE	TSSOP	DBT	30	60	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-2-260C-1 YEAR	-40 to 85	BQ7693001	Samples
BQ7693001DBTR	ACTIVE	TSSOP	DBT	30	2000	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-2-260C-1 YEAR	-40 to 85	BQ7693001	Samples
BQ7693002DBT	ACTIVE	TSSOP	DBT	30	60	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-2-260C-1 YEAR	-40 to 85	BQ7693002	Samples
BQ7693002DBTR	ACTIVE	TSSOP	DBT	30	2000	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-2-260C-1 YEAR	-40 to 85	BQ7693002	Samples
BQ7693003DBT	ACTIVE	TSSOP	DBT	30	60	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-2-260C-1 YEAR	-40 to 85	BQ7693003	Samples
BQ7693003DBTR	ACTIVE	TSSOP	DBT	30	2000	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-2-260C-1 YEAR	-40 to 85	BQ7693003	Samples
BQ7693006DBT	ACTIVE	TSSOP	DBT	30	60	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-2-260C-1 YEAR	-40 to 85	BQ7693006	Samples
BQ7693006DBTR	ACTIVE	TSSOP	DBT	30	2000	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-2-260C-1 YEAR	-40 to 85	BQ7693006	Samples
BQ7693007DBT	ACTIVE	TSSOP	DBT	30	60	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-2-260C-1 YEAR	-40 to 85	BQ7693007	Samples

包装信息

可订购器件	状态 (1)	封装类型 封装	图纸	引脚	Qty	环保方案 (2)	引脚/球焊工艺 (6)	MSL峰值温度 (3)	操作温度(°C)	设备标识 (4/5)	样品
BQ7692000PW	激活	TSSOP	PW	20	70	绿色（符合RoHS且不含锑/溴)	铜NIPDAU	二级-260C-1年	-40至85	BQ7692000	Samples
BQ7692000PWR	活跃	TSSOP	PW	20	2000	绿色（符合RoHS且不含锑/溴)	CU NIPDAU	Level-2-260C-1 年	-40 至 85	BQ7692000	Samples
BQ7692003PW	激活	TSSOP	PW	20	70	绿色（符合RoHS &不含锑/溴)	铜NIPDAU	二级-260C-1年	-40至85	BQ7692003	Samples
BQ7692003PWR	激活	TSSOP	PW	20	2000	绿色（符合RoHS 且不含Sb/Br)	铜NIPDAU | 联系TI	Level-2-260C-1 年	-40 至 85	BQ7692003	Samples
BQ7692006PW	激活	TSSOP	PW	20	70	绿色（符合RoHS &不含Sb/Br)	CU NIPDAU	二级-260C-1年	-40至85	BQ7692006	Samples
BQ7692006PWR	激活	TSSOP	PW	20	2000	绿色（无铅RoHS 且不含锑/溴)	铜NIPDAU	Level-2-260C-1 年	-40 至 85	BQ7692006	Samples
BQ7693000DBT	激活	TSSOP	DBT	30	60	绿色（符合RoHS 且不含Sb/Br)	CU NIPDAU	二级-260C-1年	-40至85	BQ7693000	Samples
BQ7693000DBTR	激活	TSSOP	DBT	30	2000	绿色（无铅RoHS 且不含锑/溴)	铜NIPDAU	Level-2-260C-1 年	-40 至 85	BQ7693000	Samples
BQ7693001DBT	活跃	TSSOP	DBT	30	60	绿色（符合RoHS &不含锑/溴)	铜NIPDAU	二级-260C-1年	-40至85	BQ7693001	Samples
BQ7693001DBTR	活跃	TSSOP	DBT	30	2000	绿色（符合RoHS标准 且不含锑/溴)	铜NIPDAU	Level-2-260C-1 年	-40 至 85	BQ7693001	Samples
BQ7693002DBT	活跃	TSSOP	DBT	30	60	绿色（符合RoHS &不含Sb/Br)	CU NIPDAU	二级-260C-1年	-40至85	BQ7693002	Samples
BQ7693002DBTR	活跃	TSSOP	DBT	30	2000	绿色（符合RoHS标准 且不含锑/溴)	CU NIPDAU	Level-2-260C-1 年	-40 至 85	BQ7693002	Samples
BQ7693003DBT	激活	TSSOP	DBT	30	60	绿色（符合RoHS &不含Sb/Br)	CU NIPDAU	二级-260C-1年	-40至85	BQ7693003	Samples
BQ7693003DBTR	激活	TSSOP	DBT	30	2000	绿色（符合RoHS且不含锑/溴)	CU NIPDAU	Level-2-260C-1年	-40 至 85	BQ7693003	Samples
BQ7693006DBT	活跃	TSSOP	DBT	30	60	绿色（符合RoHS且不含锑/溴)	CU NIPDAU	Level-2-260C-1 年	-40 至 85	BQ7693006	Samples
BQ7693006DBTR	激活	TSSOP	DBT	30	2000	绿色（符合RoHS且不含锑/溴)	CU NIPDAU	Level-2-260C-1年	-40至85	BQ7693006	Samples
BQ7693007DBT	活跃	TSSOP	DBT	30	60	绿色（符合RoHS且不含锑/溴)	CU NIPDAU	二级-260C-1年	-40至85	BQ7693007	Samples

Orderable Device	Status (1)	Package Type	Package Drawing	Pins	Package Qty	Eco Plan (2)	Lead/Ball Finish (6)	MSL Peak Temp (3)	Op Temp (°C)	Device Marking (4/5)	Samples
BQ7693007DBTR	ACTIVE	TSSOP	DBT	30	2000	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-2-260C-1 YEAR	-40 to 85	BQ7693007	Samples
BQ7694000DBT	ACTIVE	TSSOP	DBT	44	40	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-2-260C-1 YEAR	-40 to 85	BQ7694000	Samples
BQ7694000DBTR	ACTIVE	TSSOP	DBT	44	2000	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-2-260C-1 YEAR	-40 to 85	BQ7694000	Samples
BQ7694001DBT	ACTIVE	TSSOP	DBT	44	40	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-2-260C-1 YEAR	-40 to 85	BQ7694001	Samples
BQ7694001DBTR	ACTIVE	TSSOP	DBT	44	2000	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-2-260C-1 YEAR	-40 to 85	BQ7694001	Samples
BQ7694002DBT	ACTIVE	TSSOP	DBT	44	40	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-2-260C-1 YEAR	-40 to 85	BQ7694002	Samples
BQ7694002DBTR	ACTIVE	TSSOP	DBT	44	2000	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-2-260C-1 YEAR	-40 to 85	BQ7694002	Samples
BQ7694003DBT	ACTIVE	TSSOP	DBT	44	40	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-2-260C-1 YEAR	-40 to 85	BQ7694003	Samples
BQ7694003DBTR	ACTIVE	TSSOP	DBT	44	2000	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-2-260C-1 YEAR	-40 to 85	BQ7694003	Samples
BQ7694006DBT	ACTIVE	TSSOP	DBT	44	40	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-2-260C-1 YEAR	-40 to 85	BQ7694006	Samples
BQ7694006DBTR	ACTIVE	TSSOP	DBT	44	2000	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-2-260C-1 YEAR	-40 to 85	BQ7694006	Samples

(1) The marketing status values are defined as follows:
ACTIVE: Product device recommended for new designs.
LIFEBUY: TI has announced that the device will be discontinued, and a lifetime-buy period is in effect.
NRND: Not recommended for new designs. Device is in production to support existing customers, but TI does not recommend using this part in a new design.
PREVIEW: Device has been announced but is not in production. Samples may or may not be available.
OBSOLETE: TI has discontinued the production of the device.

(2) **RoHS:** TI defines "RoHS" to mean semiconductor products that are compliant with the current EU RoHS requirements for all 10 RoHS substances, including the requirement that RoHS substance do not exceed 0.1% by weight in homogeneous materials. Where designed to be soldered at high temperatures, "RoHS" products are suitable for use in specified lead-free processes. TI may reference these types of products as "Pb-Free".
RoHS Exempt: TI defines "RoHS Exempt" to mean products that contain lead but are compliant with EU RoHS pursuant to a specific EU RoHS exemption.
Green: TI defines "Green" to mean the content of Chlorine (Cl) and Bromine (Br) based flame retardants meet JS709B low halogen requirements of <=1000ppm threshold. Antimony trioxide based flame retardants must also meet the <=1000ppm threshold requirement.

(3) MSL, Peak Temp. - The Moisture Sensitivity Level rating according to the JEDEC industry standard classifications, and peak solder temperature.

可选设备	状态 (1)	包装类型 包装	图纸	引脚	Qty	环保计划 (2)	引线/球焊工艺 (6)	MSL峰值温度 (3)	开路温度 (°C)	设备标识 (4/5)	样品
BQ7693007DBTR	激活	TSSOP	DBT	30	2000	绿色（符合RoHS 且不含Sb/Br）	铜NIPDAU	二级-260C-1年	-40至85	BQ7693007	Samples
BQ7694000DBT	ACTIVE	TSSOP	DBT	44	40	绿色（符合RoHS 且不含Sb/Br）	CU NIPDAU	Level-2-260C-1 年	-40 至 85	BQ7694000	Samples
BQ7694000DBTR	激活	TSSOP	DBT	44	2000	绿色（符合RoHS 且不含Sb/Br）	CU NIPDAU	二级-260C-1年	-40至85	BQ7694000	Samples
BQ7694001DBT	ACTIVE	TSSOP	DBT	44	40	绿色 (RoHS) & 不含Sb/Br)	CU NIPDAU	Level-2-260C-1 年	-40 至 85	BQ7694001	Samples
BQ7694001DBTR	激活	TSSOP	DBT	44	2000	绿色（符合RoHS 且不含锑/溴）	铜NIPDAU	二级-260C-1年	-40至85	BQ7694001	Samples
BQ7694002DBT	ACTIVE	TSSOP	DBT	44	40	绿色（符合RoHS 且不含Sb/Br）	CU NIPDAU	Level-2-260C-1 年	-40 至 85	BQ7694002	Samples
BQ7694002DBTR	激活	TSSOP	DBT	44	2000	绿色（符合RoHS 且不含Sb/Br）	铜NIPDAU	二级-260C-1年	-40至85	BQ7694002	Samples
BQ7694003DBT	ACTIVE	TSSOP	DBT	44	40	绿色（符合Ro HS且不含锑/溴）	CU NIPDAU	Level-2-260C-1 年	-40 至 85	BQ7694003	Samples
BQ7694003DBTR	活跃	TSSOP	DBT	44	2000	绿色（符合Ro HS且不含Sb/Br）	CU NIPDAU	Level-2-260C-1 年	-40至85	BQ7694003	Samples
BQ7694006DBT	激活	TSSOP	DBT	44	40	绿色（符合Ro HS且不含锑/溴）	CU NIPDAU	Level-2-260C-1年	-40至85	BQ7694006	Samples
BQ7694006DBTR	ACTIVE	TSSOP	DBT	44	2000	绿色（符合Ro HS且不含锑/溴）	CU NIPDAU	Level-2-260C-1 年	-40 至 85	BQ7694006	Samples

(1) 市场状态值定义如下：ACTIVE：推荐用于新设计的产品设备

gns.

LIFEBUY：TI 宣布该设备将停产，并实施终身购买期。
NRND：不推荐用于新设计。该设备仍在生产以支持现有客户，但 TI 不建议在新设计中使用此元件。
预览：设备已发布但未投入生产。样品可能提供也可能不提供。
已淘汰：TI已停止该设备的生产品种。

(2) 环境无害化指令（RoHS）：TI将“RoHS”定义为符合当前欧盟RoHS指令要求的半导体产品，涵盖所有10种RoHS物质的合规性，包括RoHS物质在均匀材料中的重量含量不得超过0.1%的要求。若设计用于高温焊接，“RoHS”产品适用于指定的无铅工艺。TI可能将这些类型的产品称为“无铅”。

RoHS豁免：TI将“RoHS豁免”定义为含有铅但符合欧盟RoHS指令特定豁免要求的产品。
环保：TI将“环保”定义为氯化物（Cl）和溴化物（Br）基阻燃剂含量满足JS709B低卤素要求，即<=1000ppm阈值。三氧化二锑基阻燃剂也必须满足<=1000ppm阈值要求。

(3) MSL, 峰值温度 - 根据JEDEC行业标准分类的湿度敏感性等级，以及峰值焊料温度。

(4) There may be additional marking, which relates to the logo, the lot trace code information, or the environmental category on the device.

(5) Multiple Device Markings will be inside parentheses. Only one Device Marking contained in parentheses and separated by a "~" will appear on a device. If a line is indented then it is a continuation of the previous line and the two combined represent the entire Device Marking for that device.

(6) Lead/Ball Finish - Orderable Devices may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead/Ball Finish values may wrap to two lines if the finish value exceeds the maximum column width.

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(4) 可能存在其他标记，这可能与标志、批号追溯代码信息或设备上的环境类别有关。

(5) 多设备标记将被括号括起来。括号内只包含一个设备标记，并通过 "~" 分隔，该标记将出现在设备上。如果一行缩进，则它是前一行的一部分，这两行合并代表该设备的整个设备标记。

(6) 引脚/球面完成 - 可订购设备可能有多种材料完成选项。完成选项由垂直划线分隔。引脚/球面完成值如果超出最大列宽，可能会换到两行。

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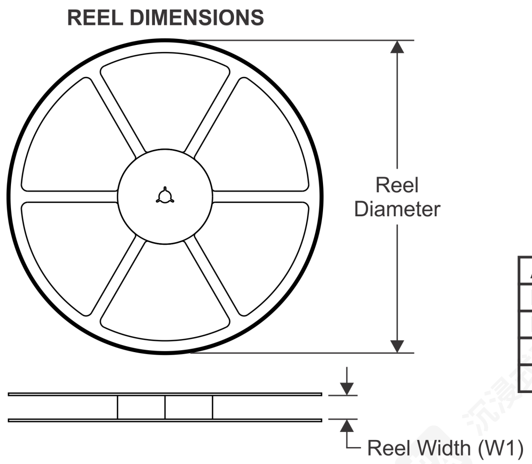
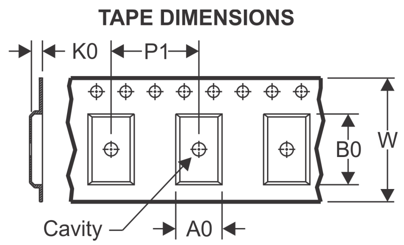
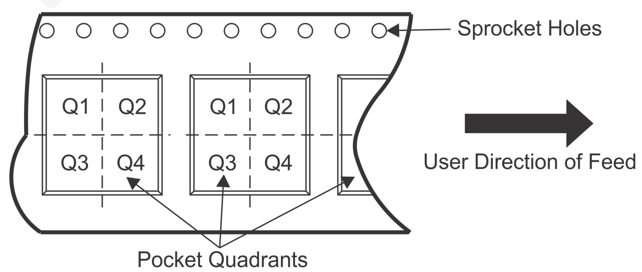
TAPE AND REEL INFORMATION





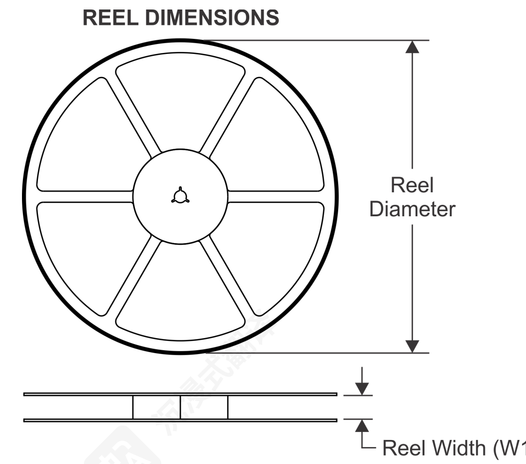
A0	Dimension designed to accommodate the component width
B0	Dimension designed to accommodate the component length
K0	Dimension designed to accommodate the component thickness
W	Overall width of the carrier tape
P1	Pitch between successive cavity centers

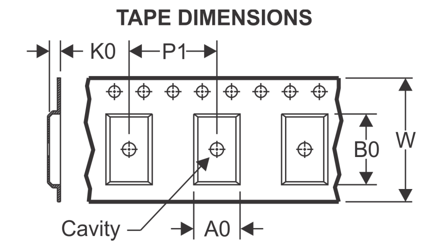


*All dimensions are nominal

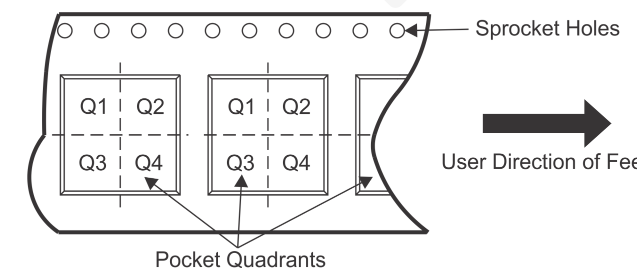
Device	Package Type	Package Drawing	Pins	SPQ	Reel Diameter (mm)	Reel Width W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	Pin1 Quadrant
BQ7692000PWR	TSSOP	PW	20	2000	330.0	16.4	6.95	7.1	1.6	8.0	16.0	Q1
BQ7692003PWR	TSSOP	PW	20	2000	330.0	16.4	6.95	7.1	1.6	8.0	16.0	Q1
BQ7692006PWR	TSSOP	PW	20	2000	330.0	16.4	6.95	7.1	1.6	8.0	16.0	Q1
BQ7693000DBTR	TSSOP	DBT	30	2000	330.0	16.4	6.95	8.3	1.6	8.0	16.0	Q1
BQ7693001DBTR	TSSOP	DBT	30	2000	330.0	16.4	6.95	8.3	1.6	8.0	16.0	Q1
BQ7693002DBTR	TSSOP	DBT	30	2000	330.0	16.4	6.95	8.3	1.6	8.0	16.0	Q1
BQ7693003DBTR	TSSOP	DBT	30	2000	330.0	16.4	6.95	8.3	1.6	8.0	16.0	Q1
BQ7693006DBTR	TSSOP	DBT	30	2000	330.0	16.4	6.95	8.3	1.6	8.0	16.0	Q1
BQ7693007DBTR	TSSOP	DBT	30	2000	330.0	16.4	6.95	8.3	1.6	8.0	16.0	Q1
BQ7694000DBTR	TSSOP	DBT	44	2000	330.0	24.4	6.8	11.7	1.6	12.0	24.0	Q1
BQ7694001DBTR	TSSOP	DBT	44	2000	330.0	24.4	6.8	11.7	1.6	12.0	24.0	Q1
BQ7694002DBTR	TSSOP	DBT	44	2000	330.0	24.4	6.8	11.7	1.6	12.0	24.0	Q1
BQ7694003DBTR	TSSOP	DBT	44	2000	330.0	24.4	6.8	11.7	1.6	12.0	24.0	Q1
BQ7694006DBTR	TSSOP	DBT	44	2000	330.0	24.4	6.8	11.7	1.6	12.0	24.0	Q1

磁带和卷轴信息





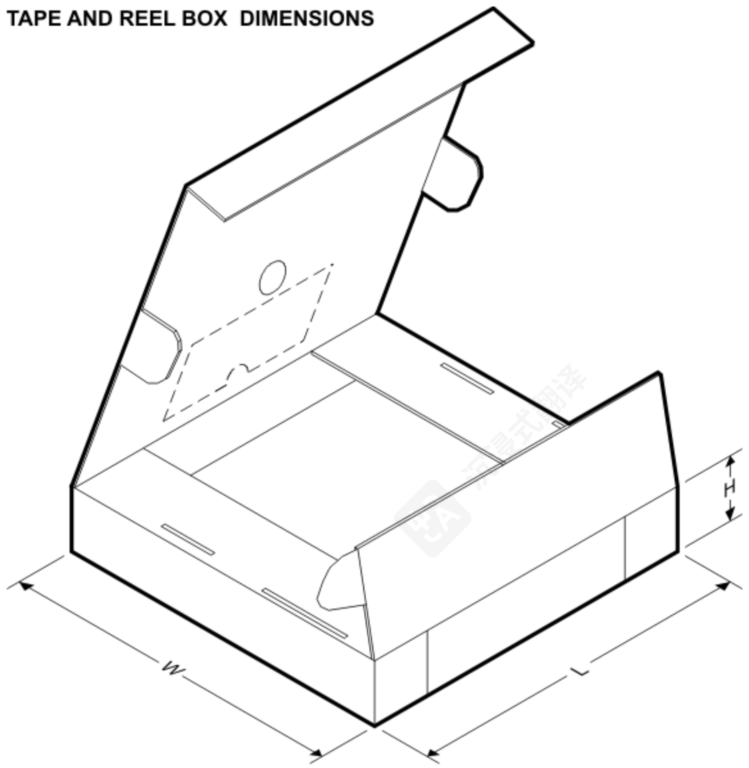
A0	Dimension designed to accommodate the component width
B0	Dimension designed to accommodate the component length
K0	Dimension designed to accommodate the component thickness
W	Overall width of the carrier tape
P1	Pitch between successive cavity centers



*所有尺寸均为标称值

设备	包装 Type	包装 图纸	Pins	SPQ	卷盘直径宽度 (mm) W1 (mm)		A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	Pin1 象限
BQ7692000PWR	TSSOP	PW	20	2000	330.0	16.4	6.95	7.1	1.6	8.0	16.0	Q1
BQ7692003PWR	TSSOP	PW	20	2000	330.0	16.4	6.95	7.1	1.6	8.0	16.0	Q1
BQ7692006PWR	TSSOP	PW	20	2000	330.0	16.4	6.95	7.1	1.6	8.0	16.0	Q1
BQ7693000DBTR	TSSOP	DBT	30	2000	330.0	16.4	6.95	8.3	1.6	8.0	16.0	Q1
BQ7693001DBTR	TSSOP	DBT	30	2000	330.0	16.4	6.95	8.3	1.6	8.0	16.0	Q1
BQ7693002DBTR	TSSOP	DBT	30	2000	330.0	16.4	6.95	8.3	1.6	8.0	16.0	Q1
BQ7693003DBTR	TSSOP	DBT	30	2000	330.0	16.4	6.95	8.3	1.6	8.0	16.0	Q1
BQ7693006DBTR	TSSOP	DBT	30	2000	330.0	16.4	6.95	8.3	1.6	8.0	16.0	Q1
BQ7693007DBTR	TSSOP	DBT	30	2000	330.0	16.4	6.95	8.3	1.6	8.0	16.0	Q1
BQ7694000DBTR	TSSOP	DBT	44	2000	330.0	24.4	6.8	11.7	1.6	12.0	24.0	Q1
BQ7694001DBTR	TSSOP	DBT	44	2000	330.0	24.4	6.8	11.7	1.6	12.0	24.0	Q1
BQ7694002DBTR	TSSOP	DBT	44	2000	330.0	24.4	6.8	11.7	1.6	12.0	24.0	Q1
BQ7694003DBTR	TSSOP	DBT	44	2000	330.0	24.4	6.8	11.7	1.6	12.0	24.0	Q1
BQ7694006DBTR	TSSOP	DBT	44	2000	330.0	24.4	6.8	11.7	1.6	12.0	24.0	Q1

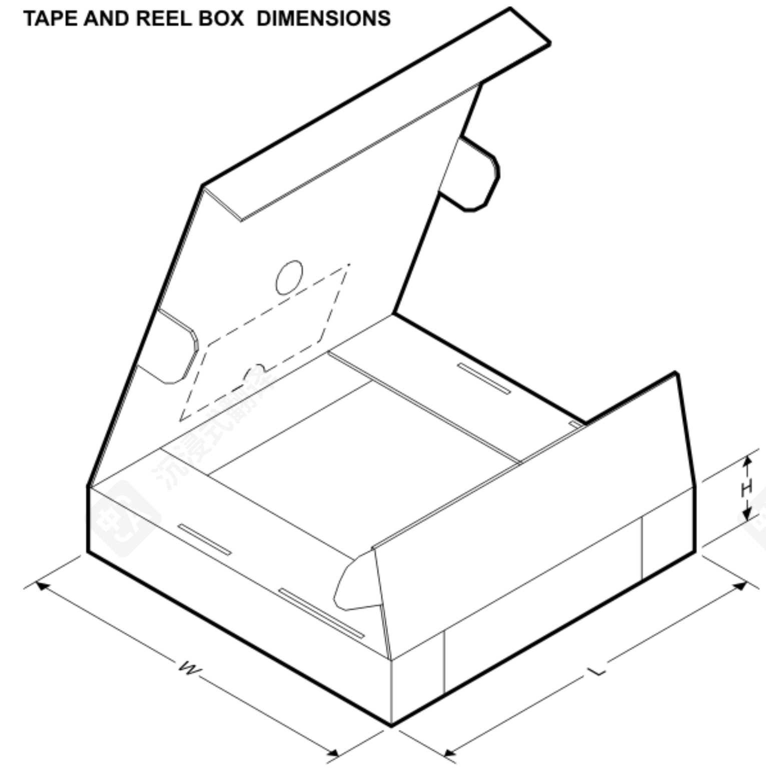
TAPE AND REEL BOX DIMENSIONS



*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Length (mm)	Width (mm)	Height (mm)
BQ7692000PWR	TSSOP	PW	20	2000	350.0	350.0	43.0
BQ7692003PWR	TSSOP	PW	20	2000	350.0	350.0	43.0
BQ7692006PWR	TSSOP	PW	20	2000	350.0	350.0	43.0
BQ7693000DBTR	TSSOP	DBT	30	2000	350.0	350.0	43.0
BQ7693001DBTR	TSSOP	DBT	30	2000	350.0	350.0	43.0
BQ7693002DBTR	TSSOP	DBT	30	2000	350.0	350.0	43.0
BQ7693003DBTR	TSSOP	DBT	30	2000	350.0	350.0	43.0
BQ7693006DBTR	TSSOP	DBT	30	2000	350.0	350.0	43.0
BQ7693007DBTR	TSSOP	DBT	30	2000	350.0	350.0	43.0
BQ7694000DBTR	TSSOP	DBT	44	2000	350.0	350.0	43.0
BQ7694001DBTR	TSSOP	DBT	44	2000	350.0	350.0	43.0
BQ7694002DBTR	TSSOP	DBT	44	2000	350.0	350.0	43.0
BQ7694003DBTR	TSSOP	DBT	44	2000	350.0	350.0	43.0
BQ7694006DBTR	TSSOP	DBT	44	2000	350.0	350.0	43.0

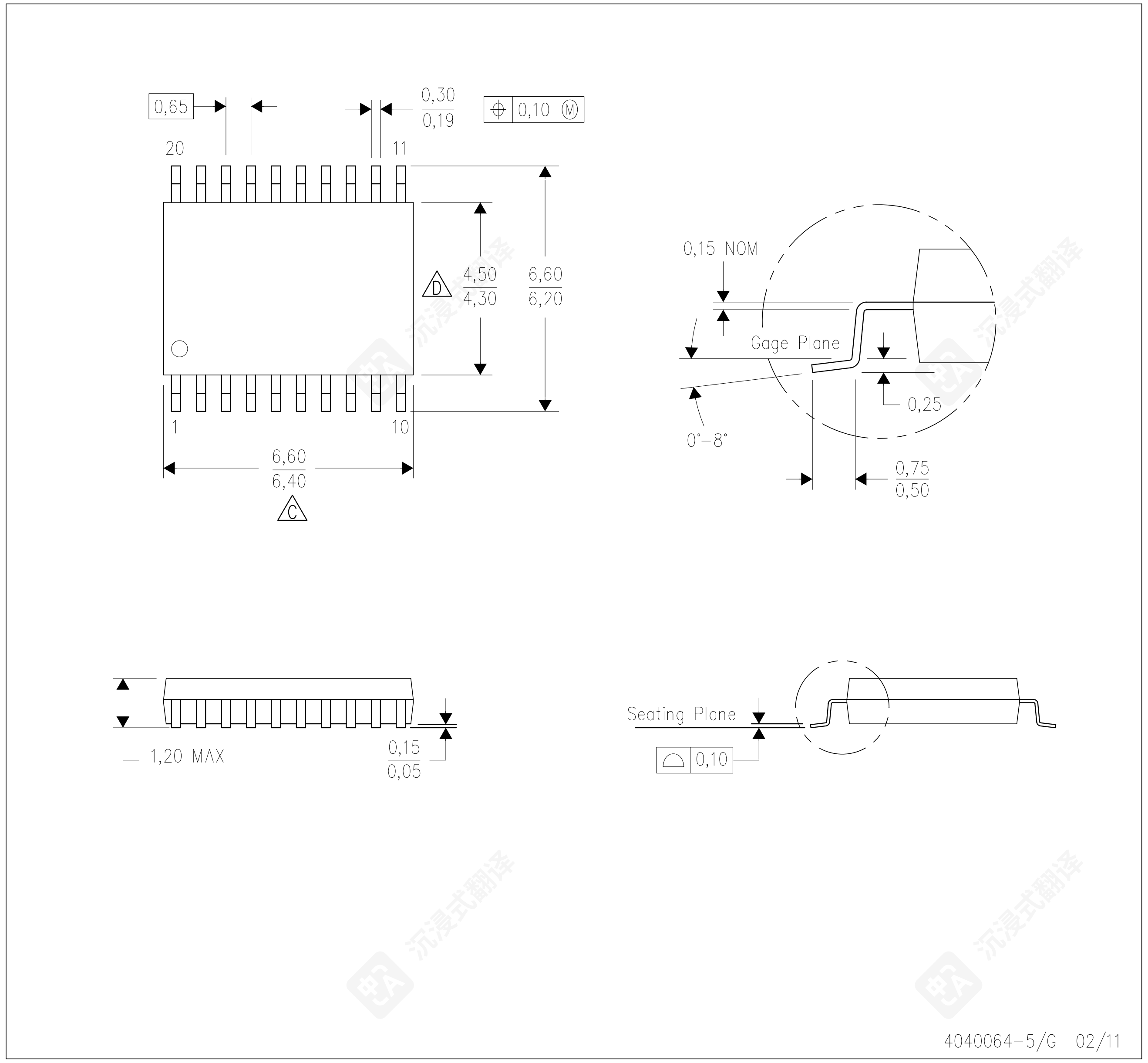
TAPE AND REEL BOX DIMENSIONS



*所有尺寸均为标称值

设备	封装类型	封装图纸	Pins	SPQ	长度 (mm)	宽度 (mm)	高度 (mm)
BQ7692000PWR	TSSOP	PW	20	2000	350.0	350.0	43.0
BQ7692003PWR	TSSOP	PW	20	2000	350.0	350.0	43.0
BQ7692006PWR	TSSOP	PW	20	2000	350.0	350.0	43.0
BQ7693000DBTR	TSSOP	DBT	30	2000	350.0	350.0	43.0
BQ7693001DBTR	TSSOP	DBT	30	2000	350.0	350.0	43.0
BQ7693002DBTR	TSSOP	DBT	30	2000	350.0	350.0	43.0
BQ7693003DBTR	TSSOP	DBT	30	2000	350.0	350.0	43.0
BQ7693006DBTR	TSSOP	DBT	30	2000	350.0	350.0	43.0
BQ7693007DBTR	TSSOP	DBT	30	2000	350.0	350.0	43.0
BQ7694000DBTR	TSSOP	DBT	44	2000	350.0	350.0	43.0
BQ7694001DBTR	TSSOP	DBT	44	2000	350.0	350.0	43.0
BQ7694002DBTR	TSSOP	DBT	44	2000	350.0	350.0	43.0
BQ7694003DBTR	TSSOP	DBT	44	2000	350.0	350.0	43.0
BQ7694006DBTR	TSSOP	DBT	44	2000	350.0	350.0	43.0

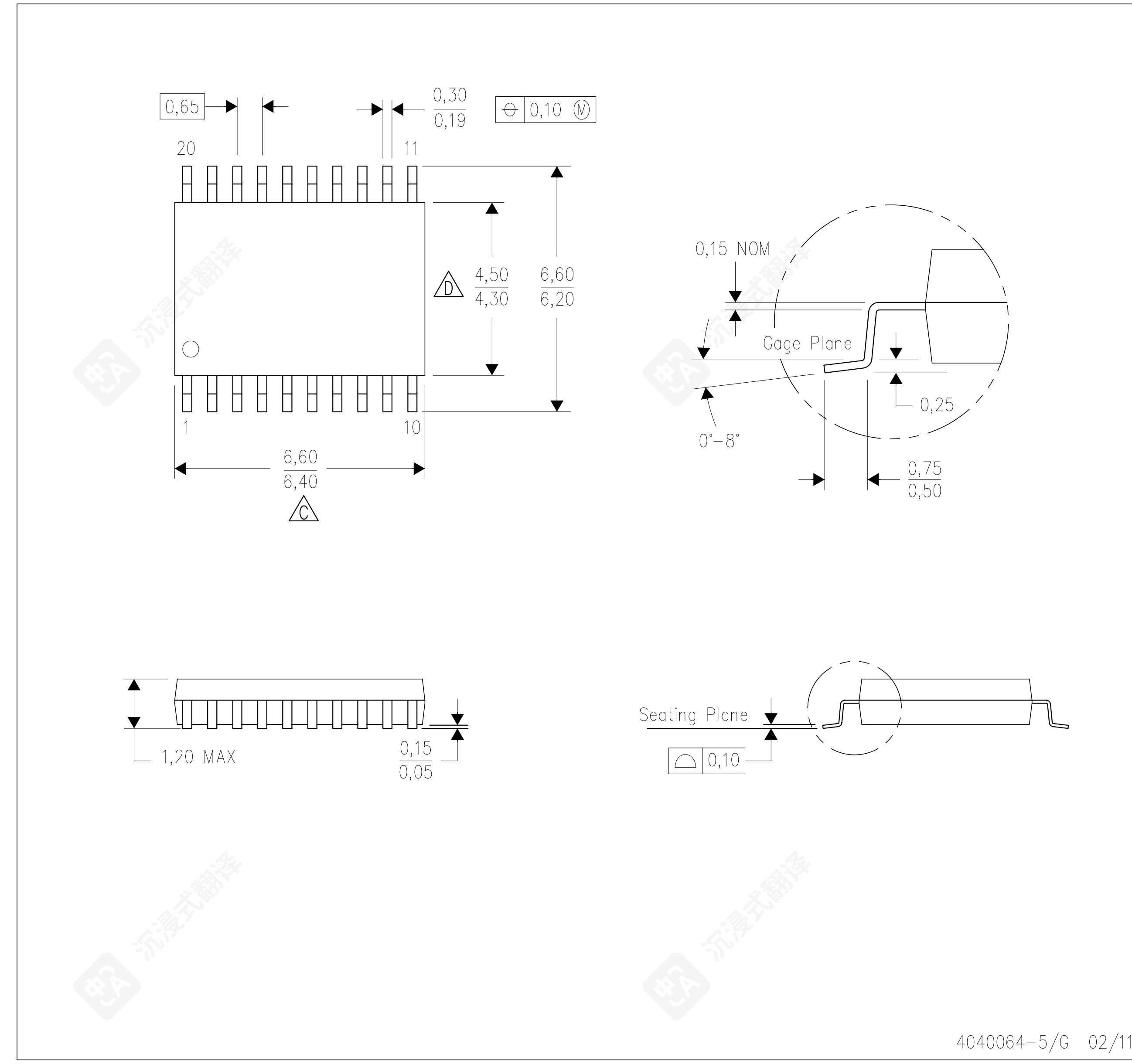
PW (R-PDSO-G20) PLASTIC SMALL OUTLINE



NOTES: A. All linear dimensions are in millimeters. Dimensioning and tolerancing per ASME Y14.5M–1994.
B. This drawing is subject to change without notice.
C. Body length does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0,15 each side.
D. Body width does not include interlead flash. Interlead flash shall not exceed 0,25 each side.
E. Falls within JEDEC MO-153

MECHANICAL DATA

PW (R-PDSO-G20) PLASTIC SMALL OUTLINE



NOTES: A. All linear dimensions are in millimeters. Dimensioning and tolerancing per ASME Y14.5M–1994.
B. This drawing is subject to change without notice.
C. Body length does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0,15 each side.
D. Body width does not include interlead flash. Interlead flash shall not exceed 0,25 each side.
E. Falls within JEDEC MO-153

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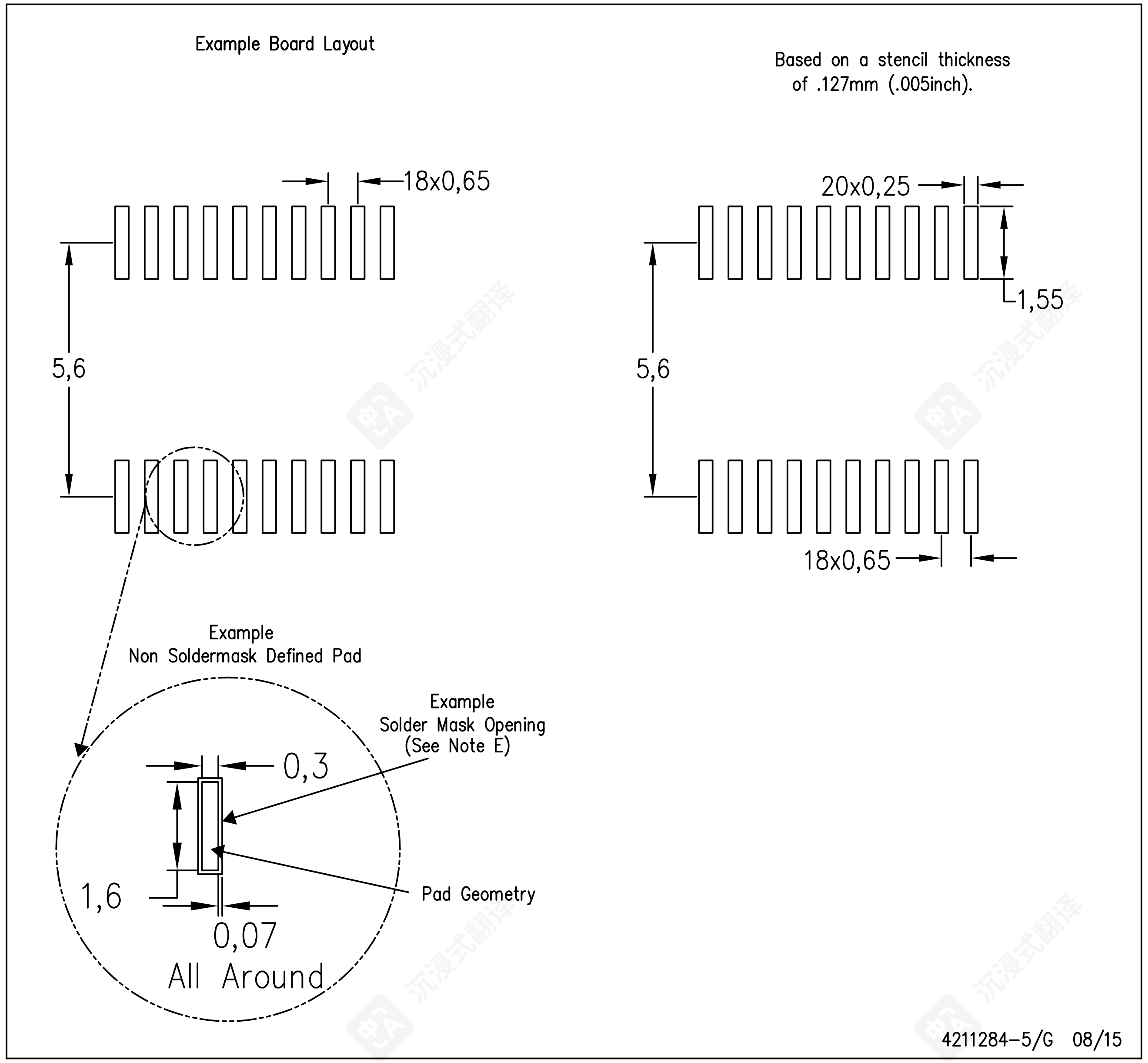
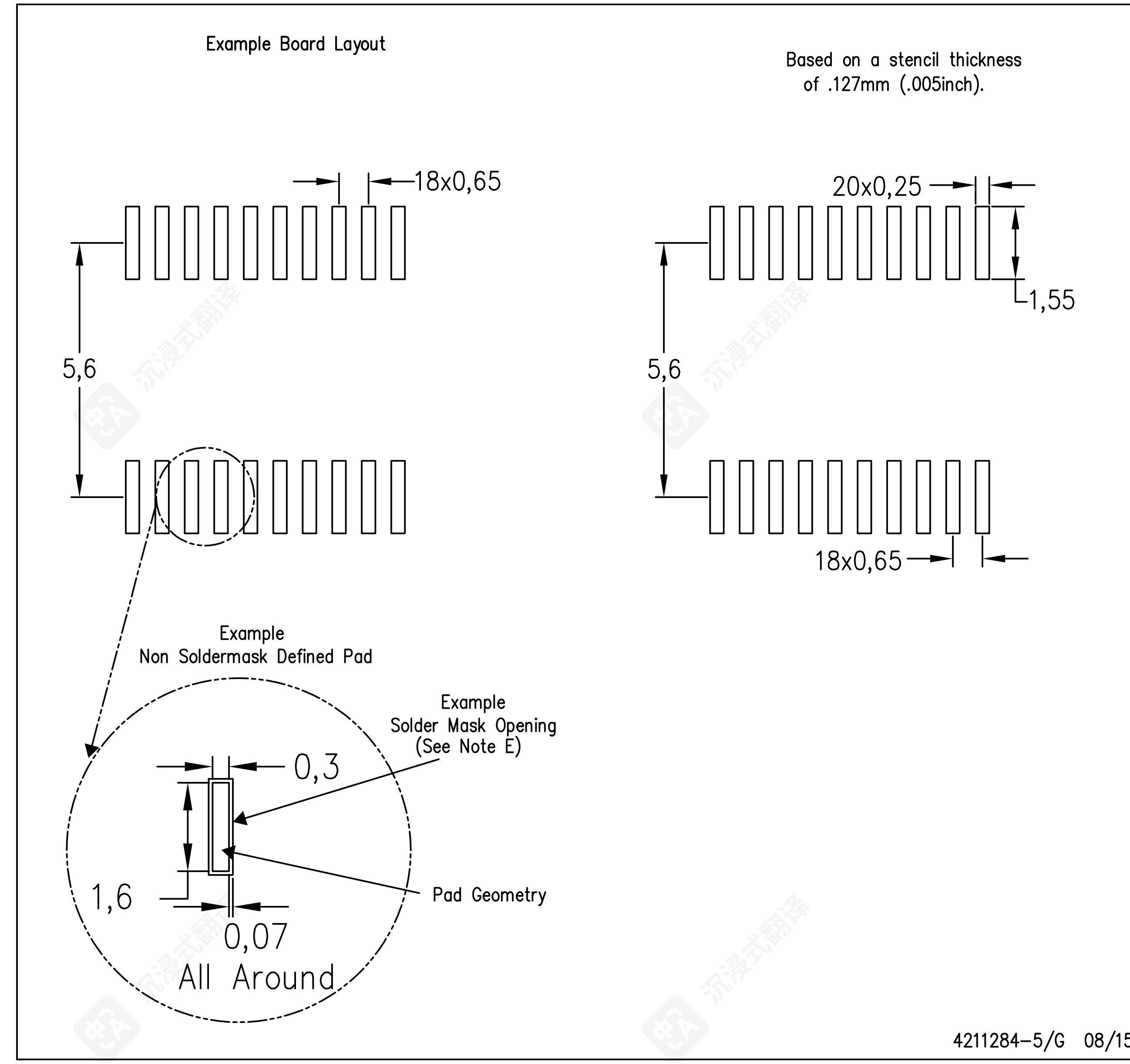
PW (R–PDSO–G20) PLASTIC SMALL OUTLINE



NOTES:
A. All linear dimensions are in millimeters.
B. This drawing is subject to change without notice.
C. Publication IPC–7351 is recommended for alternate design.
D. Laser cutting apertures with trapezoidal walls and also rounding corners will offer better paste release. Customers should contact their board assembly site for stencil design recommendations. Refer to IPC–7525 for other stencil recommendations.
E. Customers should contact their board fabrication site for solder mask tolerances between and around signal pads.

LAND PATTERN DATA

PW (R–PDSO–G20) PLASTIC SMALL OUTLINE



NOTES:
A. All linear dimensions are in millimeters.
B. This drawing is subject to change without notice.
C. Publication IPC–7351 is recommended for alternate design.
D. Laser cutting apertures with trapezoidal walls and also rounding corners will offer better paste release. Customers should contact their board assembly site for stencil design recommendations. Refer to IPC–7525 for other stencil recommendations.
E. Customers should contact their board fabrication site for solder mask tolerances between and around signal pads.

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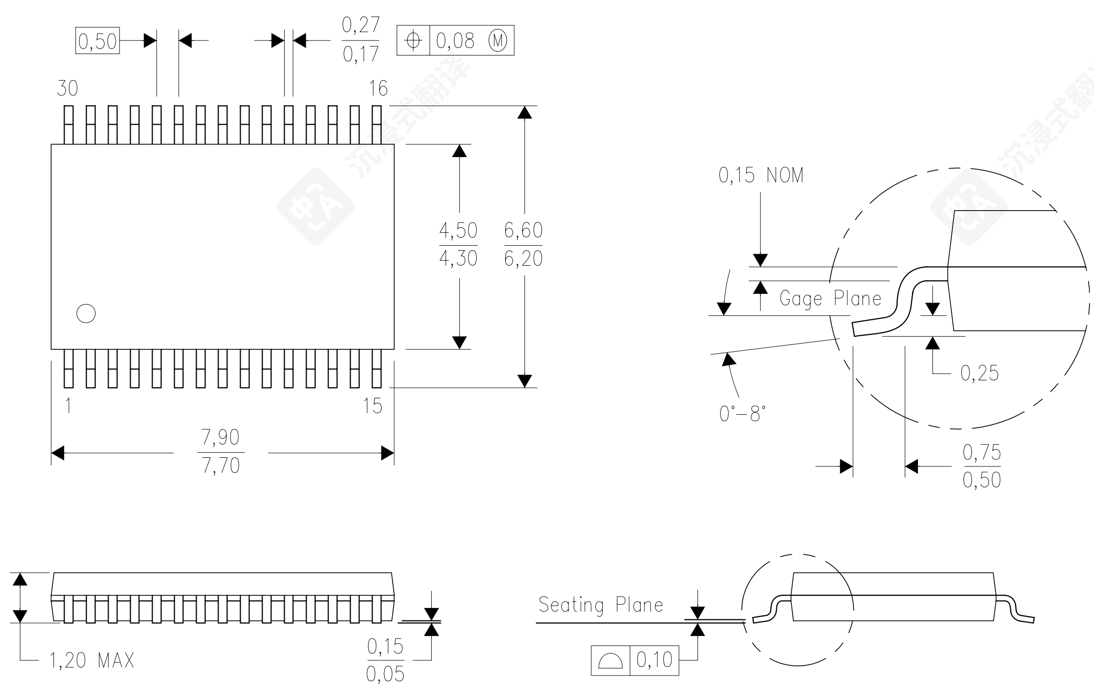
DBT (R–PDSO–G30) PLASTIC SMALL OUTLINE

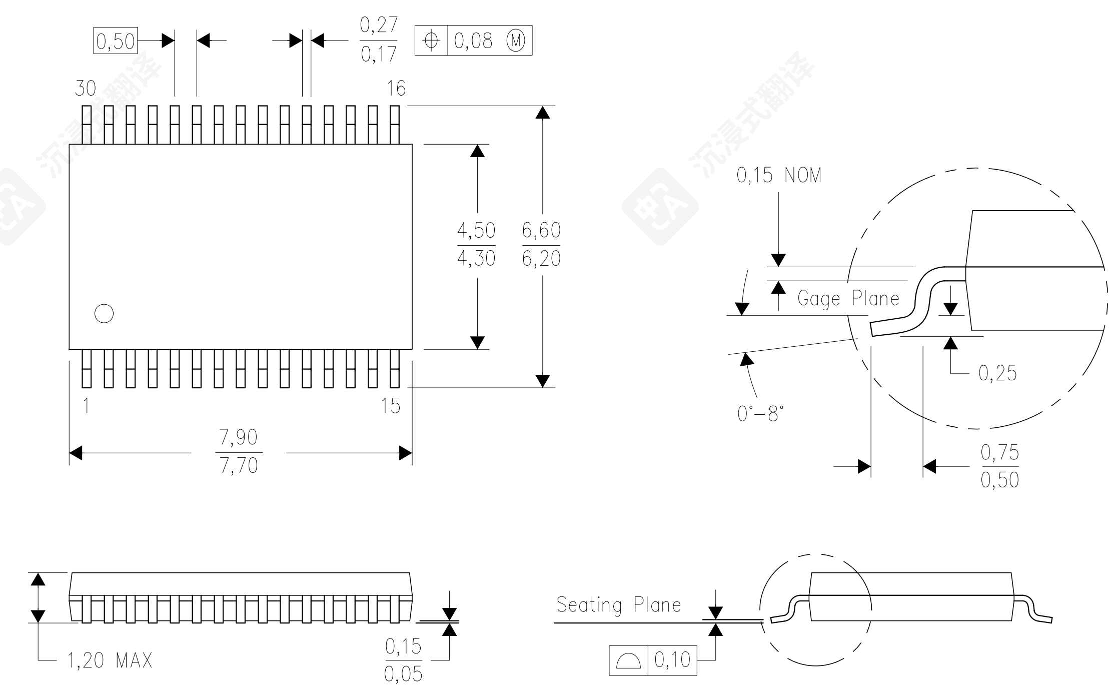


4073252-5/I 03/11

NOTES:
A. All linear dimensions are in millimeters. Dimensioning and tolerancing per ASME Y14.5M–1994.
B. This drawing is subject to change without notice.
C. Body dimensions do not include mold flash or protrusion.
D. Falls within JEDEC MO–153.

MECHANICAL DATA

DBT (R–PDSO–G30) PLASTIC SMALL OUTLINE



4073252-5/I 03/11

NOTES:
A. All linear dimensions are in millimeters. Dimensioning and tolerancing per ASME Y14.5M–1994.
B. This drawing is subject to change without notice.
C. Body dimensions do not include mold flash or protrusion.
D. Falls within JEDEC MO–153.

TEXAS INSTRUMENTS
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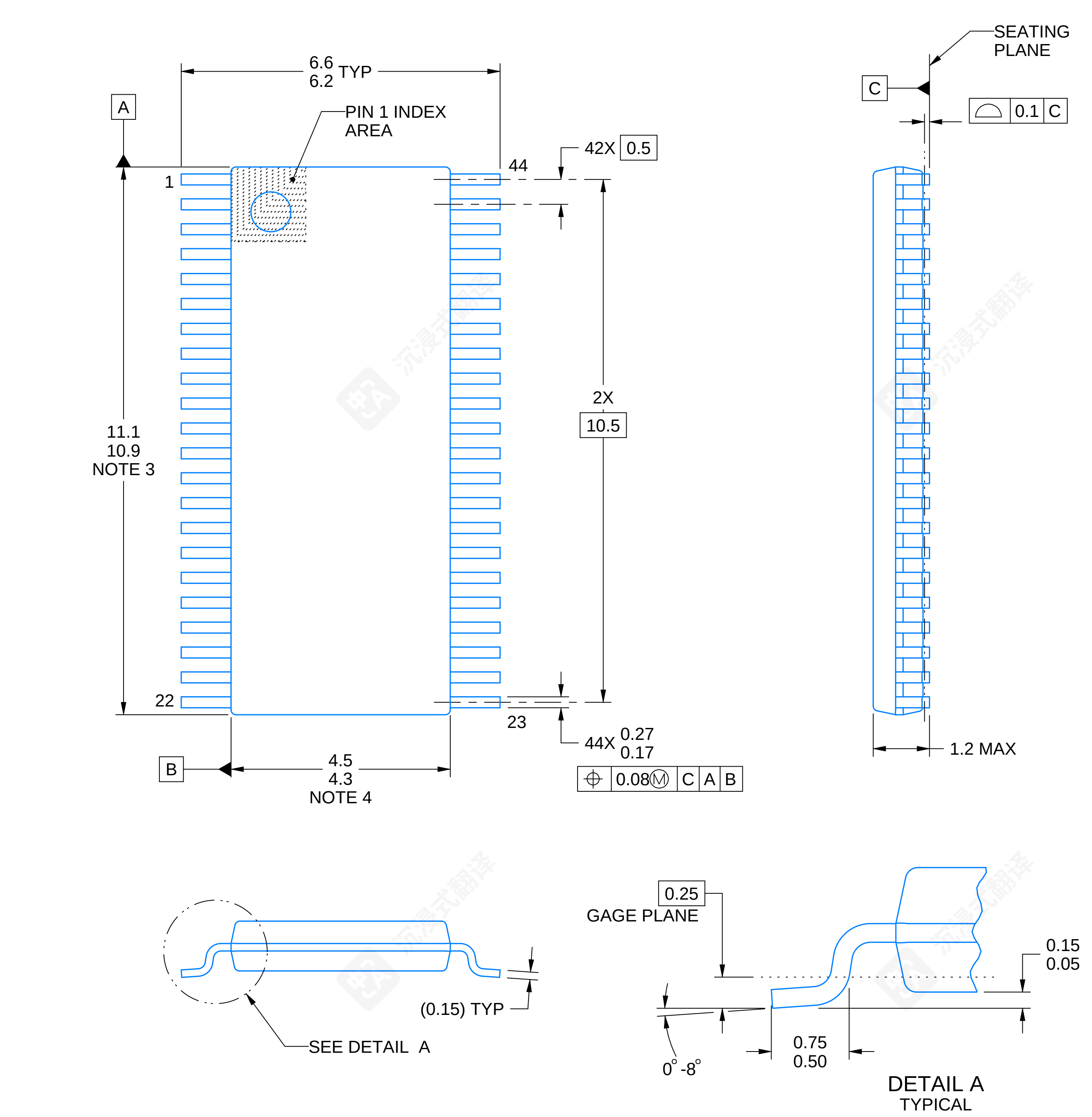
PACKAGE OUTLINE

TSSOP - 1.2 mm max height

SMALL OUTLINE PACKAGE



4220223/A 02/2017

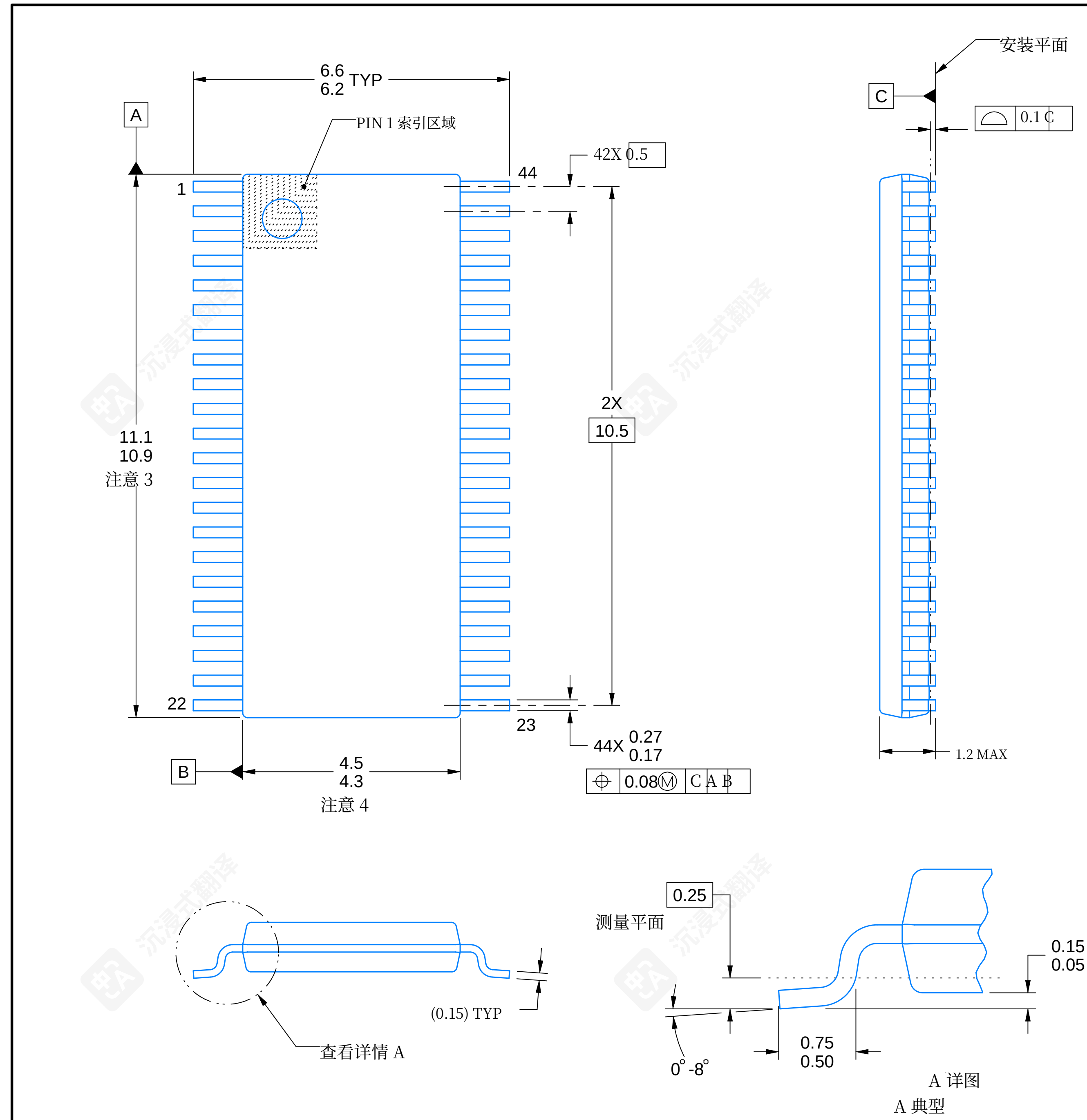
NOTES:

1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm per side.
4. This dimension does not include interlead flash. Interlead flash shall not exceed 0.25 mm per side.

封装轮廓

TSSOP - 最大高度 1.2 毫米

小型外型封装



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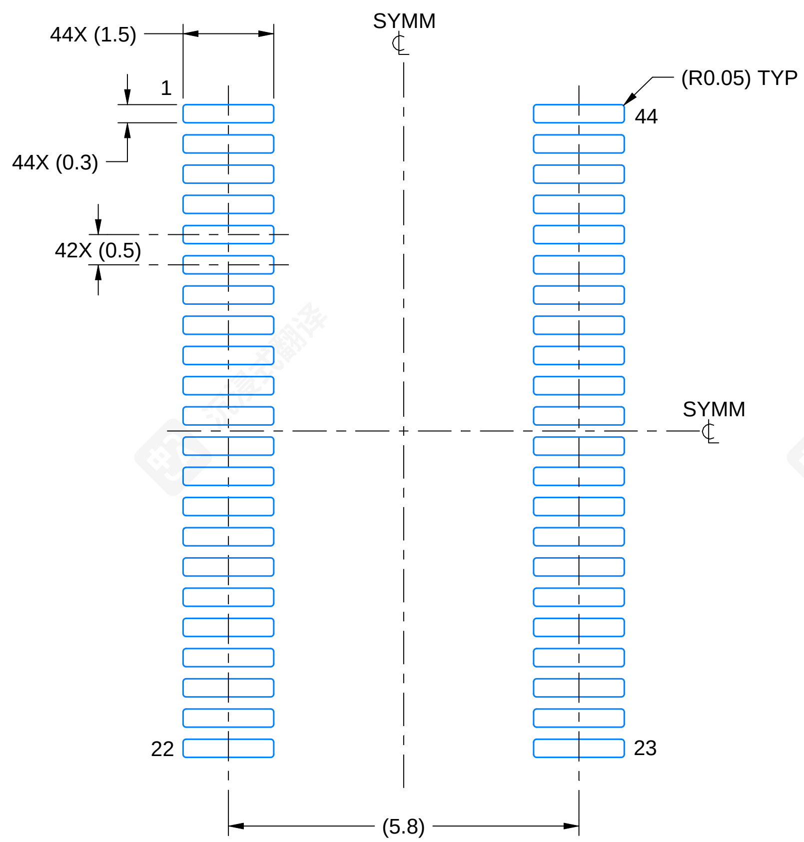
备注:

1. 所有线性尺寸以毫米为单位。括号中的尺寸仅供参考。尺寸和公差遵照 ASME Y14.5M 标准。
2. 本图纸如有更改恕不另行通知。
3. 此尺寸不包括模压毛刺、凸起或流道毛刺。模压毛刺、凸起或流道毛刺每侧不得超过 0.15 毫米。4. 此尺寸不包括引脚间毛刺。引脚间毛刺每侧不得超过 0.25 毫米。

DBT0044A **TSSOP - 1.2 mm max height**

SMALL OUTLINE PACKAGE



LAND PATTERN EXAMPLE
EXPOSED METAL SHOWN
SCALE: 8X

NON-SOLDER MASK DEFINED (PREFERRED)

SOLDER MASK DEFINED

SOLDER MASK DETAILS

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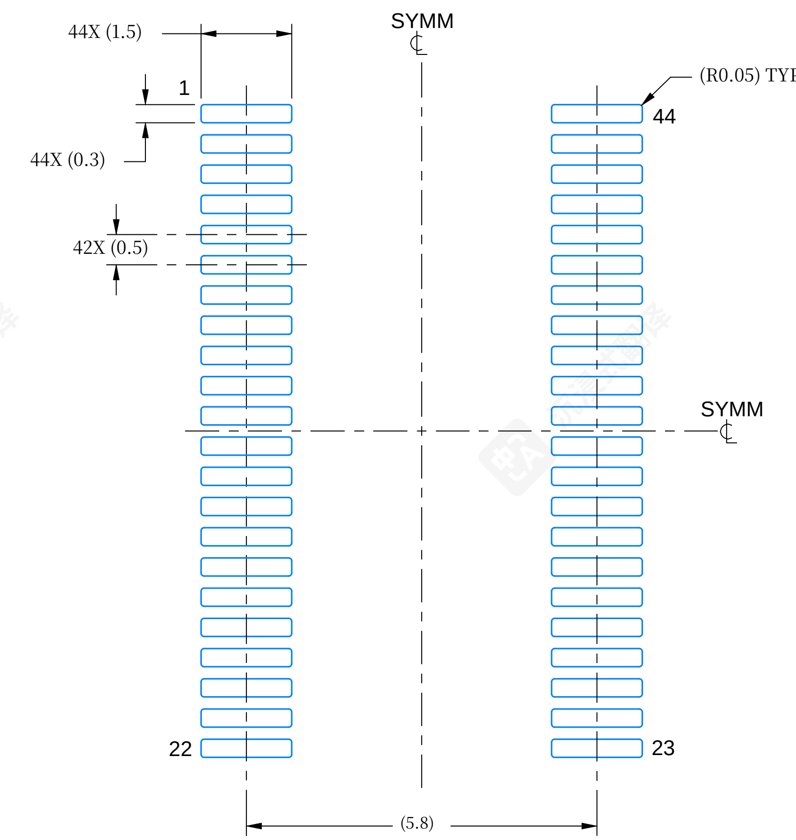
NOTES: (continued)

5. Publication IPC-7351 may have alternate designs.
6. Solder mask tolerances between and around signal pads can vary based on board fabrication site.

DBT0044A TSSOP - 1.2 mm 最大高度

小型外型封装



焊盘图形示例：暴露金属所示比例：
8X

非焊料遮蔽层定义
（首选）

焊料遮蔽层定义

焊盘细节

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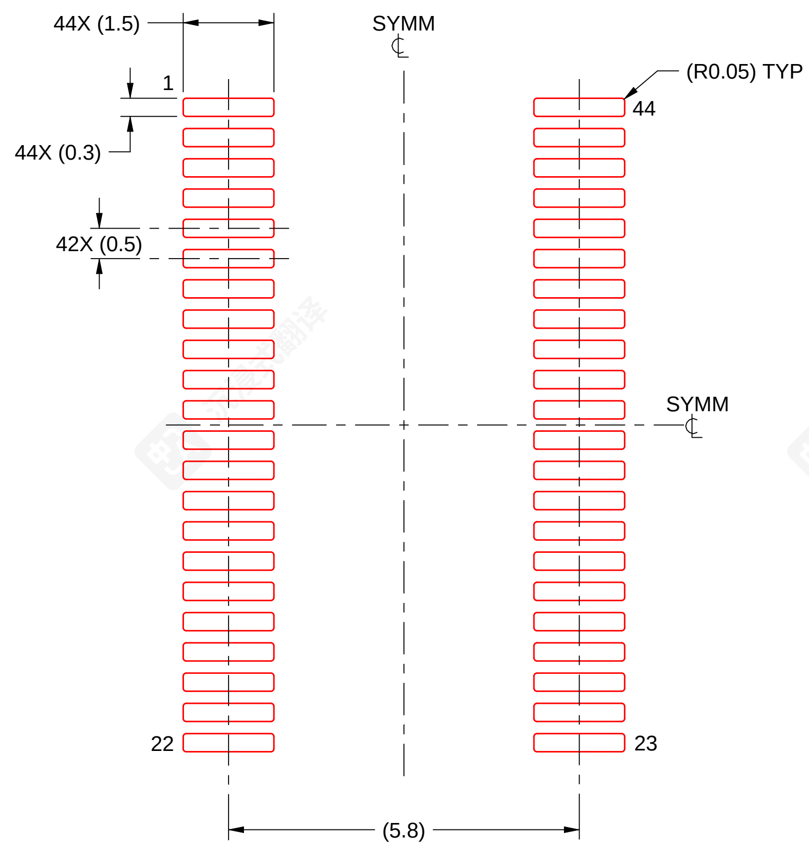
注释：（续）

5. 出版物IPC-7351可能有替代设计。
6. 焊膏层公差可在信号焊盘之间和周围根据电路板制造地点而变化。



SOLDER PASTE EXAMPLE
BASED ON 0.125 mm THICK STENCIL
SCALE: 8X

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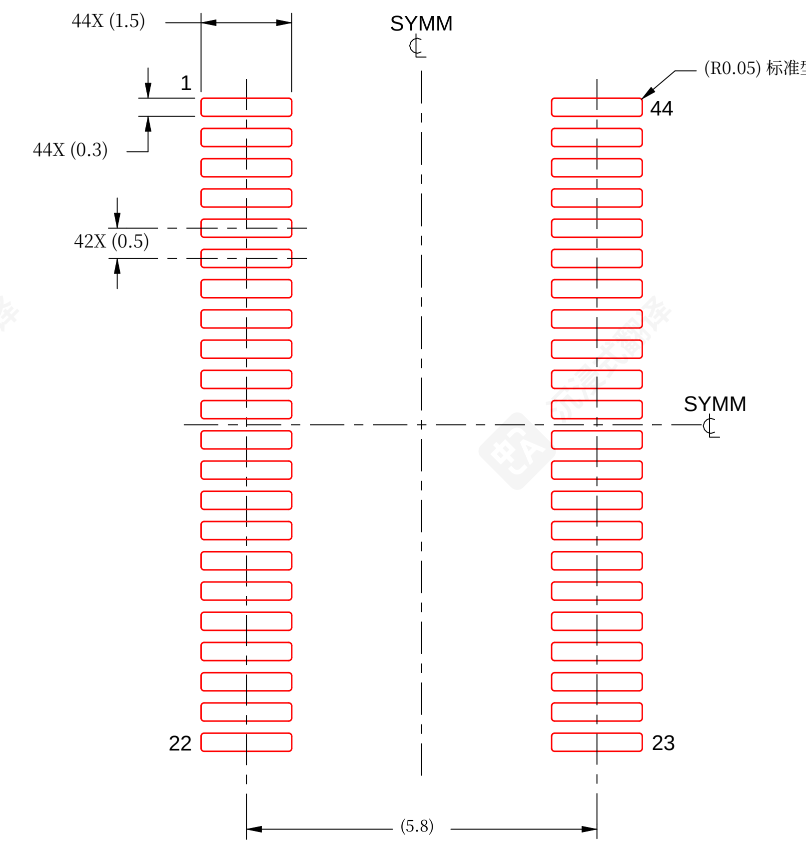
NOTES: (continued)

7. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
8. Board assembly site may have different recommendations for stencil design.



焊膏示例
基于 0.125 毫米厚的模板
比例：8X

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备注：继续

7. 具有梯形壁和圆角的激光切割孔可能提供更好的贴片释放效果。IPC-7525 可能存在其他设计建议。
8. PCB 组装现场可能对模板设计有不同建议。

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